

Physical Layer

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Introduction to Physical Layer

What is the Physical Layer?

The Physical Layer is the lowest layer of the OSI model

It is responsible for transmitting raw bits (0s and 1s) over a communication channel.

Defines how data is physically sent from one device to another.

Key Focus

Transmission of bits

Physical connection between devices

 **Example:**

Sending binary data through a cable or wireless signal

Functions of the Physical Layer

Main Functions

Defines bit representation (0s and 1s)

Controls data rate (speed of transmission)

Defines transmission mode

- Simplex

- Half-duplex

- Full-duplex

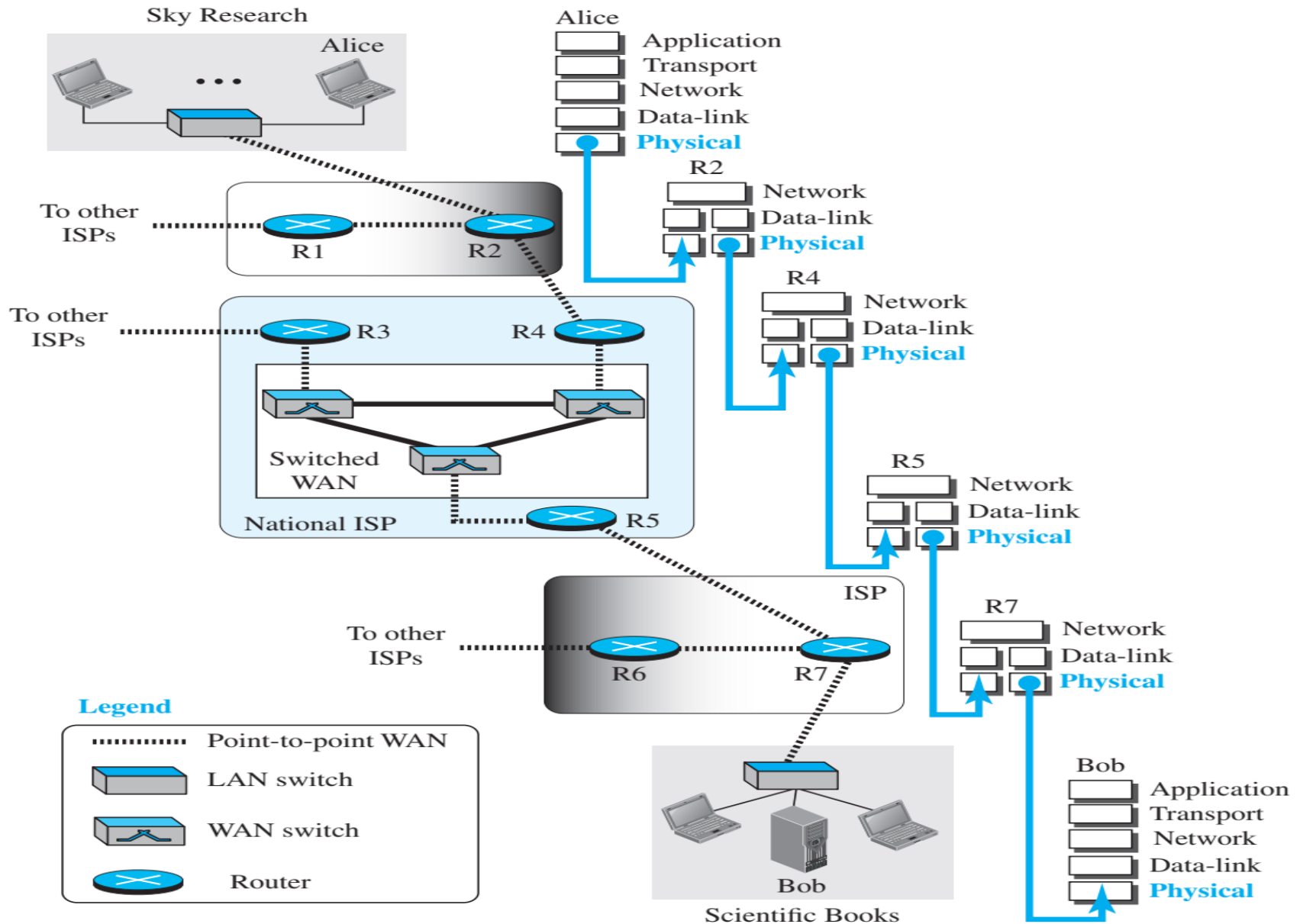
Specifies physical media and connectors

Handles signal transmission (analog & digital)

Important Point:

Physical Layer does not understand data, it only transmits bits.

Communication at the physical layer



Note

**Communication at application, transport, network, and data-link layer is logical;
communication at the physical layer is physical.**



Note

To be transmitted, data must be transformed to electromagnetic signals.

3-1 ANALOG AND DIGITAL

- **Data can be analog or digital.**

The term analog data refers to information that is continuous; digital data refers to information that has discrete states.

For example, an analog clock that has hour, minute, and second hands gives information in a continuous form; the movements of the hands are continuous. On the other hand, a digital clock that reports the hours and the minutes will change suddenly from 8:05 to 8:06.

Topics discussed in this section:

Analog and Digital Data

Analog and Digital Signals

Periodic and Nonperiodic Signals

Note

**Data can be analog or digital.
Analog data are continuous and take
continuous values.
Digital data have discrete states and
take discrete values.**

Note

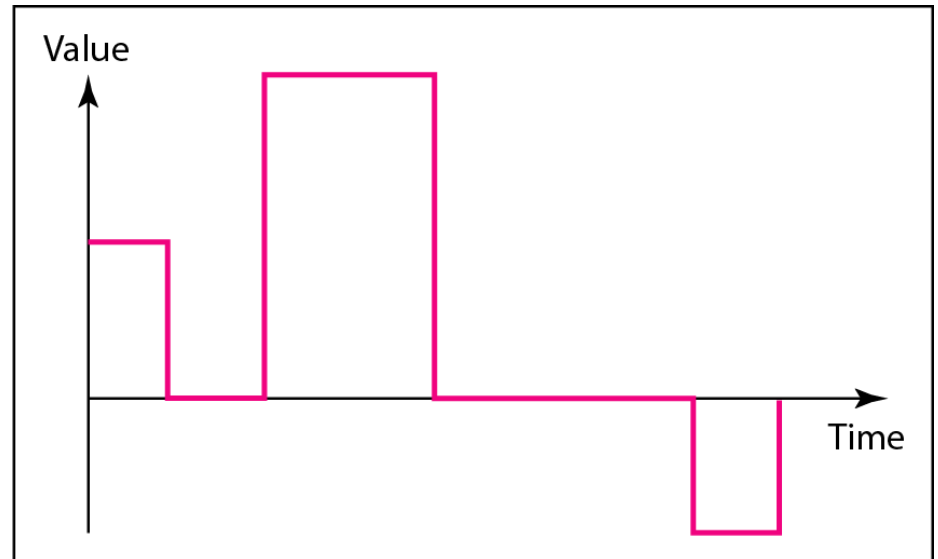
Signals can be analog or digital. Analog signals can have an infinite number of values in a range; digital signals can have only a limited number of values.

Figure 3.1 *Comparison of analog and digital signals*



a. Analog signal

An **analog signal** has infinitely many levels of intensity over a period of time. As the wave moves from value A to value B , it passes through and includes an infinite number of values along its path.



b. Digital signal

A **digital signal**, on the other hand, can have only a limited number of defined values. Although each value can be any number, it is often as simple as 1 and 0.

Analog Signal

An **analog signal** is a **continuous signal** that varies smoothly with time and represents physical quantities such as sound or light.

Key Points (Analog):

- Continuous by nature
- Uses curved (sine) waveforms
- Commonly used for audio and video signals
- More affected by noise and interference
- Higher error rate
- Suitable for short-distance transmission
- Uses amplifiers, which increase signal strength but also amplify noise
- Requires large bandwidth compared to digital

Example:

- Traditional telephone lines
- Radio and TV broadcasting

Digital Signal

A **digital signal** is a **discrete signal** that represents data in the form of **binary bits (0 and 1)**.

Key Points (Digital):

Discrete by nature

Uses square waveforms

Used for data, file transfer, and computer communication

Low error rate

Less affected by noise

Suitable for long-distance transmission

Uses repeaters, which regenerate and correct signals

Requires low bandwidth

Example:

Computer networks

Internet communication

Mobile communication

Note

In data communications, we commonly use periodic analog signals and nonperiodic digital signals.

3-2 PERIODIC ANALOG SIGNALS

Periodic analog signals can be classified as **simple** or **composite**. A simple periodic analog signal, a **sine wave**, cannot be decomposed into simpler signals. A composite periodic analog signal is composed of multiple sine waves.

Topics discussed in this section:

Sine Wave

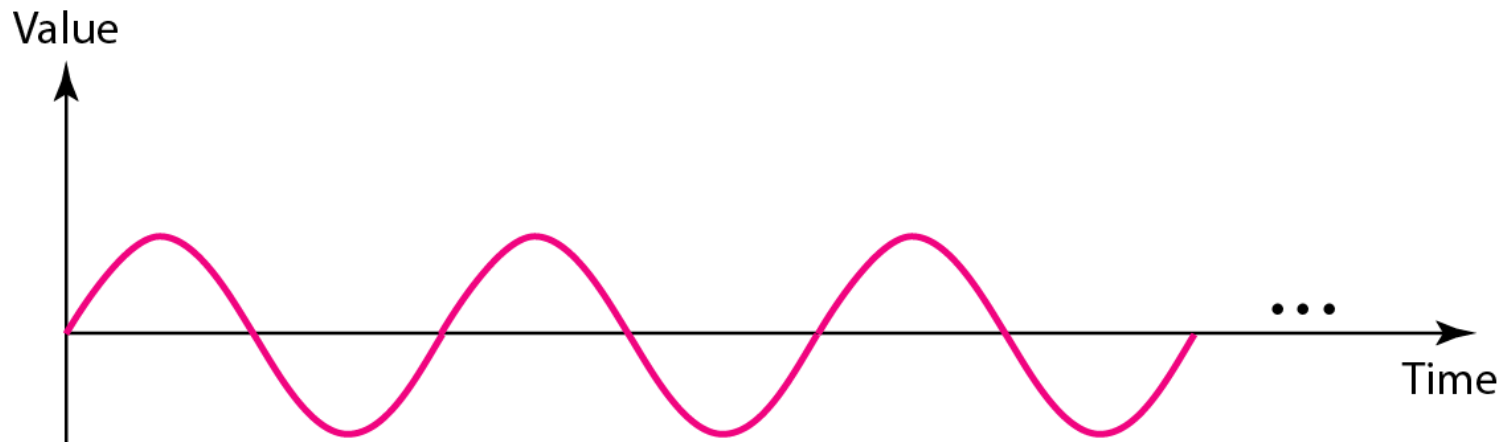
Wavelength

Time and Frequency Domain

Composite Signals

Bandwidth

Figure 3.3 *A sine wave*



A sine wave can be represented by three parameters: *the peak amplitude, the frequency, and the phase.*



Example 3.1

The sine wave is the fundamental periodic signal. A general sine wave can be represented by three parameters:

peak amplitude (A) - The peak amplitude of a signal is the absolute value of its highest intensity, proportional to the energy it carries. For electric signals, peak amplitude is normally measured in volts.

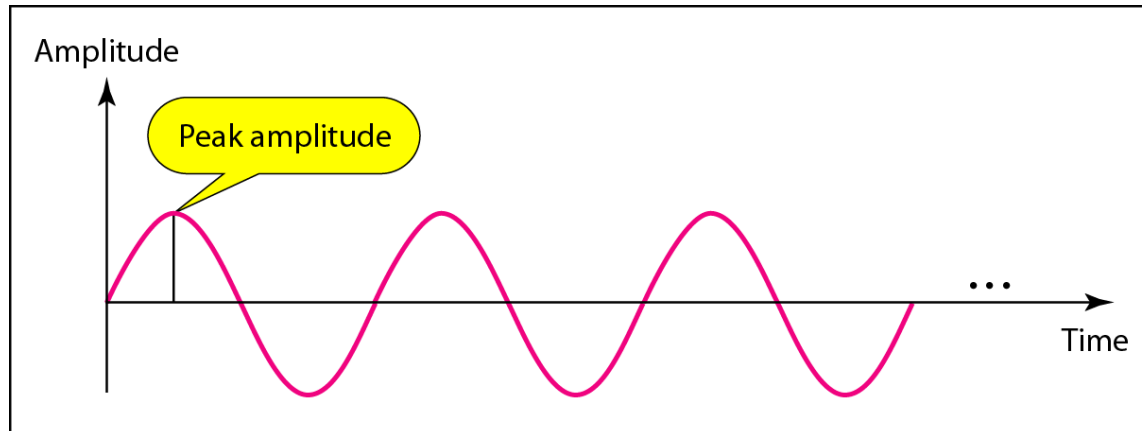
frequency (f) - the rate [in cycles per second, or Hertz (Hz)] at which the signal repeats. An equivalent parameter is the period (T) of a signal, so $T = 1/f$.

phase (ϕ) - measure of relative position in time within a single period of a signal, illustrated subsequently

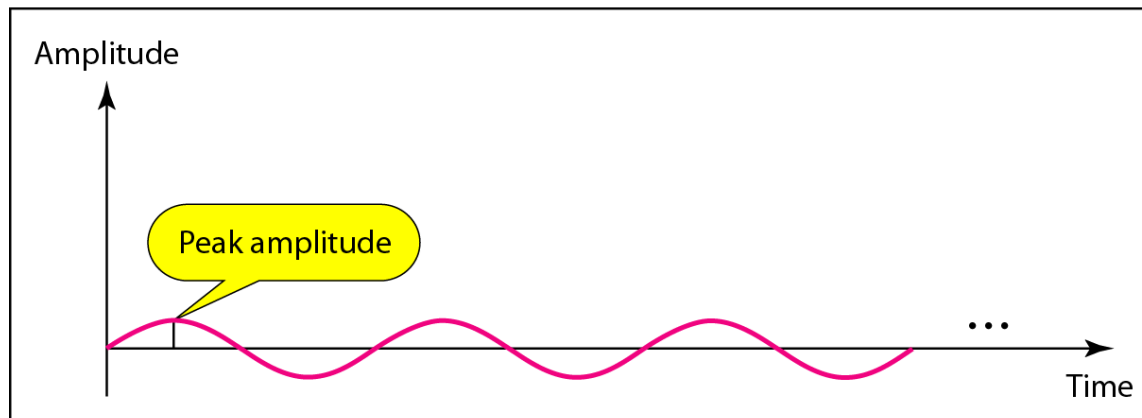
Period (T) - Period refers to the amount of time, in seconds, a signal needs to complete 1 cycle.

Frequency and period are the inverse of each other.

Figure 3.4 *Two signals with the same phase and frequency, but different amplitudes*



a. A signal with high peak amplitude



b. A signal with low peak amplitude



Example 3.2

The voltage of a battery is a constant; this constant value can be considered a sine wave, as we will see later. For example, the peak value of an AA battery is normally 1.5 V.



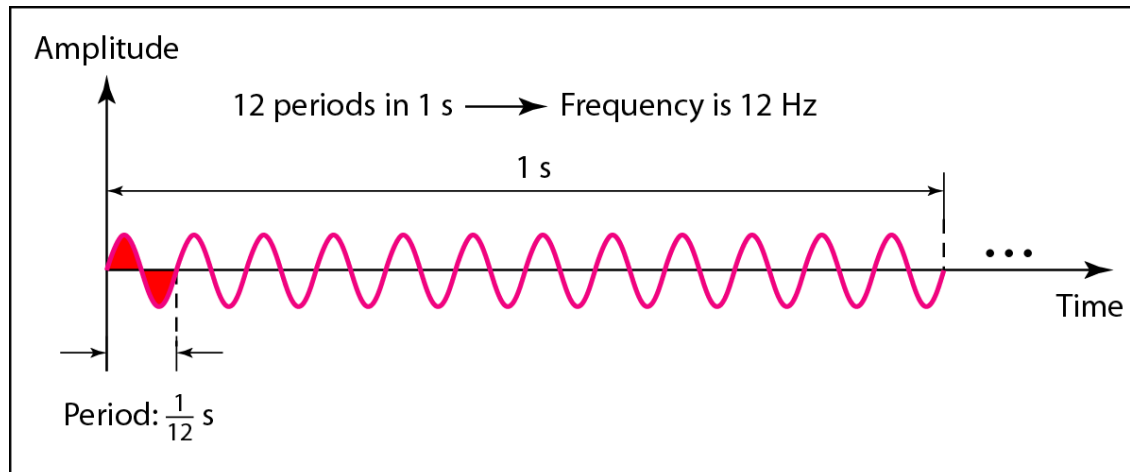
Frequency and Period

Note

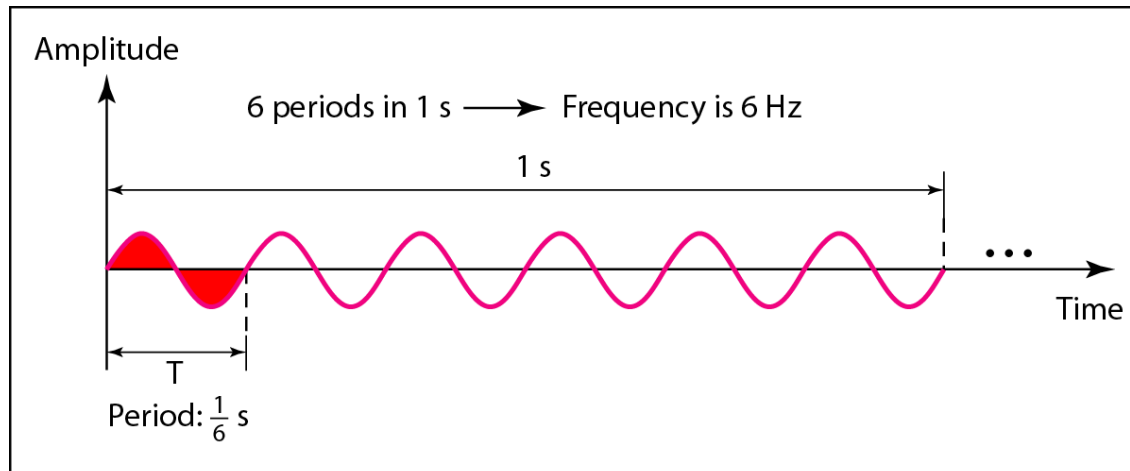
Frequency and period are the inverse of each other.

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

Figure 3.5 *Two signals with the same amplitude and phase, but different frequencies*



a. A signal with a frequency of 12 Hz



b. A signal with a frequency of 6 Hz

Table 3.1 *Units of period and frequency*

<i>Unit</i>	<i>Equivalent</i>	<i>Unit</i>	<i>Equivalent</i>
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^3 Hz
Microseconds (μ s)	10^{-6} s	Megahertz (MHz)	10^6 Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^9 Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz



Example 3.3

The power we use at home has a frequency of 60 Hz. The period of this sine wave can be determined as follows:



Example 3.3

The power we use at home has a frequency of 60 Hz. The period of this sine wave can be determined as follows:

$$T = \frac{1}{f} = \frac{1}{60} = 0.0166 \text{ s} = 0.0166 \times 10^3 \text{ ms} = 16.6 \text{ ms}$$



Example 3.4

Express a period of 100 ms in microseconds.



Example 3.4

Express a period of 100 ms in microseconds.

Solution

From Table 3.1 we find the equivalents of 1 ms (1 ms is 10^{-3} s) and 1 s (1 s is 10^6 μ s). We make the following substitutions:.

$$100 \text{ ms} = 100 \times 10^{-3} \text{ s} = 100 \times 10^{-3} \times 10^6 \mu\text{s} = 10^2 \times 10^{-3} \times 10^6 \mu\text{s} = 10^5 \mu\text{s}$$



Example 3.5

The period of a signal is 100 ms. What is its frequency in kilohertz?

Solution

First we change 100 ms to seconds, and then we calculate the frequency from the period ($1 \text{ Hz} = 10^{-3} \text{ kHz}$).

$$100 \text{ ms} = 100 \times 10^{-3} \text{ s} = 10^{-1} \text{ s}$$
$$f = \frac{1}{T} = \frac{1}{10^{-1}} \text{ Hz} = 10 \text{ Hz} = 10 \times 10^{-3} \text{ kHz} = 10^{-2} \text{ kHz}$$

Note

Frequency is the rate of change with respect to time.

Change in a short span of time means high frequency.

Change over a long span of time means low frequency.

Note

If a signal does not change at all, its frequency is zero.

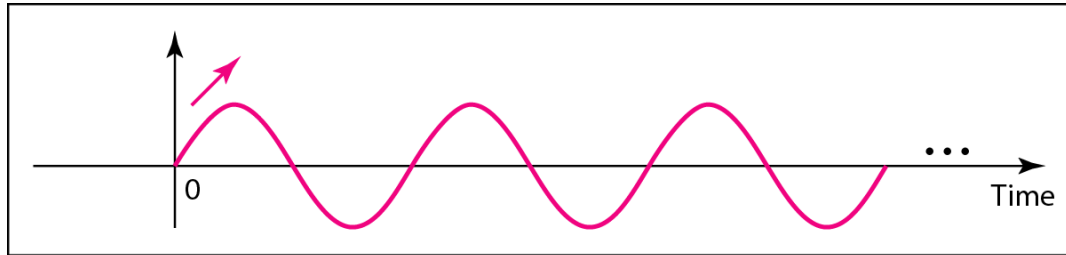
If a signal changes instantaneously, its frequency is infinite.



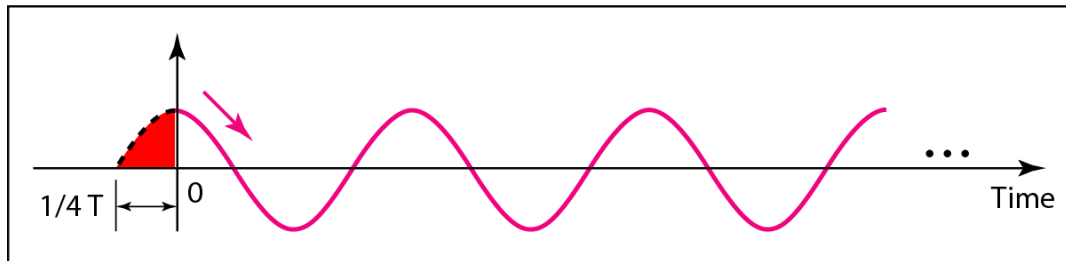
Note

Phase describes the position of the waveform relative to time 0.

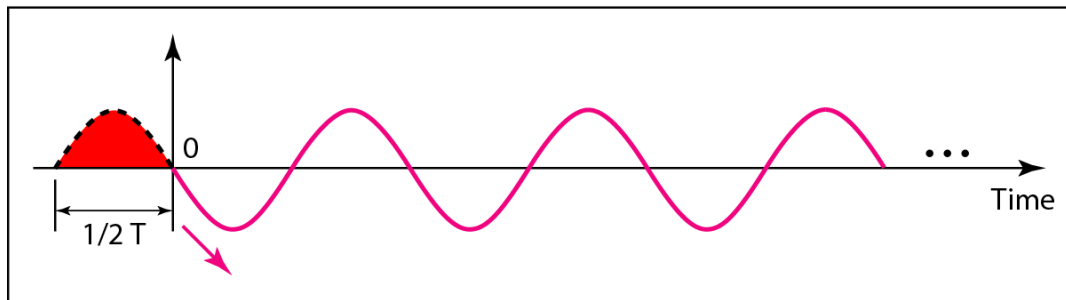
Figure 3.6 *Three sine waves with the same amplitude and frequency, but different phases*



a. 0 degrees



b. 90 degrees



c. 180 degrees

Looking at Figure 3.6, we can say that

- a.** A sine wave with a phase of 0° starts at time 0 with a zero amplitude. The amplitude is increasing.
- b.** A sine wave with a phase of 90° starts at time 0 with a peak amplitude. The amplitude is decreasing.
- c.** A sine wave with a phase of 180° starts at time 0 with a zero amplitude. The amplitude is decreasing.

Another way to look at the phase is in terms of shift or offset. We can say that

- a.** A sine wave with a phase of 0° is not shifted.
- b.** A sine wave with a phase of 90° is shifted to the left by cycle. However, note that the signal does not really exist before time 0.
- c.** A sine wave with a phase of 180° is shifted to the left by cycle. However, note that the signal does not really exist before time 0.



Example

A sine wave is offset cycle with respect to time 0. What is its phase in degrees and radians?



Example

A sine wave is offset cycle with respect to time 0. What is its phase in degrees and radians?

Solution

We know that 1 complete cycle is 360°. Therefore, cycle is

$$\frac{1}{6} \times 360 = 60^\circ = 60 \times \frac{2\pi}{360} \text{ rad} = \frac{\pi}{3} \text{ rad} = 1.046 \text{ rad}$$

Wavelength

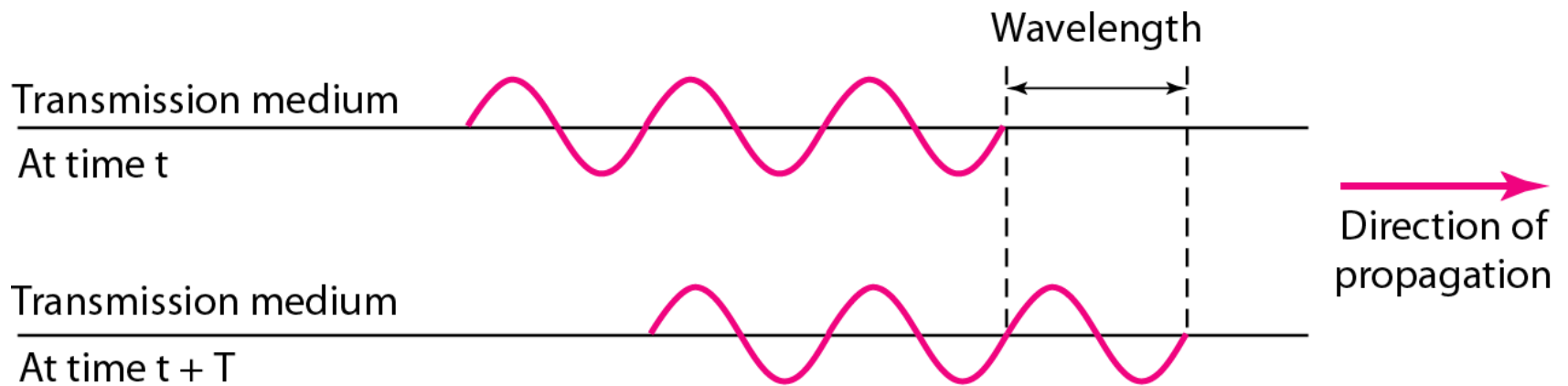
Wavelength is another characteristic of a signal traveling through a transmission medium. Wavelength binds the period or the frequency of a simple sine wave to the **propagation speed** of the medium.

While the frequency of a signal is independent of the medium, the wavelength depends on both the frequency and the medium. Wavelength is a property of any type of signal. In data communications, we often use wavelength to describe the transmission of light in an optical fiber. The wavelength is the distance a simple signal can travel in one period.

$$\text{Wavelength} = (\text{propagation speed}) \times \text{period} = \frac{\text{propagation speed}}{\text{frequency}}$$

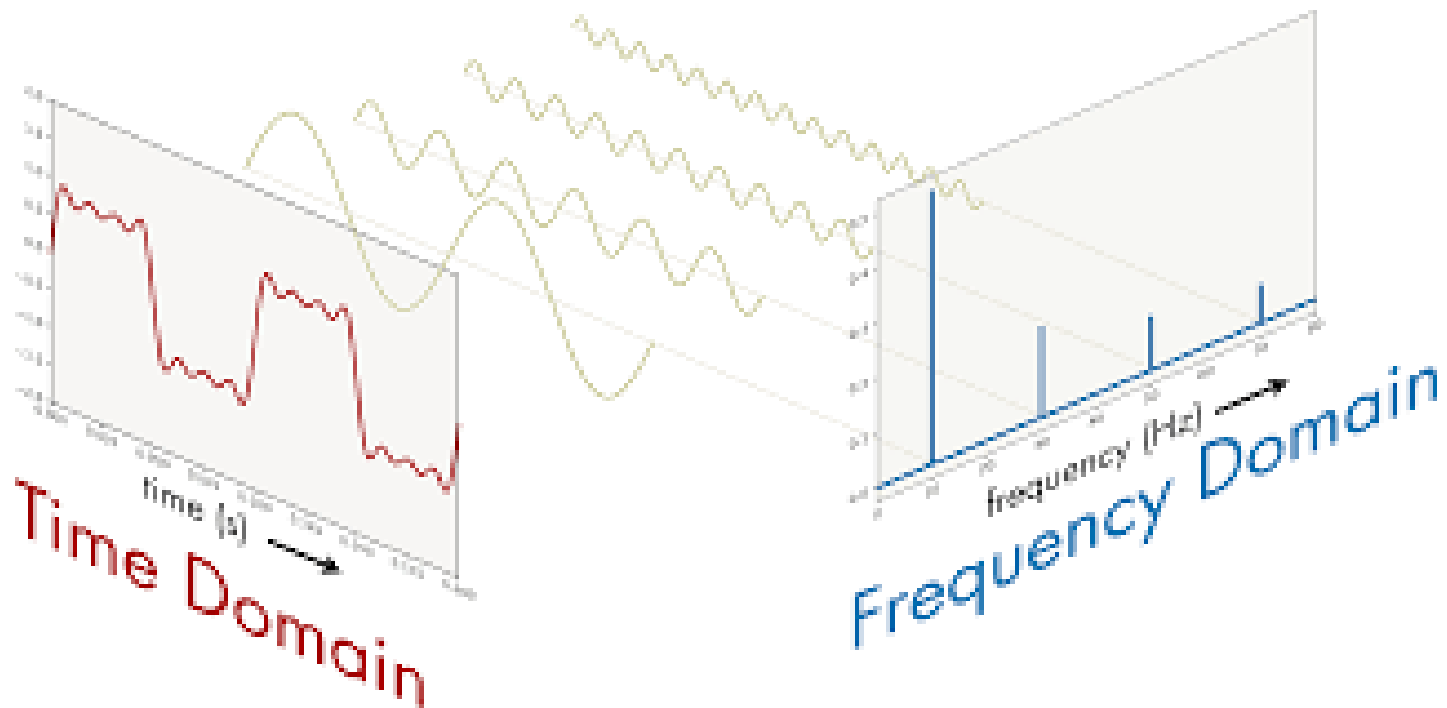
$$\lambda = \frac{c}{f}$$

Figure 3.6 *Wavelength and period*



Note

Time and Frequency Domains.



Time and Frequency Domains.

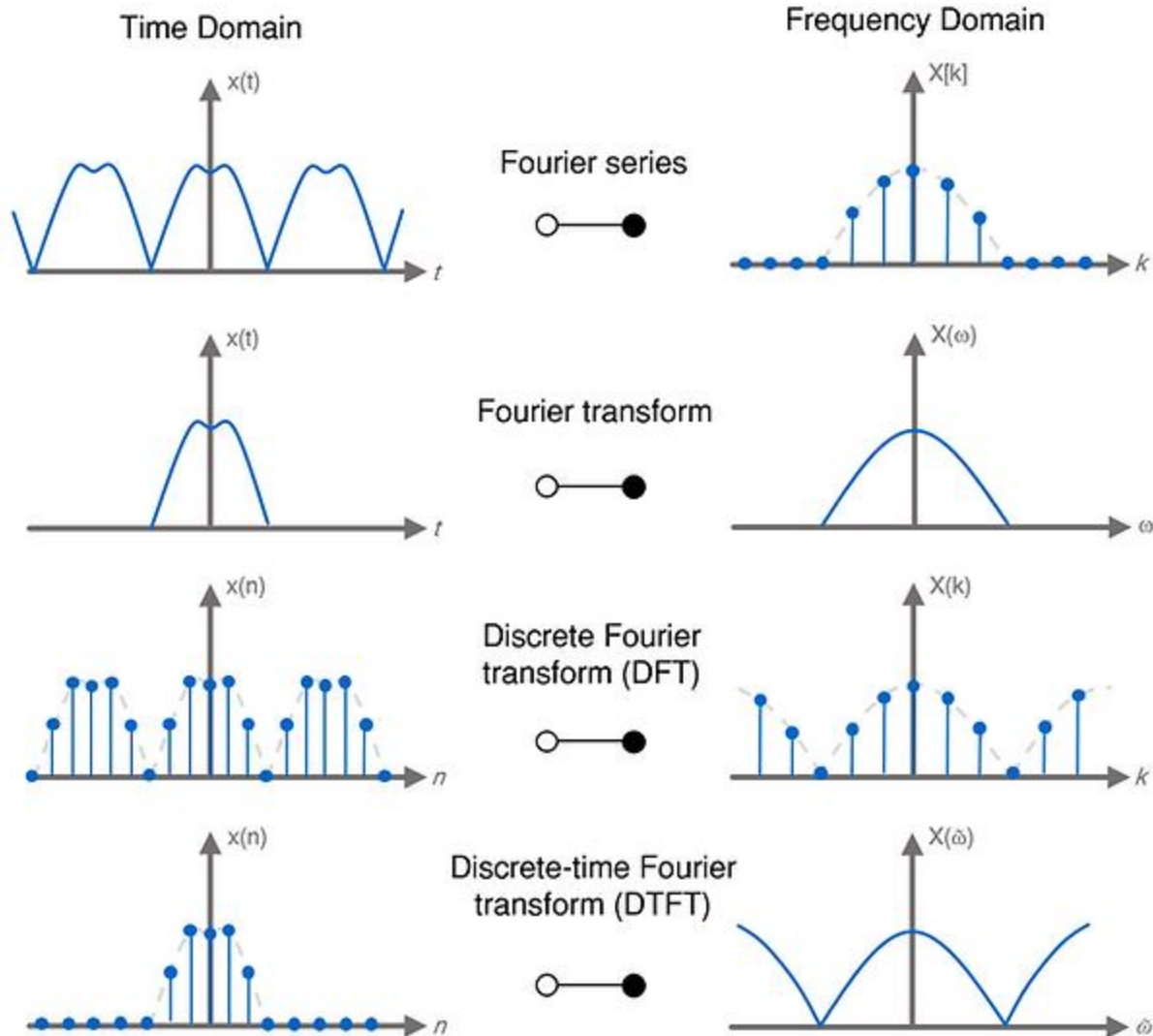
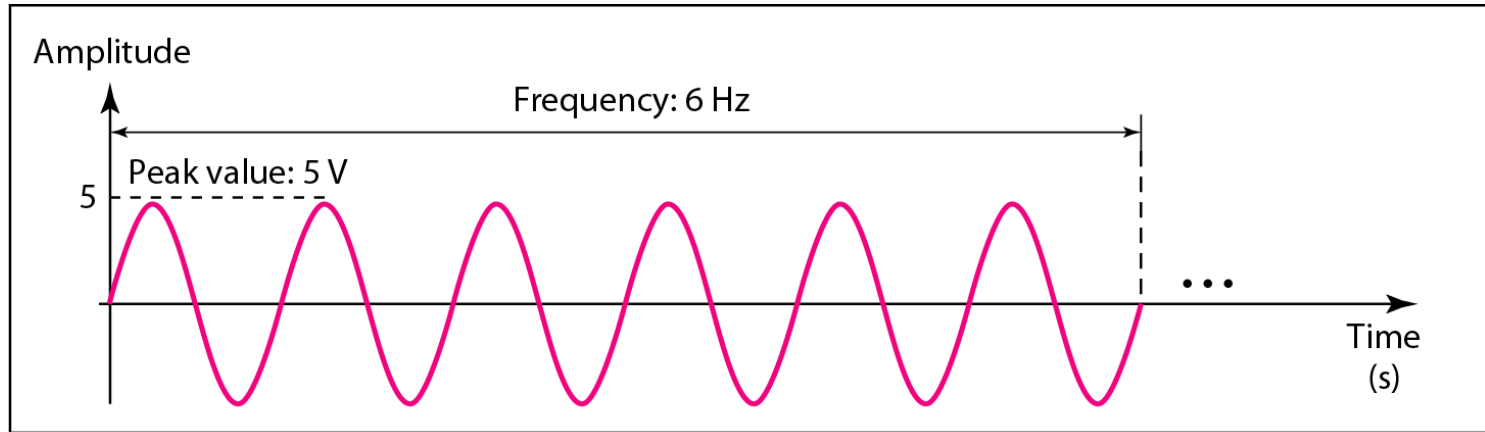
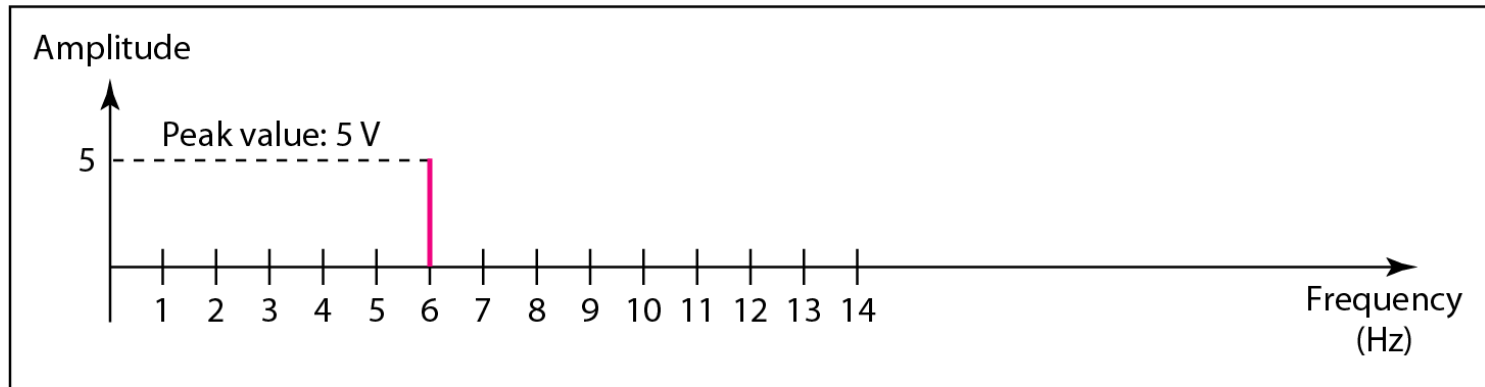


Figure 3.7 *The time-domain and frequency-domain plots of a sine wave*



a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)



b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)



Note

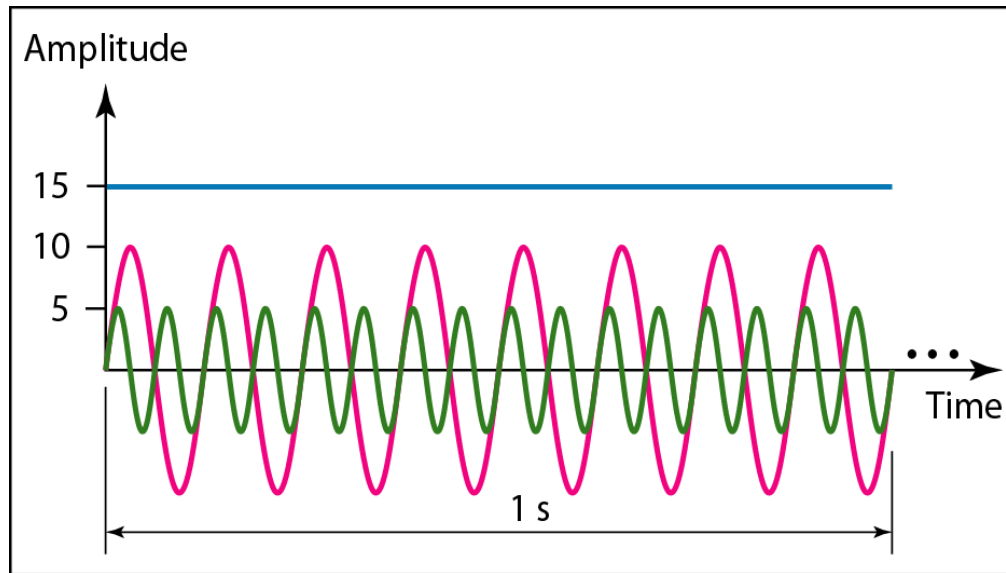
A complete sine wave in the time domain can be represented by one single spike in the frequency domain.



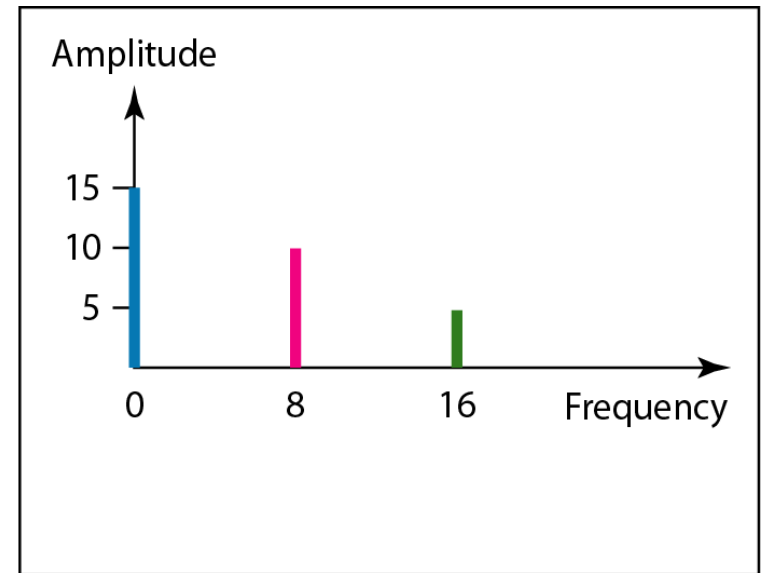
Example 3.7

The frequency domain is more compact and useful when we are dealing with more than one sine wave. For example, Figure 3.8 shows three sine waves, each with different amplitude and frequency. All can be represented by three spikes in the frequency domain.

Figure 3.8 *The time domain and frequency domain of three sine waves*



a. Time-domain representation of three sine waves with frequencies 0, 8, and 16



b. Frequency-domain representation of the same three signals

Note

A single-frequency sine wave is not useful in data communications; we need to send a composite signal, a signal made of many simple sine waves.

Note

According to Fourier analysis, any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases.



Note

**If the composite signal is periodic, the decomposition gives a series of signals with discrete frequencies;
if the composite signal is nonperiodic, the decomposition gives a combination of sine waves with continuous frequencies.**

Figure 3.9 *A composite periodic signal*

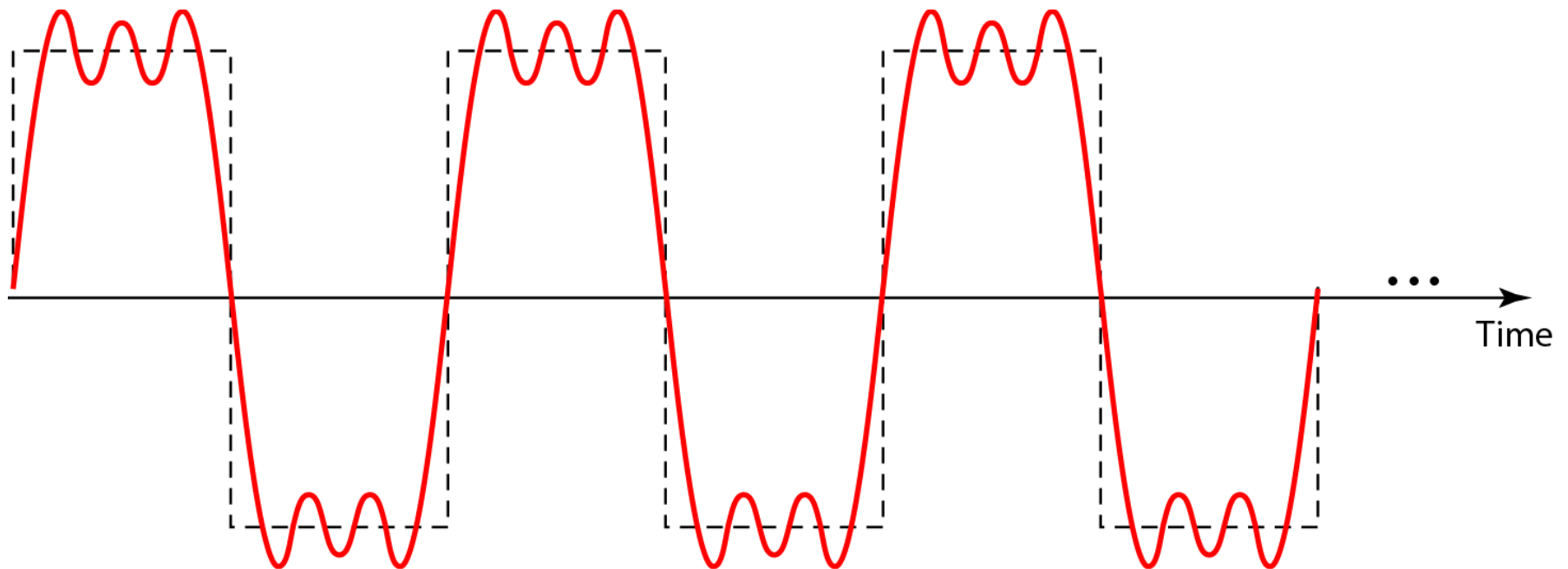
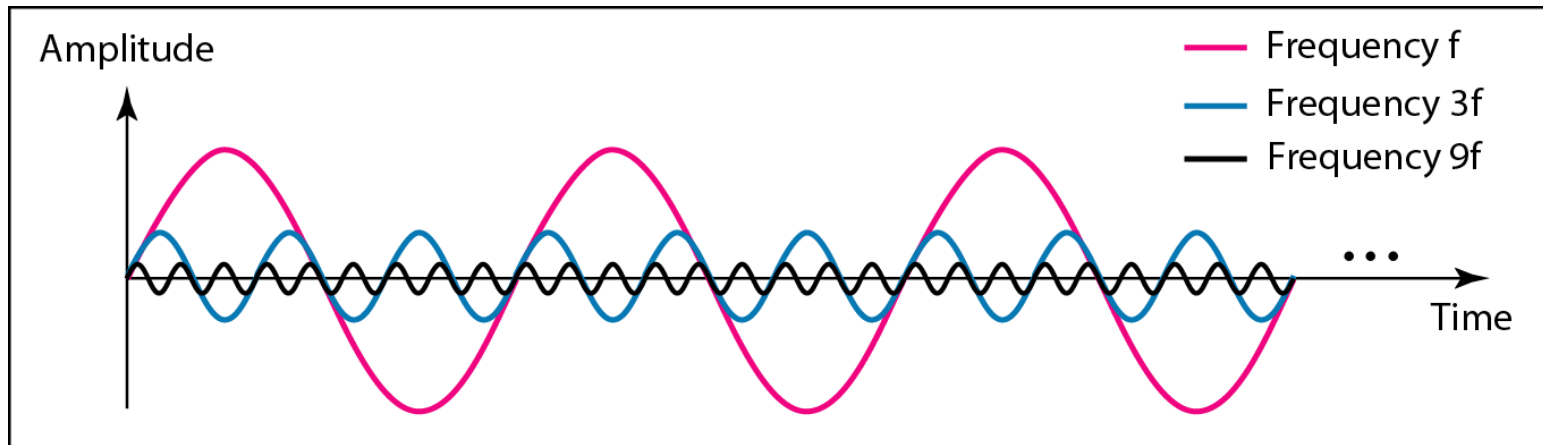
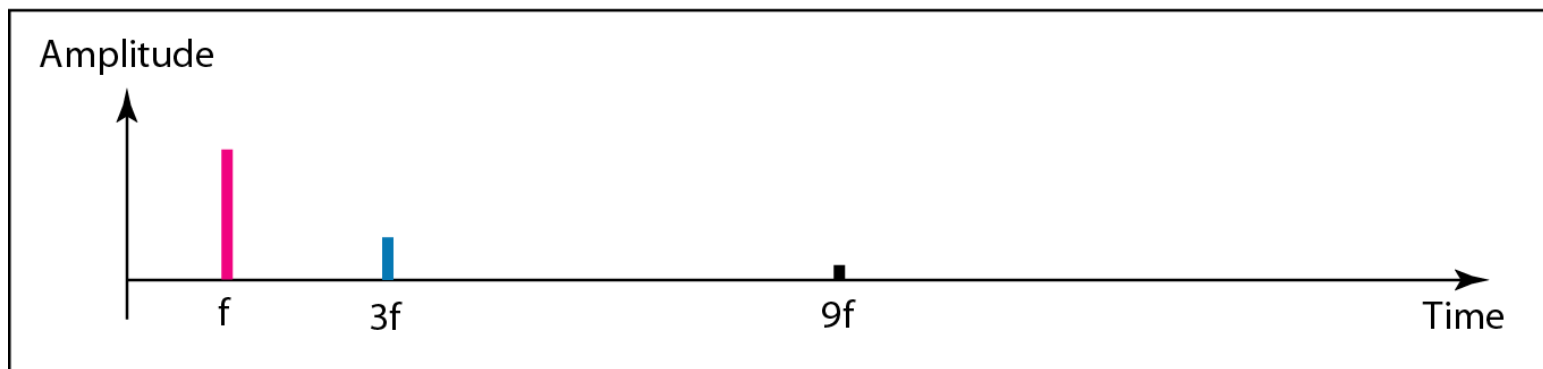


Figure 3.10 *Decomposition of a composite periodic signal in the time and frequency domains*



a. Time-domain decomposition of a composite signal



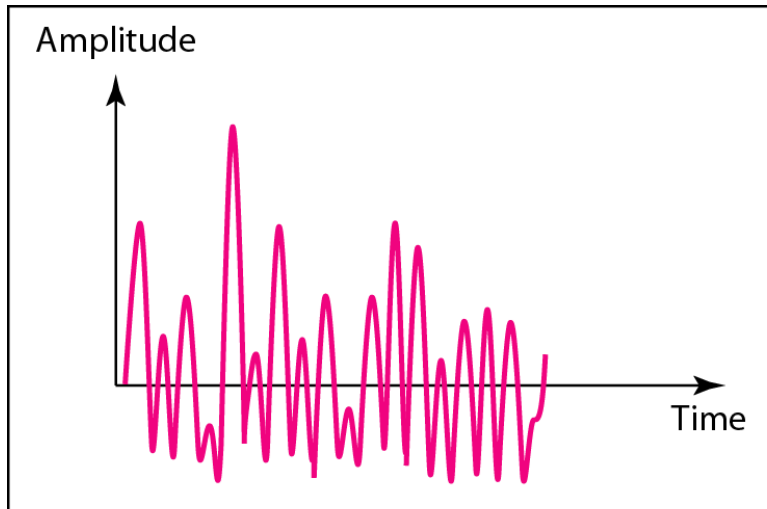
b. Frequency-domain decomposition of the composite signal



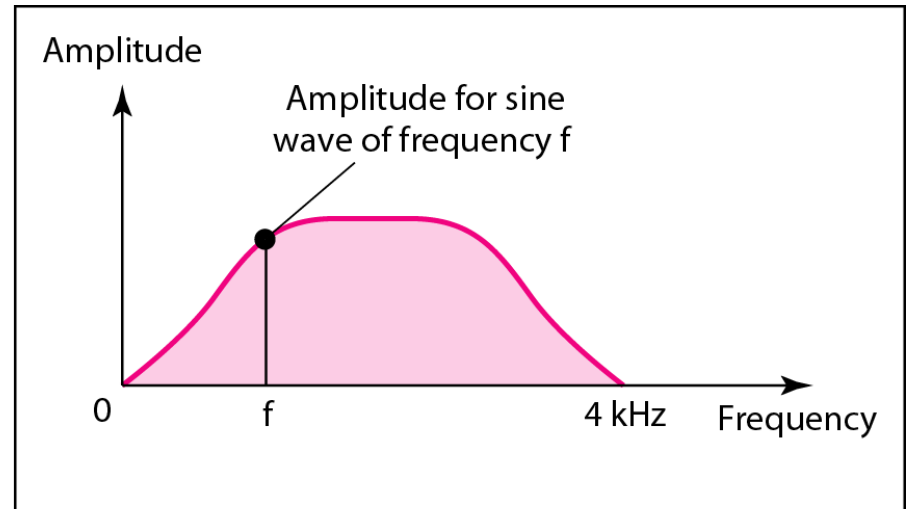
Example 3.9

Figure 3.11 shows a nonperiodic composite signal. It can be the signal created by a microphone or a telephone set when a word or two is pronounced. In this case, the composite signal cannot be periodic, because that implies that we are repeating the same word or words with exactly the same tone.

Figure 3.11 *The time and frequency domains of a nonperiodic signal*



a. Time domain

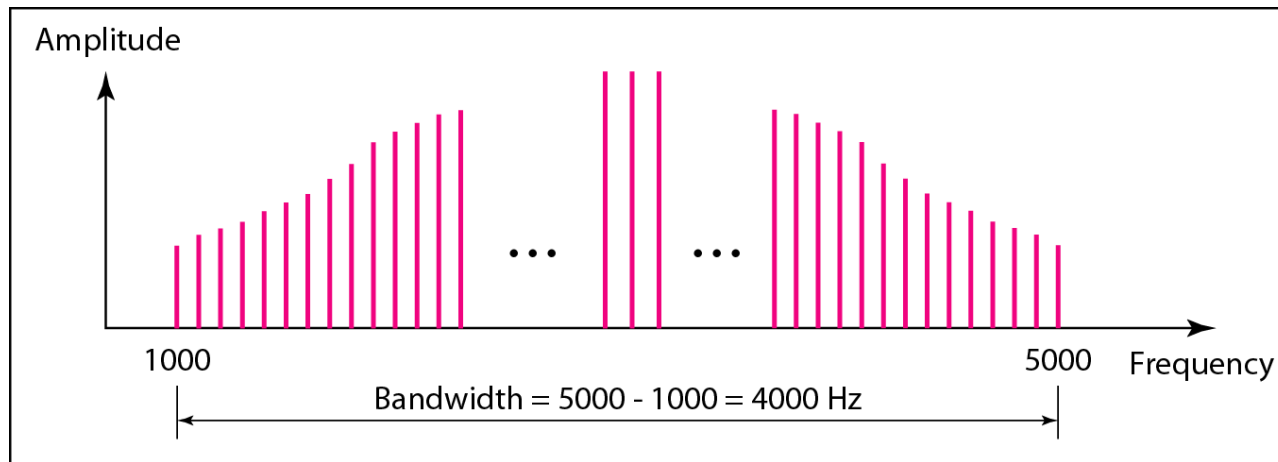


b. Frequency domain

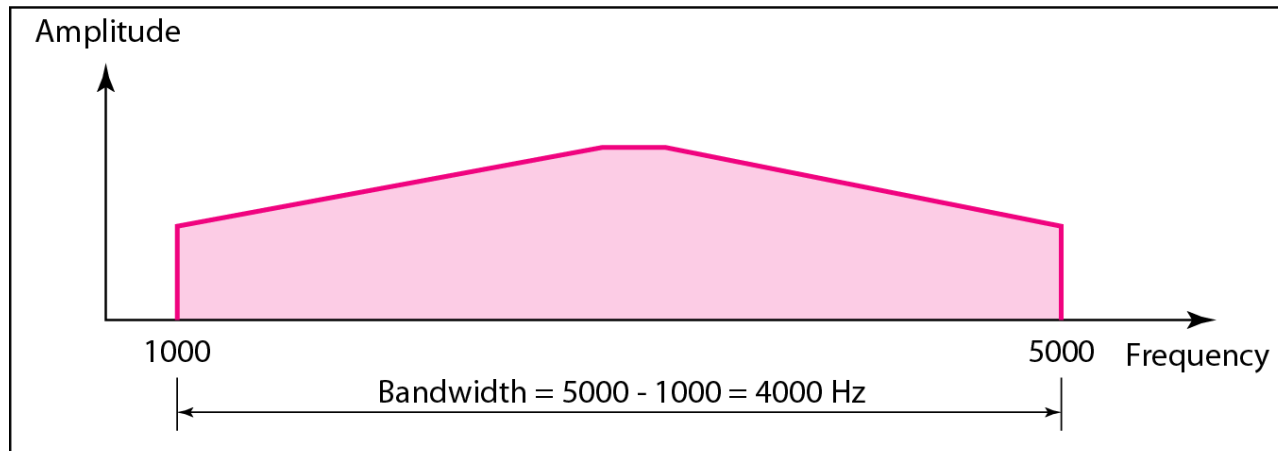
Note

The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal.

Figure 3.12 *The bandwidth of periodic and nonperiodic composite signals*



a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal



Example 3.10

If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.

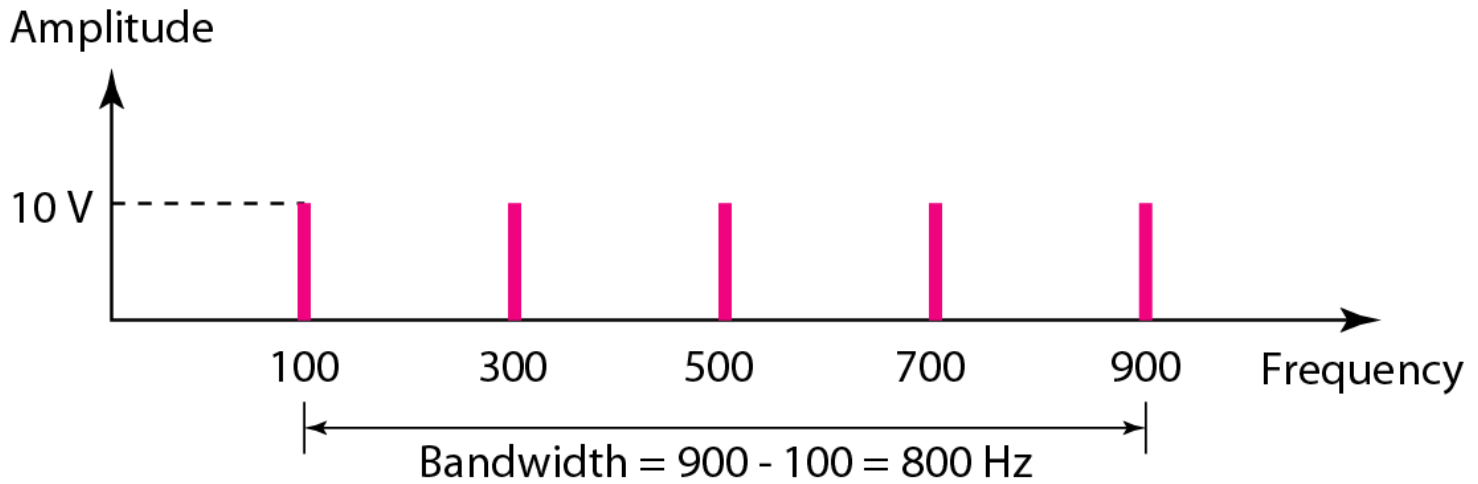
Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l = 900 - 100 = 800 \text{ Hz}$$

The spectrum has only five spikes, at 100, 300, 500, 700, and 900 Hz (see Figure 3.13).

Figure 3.13 *The bandwidth for Example 3.10*





Example 3.11

A periodic signal has a bandwidth of 20 Hz. The highest frequency is 60 Hz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of the same amplitude.

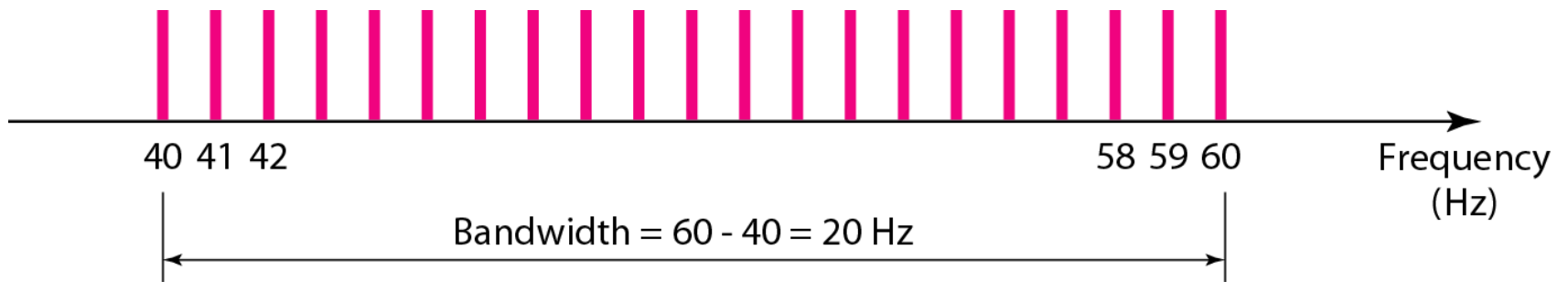
Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l \Rightarrow 20 = 60 - f_l \Rightarrow f_l = 60 - 20 = 40 \text{ Hz}$$

The spectrum contains all integer frequencies. We show this by a series of spikes (see Figure 3.14).

Figure 3.14 *The bandwidth for Example 3.11*





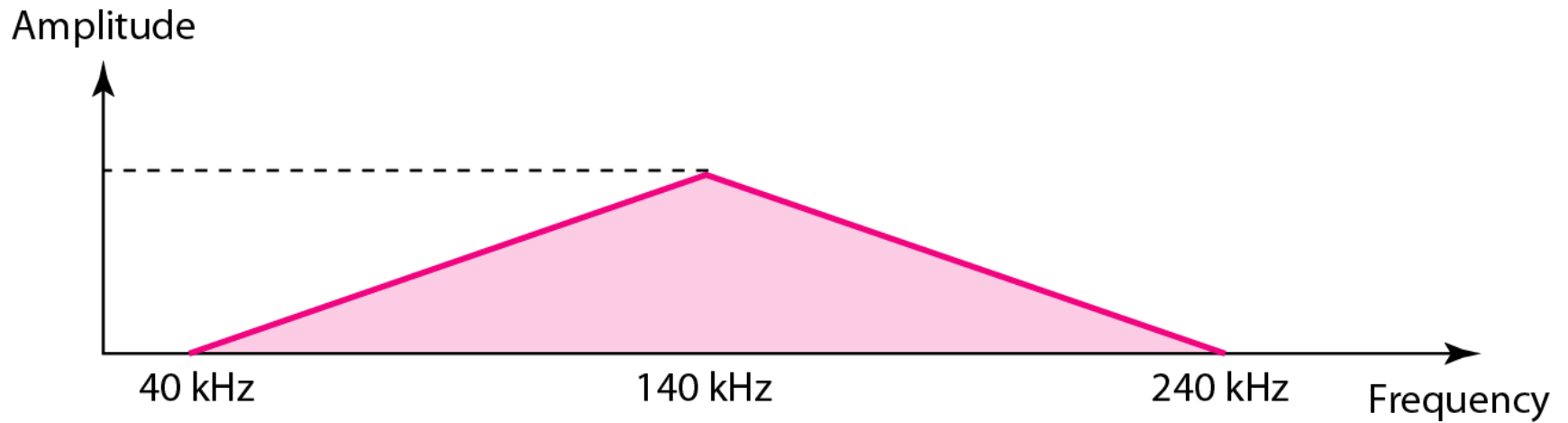
Example 3.12

A nonperiodic composite signal has a bandwidth of 200 kHz, with a middle frequency of 140 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal.

Solution

The lowest frequency must be at 40 kHz and the highest at 240 kHz. Figure 3.15 shows the frequency domain and the bandwidth.

Figure 3.15 *The bandwidth for Example 3.12*



3-3 DIGITAL SIGNALS

In addition to being represented by an analog signal, information can also be represented by a **digital signal**. For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels. In this case, we can send more than 1 bit for each level.

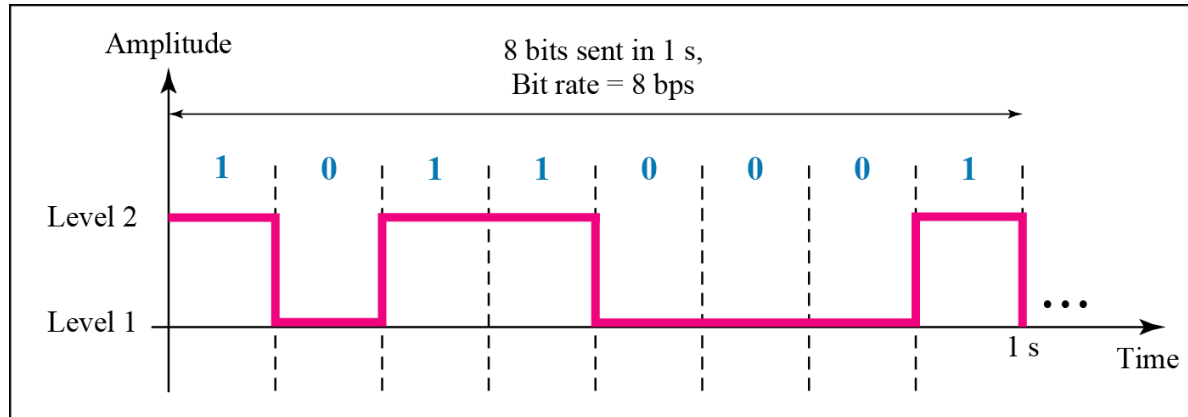
Topics discussed in this section:

Bit Rate

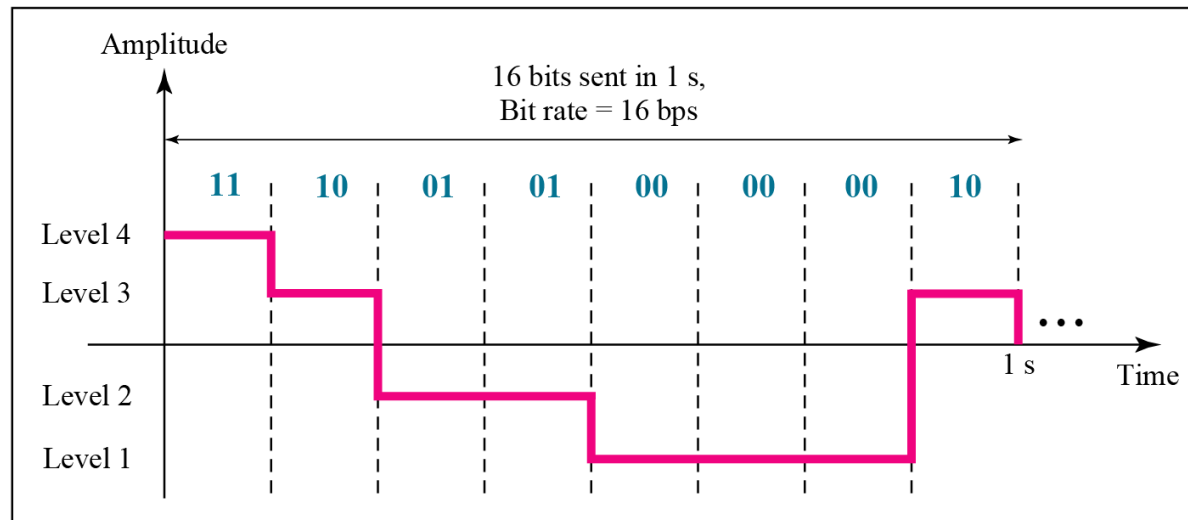
Bit Length

Digital Signal as a Composite Analog Signal

Figure 3.16 *Two digital signals: one with two signal levels and the other with four signal levels*



a. A digital signal with two levels



b. A digital signal with four levels



Example 3.16

A digital signal has eight levels. How many bits are needed per level? We calculate the number of bits from the formula

$$\text{Number of bits per level} = \log_2 8 = 3$$

Each signal level is represented by 3 bits.



Example 3.17

A digital signal has nine levels. How many bits are needed per level?



Example 3.17

A digital signal has nine levels. How many bits are needed per level?

We calculate the number of bits by using the formula. Each signal level is represented by 3.17 bits. However, this answer is not realistic. The number of bits sent per level needs to be an integer as well as a power of 2. For this example, 4 bits can represent one level.



Bit rate or bits per second (bps)

Most digital signals are nonperiodic, and thus period and frequency are not appropriate characteristics.

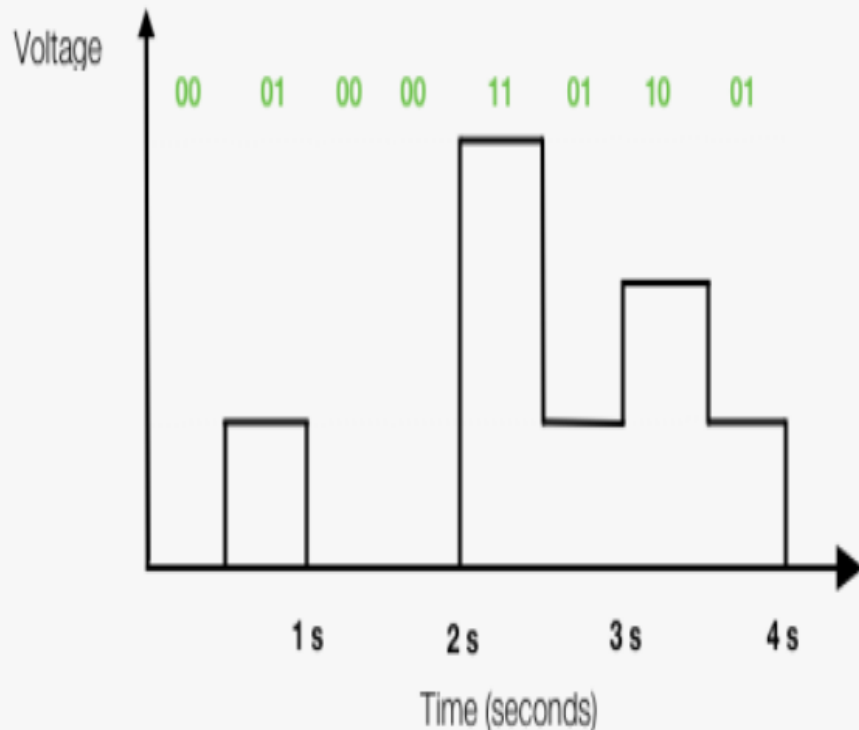
Another term: **bit rate (instead of frequency) is used to describe digital signals. The bit rate is the number of bits sent in 1s, expressed in bits per second (bps).**



Baud rate (bd)

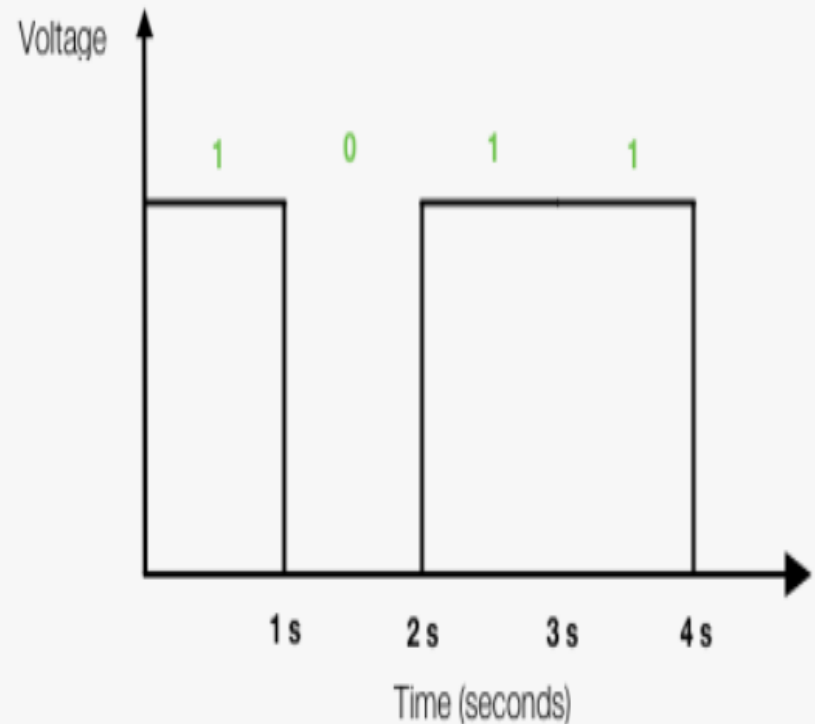
Baud rate is the number of signal changes that can be sent or received in one second.

Baud Rate vs Bit Rate



Bit rate = 4 bits per second
Baud rate = 2 baud per second

We can send 4 bits per second. The baud rate is 2 because there can be 2 pulse changes per second.



Bit rate = 1 bits per second
Baud rate = 1 baud per second

The bit rate is 1 bit per second. The baud rate is also 1 because there can be one pulse change per second.



Example 3.18

Assume we need to download text documents at the rate of 100 pages per minute. What is the required bit rate of the channel?



Example 3.18

Assume we need to download text documents at the rate of 100 pages per minute. What is the required bit rate of the channel?

Solution

A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires 8 bits, the bit rate is

$$100 \text{ pages/min} = \frac{100}{60} = 1.67 \text{ pages/sec}$$

$$\text{Bits per page} = 2400 \times 8 = 19,200 \text{ bits}$$

$$\text{Bit rate} = 1.67 \times 19,200 \approx 32,064 \text{ bps}$$

Answer:

Required channel bit rate $\approx 32 \text{ kbps}$



Example 3.19

A digitized voice channel is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per cycle). We assume that each sample requires 8 bits. What is the required bit rate?

Solution

The bit rate can be calculated as

$$2 \times 4000 \times 8 = 64,000 \text{ bps} = 64 \text{ kbps}$$



Example 3.19

A digitized voice channel is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per cycle). We assume that each sample requires 8 bits. What is the required bit rate?



Example 3.20

What is the bit rate for high-definition TV (HDTV)?



Example 3.20

What is the bit rate for high-definition TV (HDTV)?

Solution

HDTV uses digital signals to broadcast high quality video signals. The HDTV screen is normally a ratio of 16 : 9. There are 1920 by 1080 pixels per screen, and the screen is renewed 30 times per second. Twenty-four bits represents one color pixel.

$$1920 \times 1080 \times 30 \times 24 = 1,492,992,000 \text{ or } 1.5 \text{ Gbps}$$

The TV stations reduce this rate to 20 to 40 Mbps through compression.



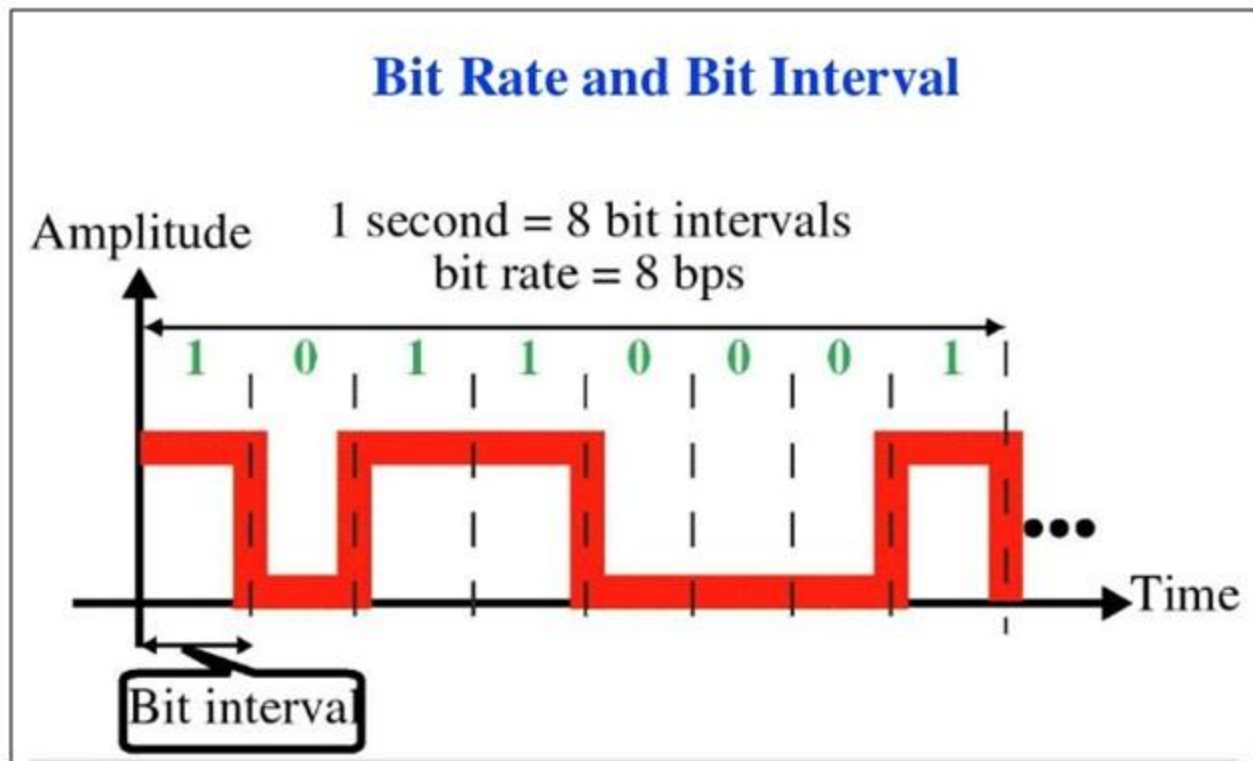
Bit Length

The concept of the **wavelength for an analog** signal: the distance one cycle occupies on the transmission medium.

We can define something similar for a digital signal: the bit length. The bit length is the distance one bit occupies on the transmission medium.

Bit Length

Bit length = propagation speed $\times$ bit duration



Periodic Signal	Aperiodic Signal
☐ A signal which repeats itself after a specific interval of time is called periodic signal.	☐ A signal which does not repeat itself after a specific interval of time is called aperiodic signal.
☐ A signal that repeats its pattern over a period is called periodic signal	☐ A signal that does not repeat its pattern over a period is called aperiodic signal or non periodic.
☐ They can be represented by a mathematical equation	☐ They cannot be represented by any mathematical equation
☐ Their value can be determined at any point of time	☐ Their value cannot be determined with certainty at any given point of time
☐ They are deterministic signals	☐ They are random signals
☐ Example: sine cosine square sawtooth etc	☐ Example: sound signals from radio , all types of noise signals
☐ Figure:	☐ Figure:

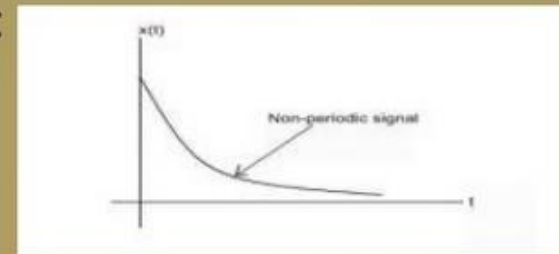
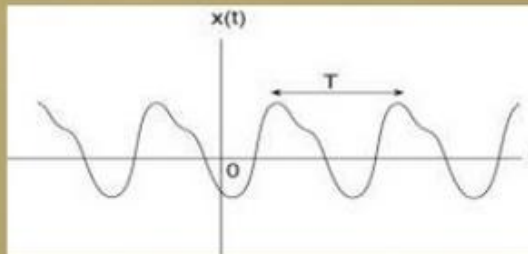
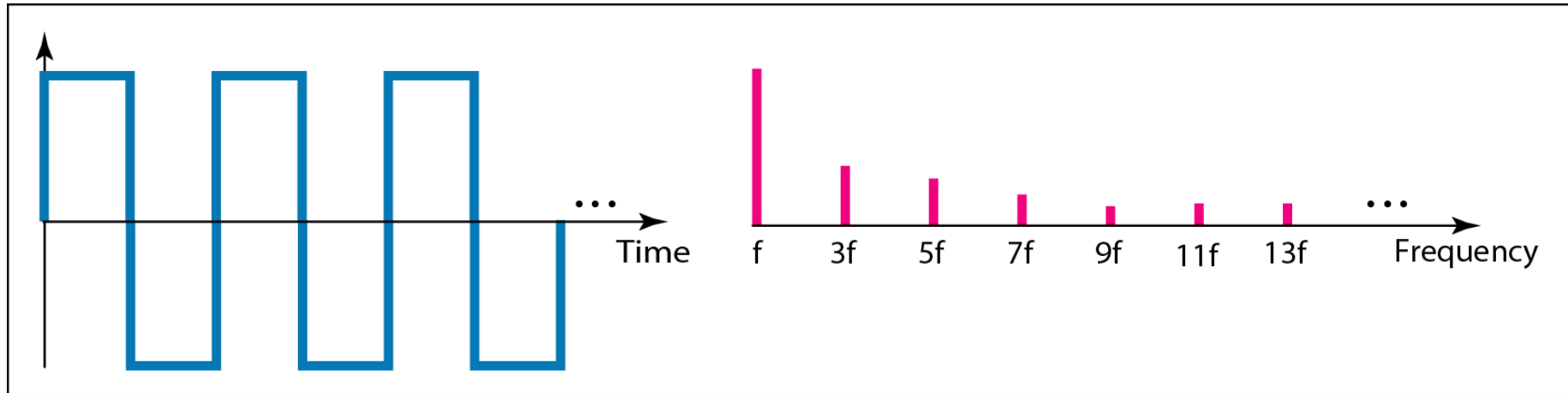
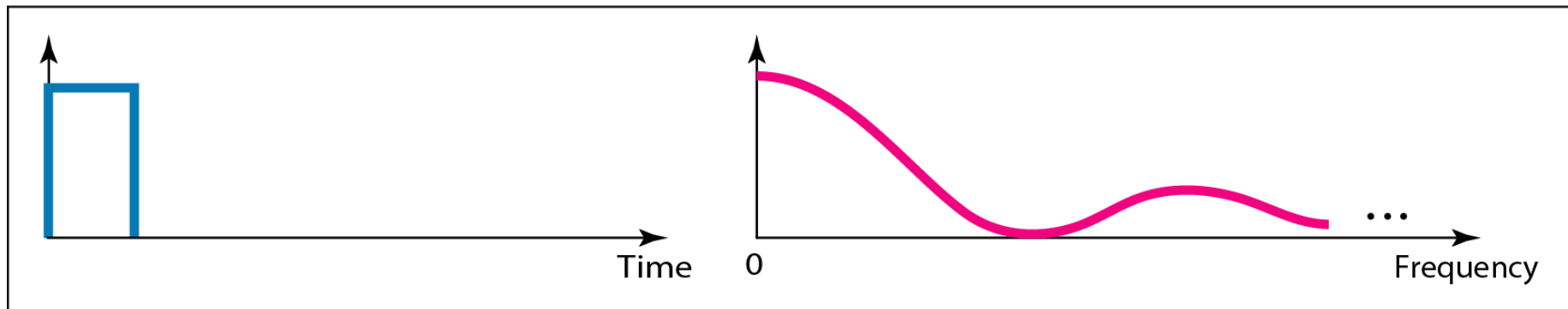


Figure 3.17 *The time and frequency domains of periodic and nonperiodic digital signals*



a. Time and frequency domains of periodic digital signal



b. Time and frequency domains of nonperiodic digital signal

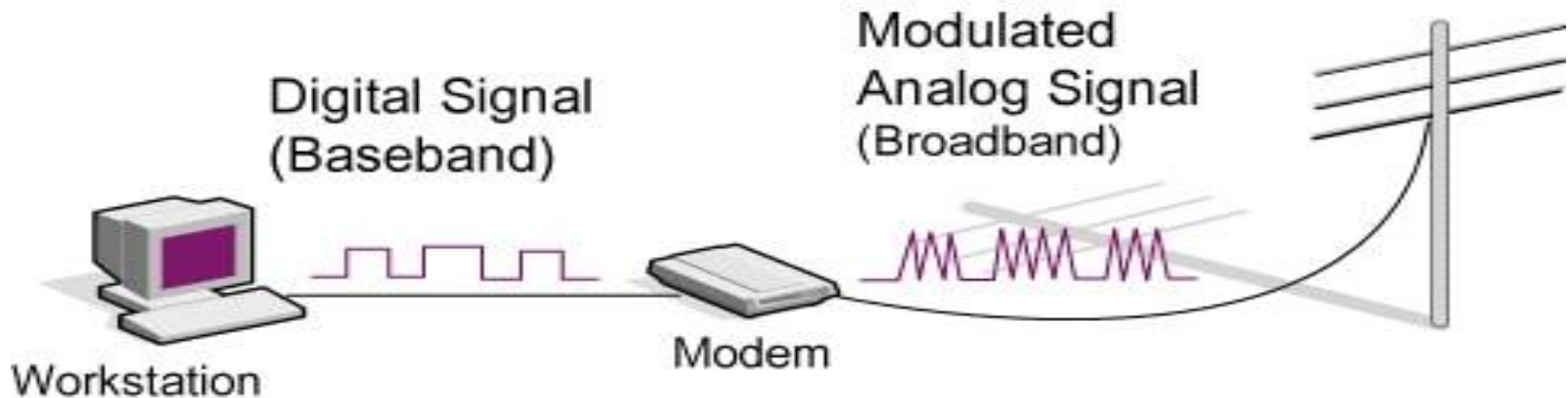
Transmission of Digital Signals

- We can transmit a digital signal by using one of two different approaches: baseband transmission or broadband transmission (using modulation).

A digital signal is a composite analog signal with an infinite bandwidth.

Baseband v/s Broadband

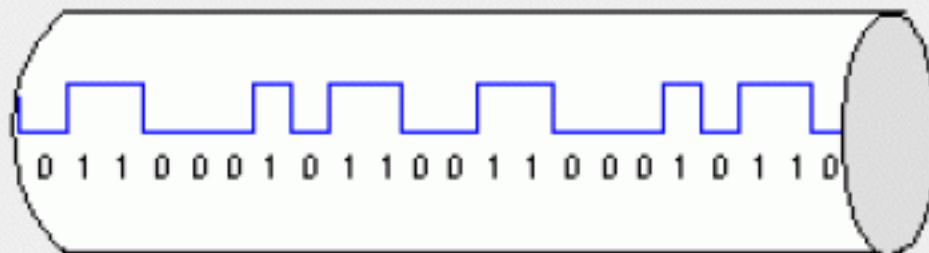
- In the **baseband** transmission the whole bandwidth of the cable is utilized by a single signal.
- Conversely, in the **broadband** transmission, multiple signals are sent on multiple frequencies simultaneously using a single channel.



BASEBAND & BROADBAND (TDM & FDM)

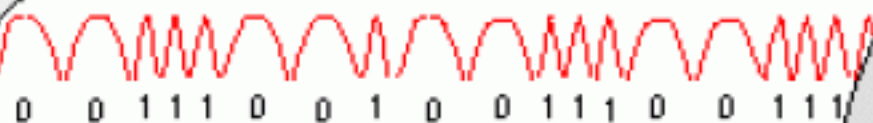
BASEBAND (TDM, interleaves digital signals)

**Data,
Voice,
Video**



BROADBAND (FDM, true simultaneous)

Data



Voice



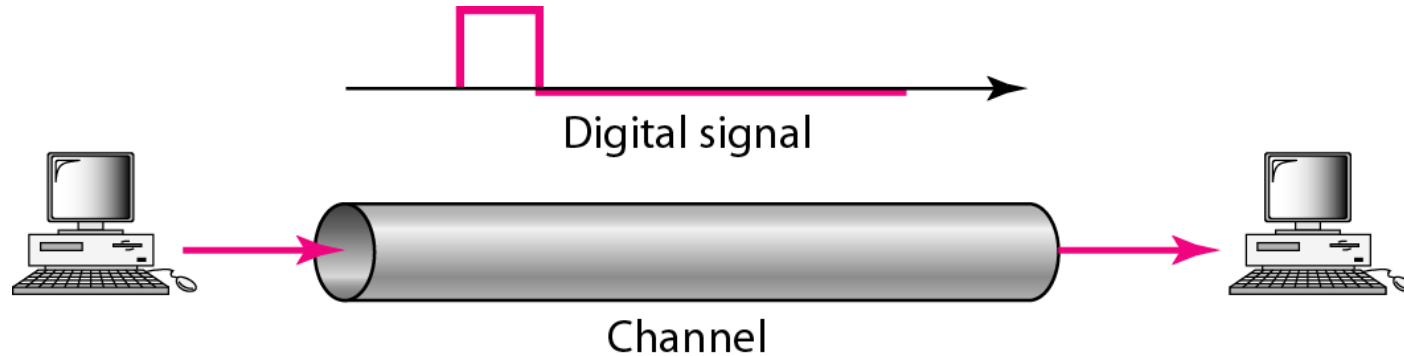
Video



Baseband Transmission

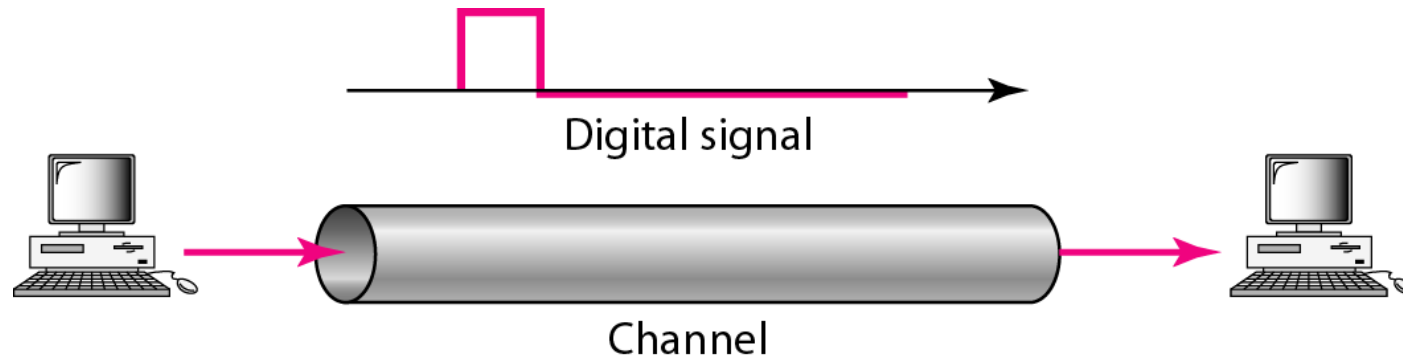
- Baseband transmission means sending a digital signal over a channel without changing the digital signal to an analog signal.

Figure 3.19 *Baseband transmission*



Baseband transmission requires that we have a **low-pass channel**, a channel with a bandwidth that starts from zero.

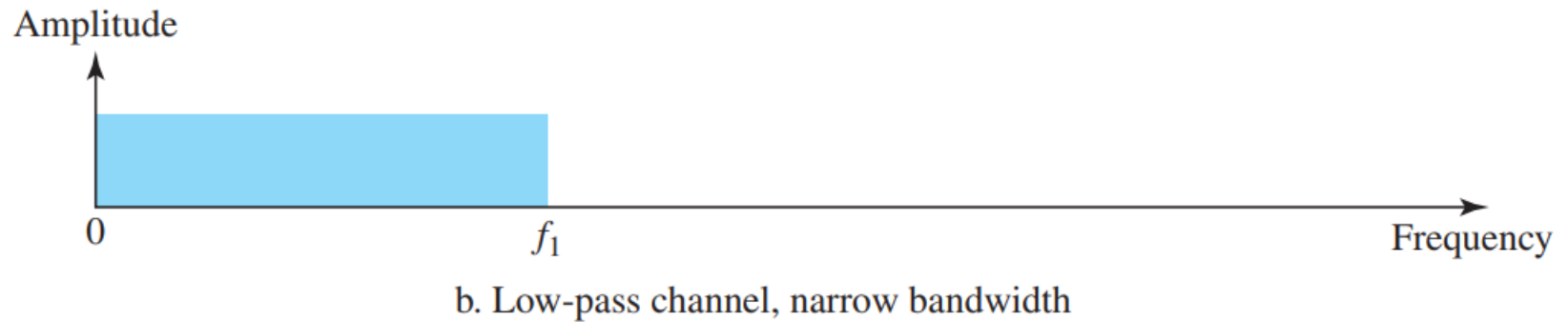
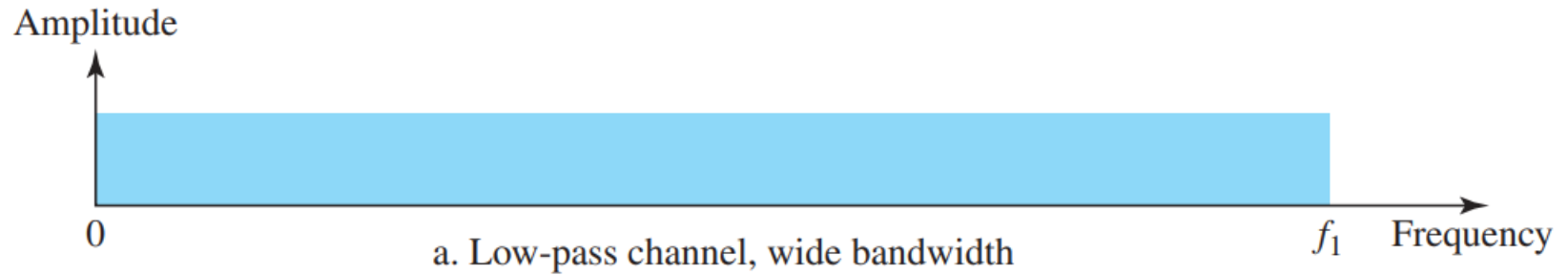
Figure 3.19 *Baseband transmission*



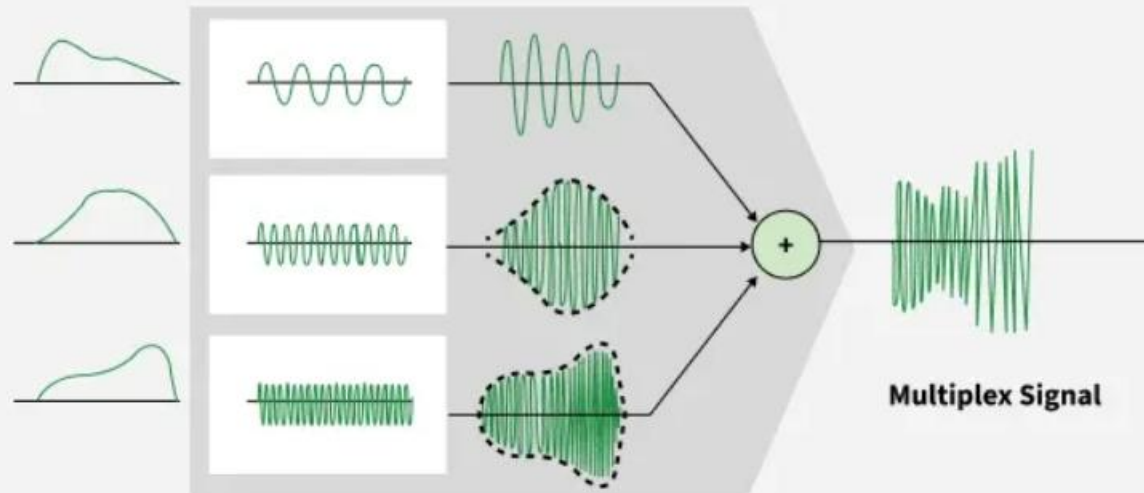
Baseband transmission requires that we have a **low-pass channel**, a channel with a bandwidth that starts from zero.

- We need to remember that a low-pass channel with infinite bandwidth is ideal, but we cannot have such a channel in real life. However, we can get close.

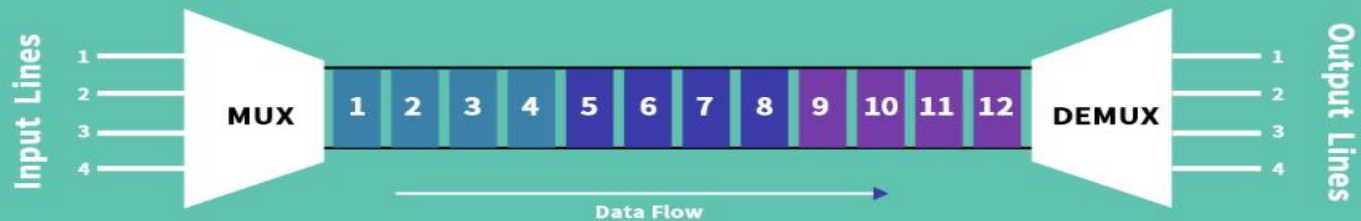
Figure 3.20 *Bandwidths of two low-pass channels*



A Conceptual illustration of multiplexing Process



Time Division Multiplexing (TDM)



Frequency Division Multiplexing (FDM)

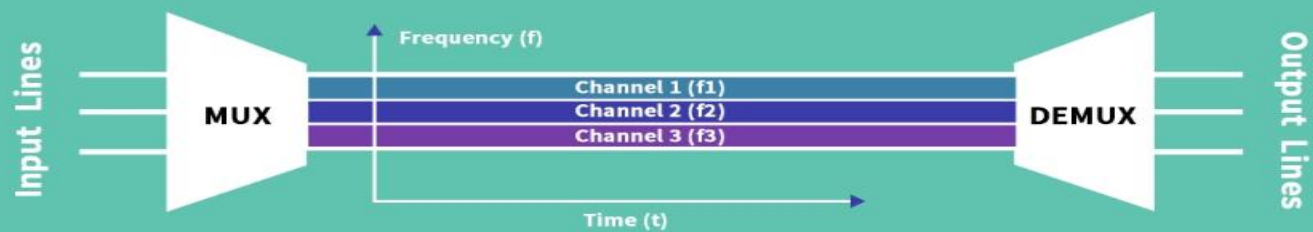


Figure 3.20 *Baseband transmission using a dedicated medium*

Case 1: Low-Pass Channel with Wide Bandwidth

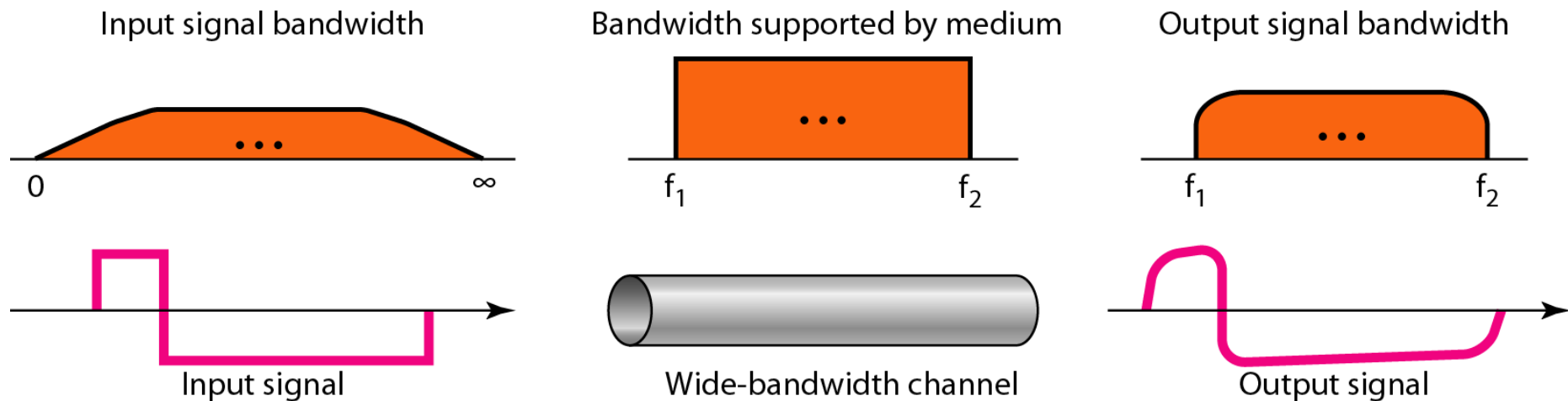
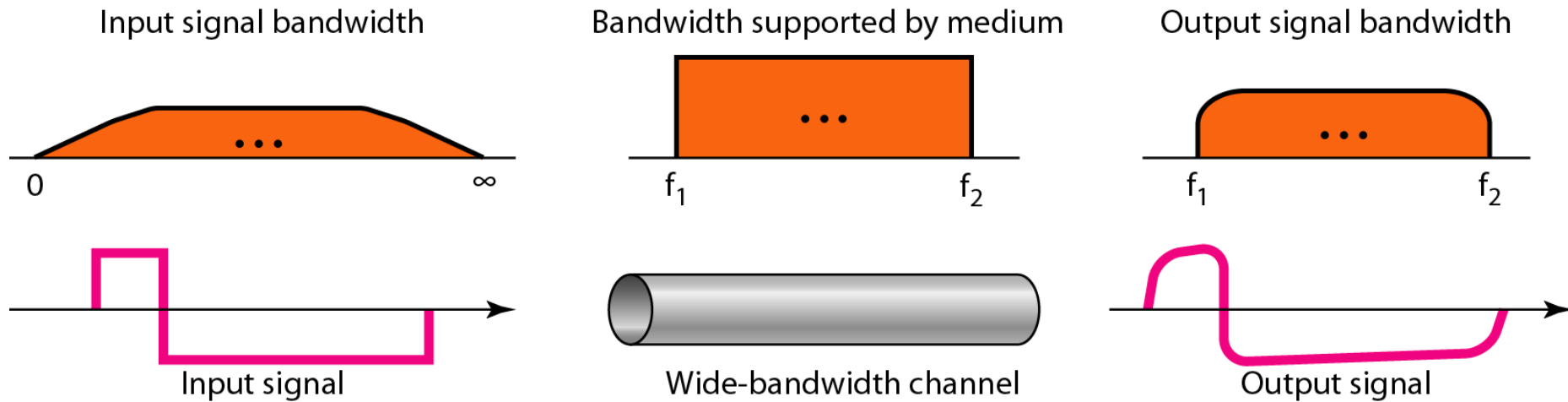
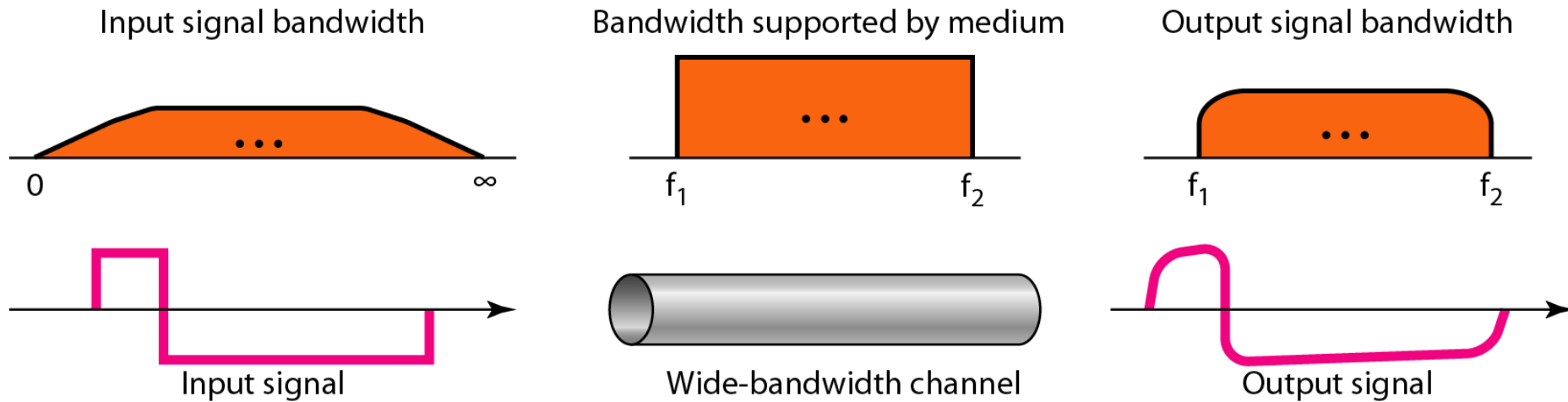


Figure 3.20 *Baseband transmission using a dedicated medium*



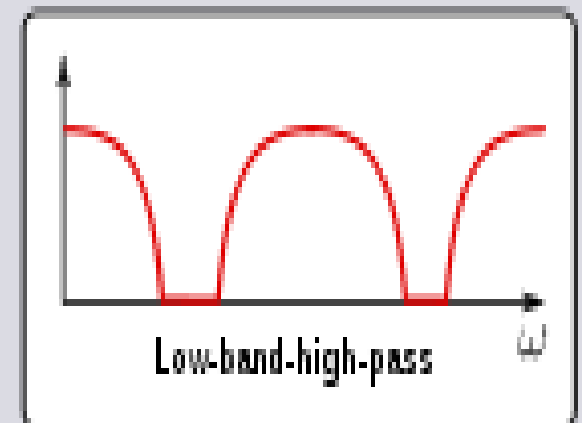
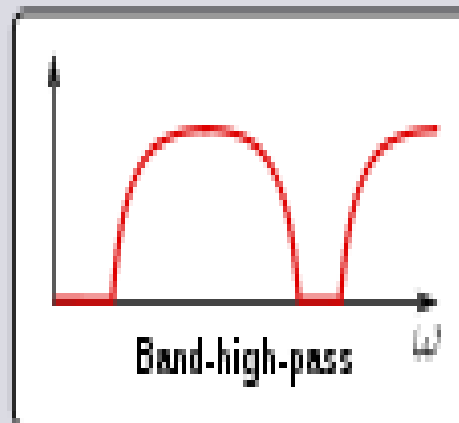
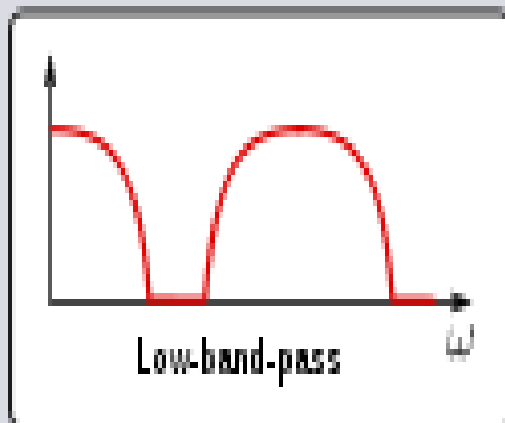
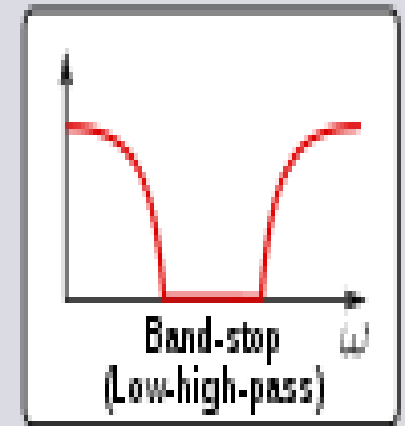
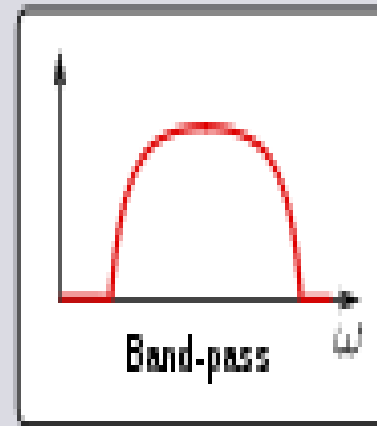
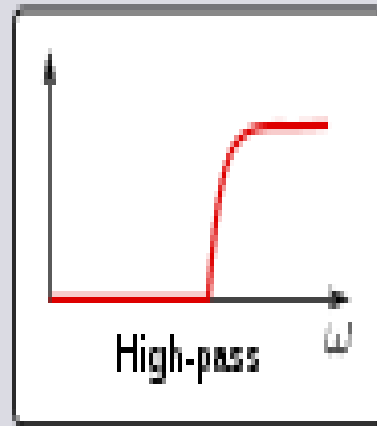
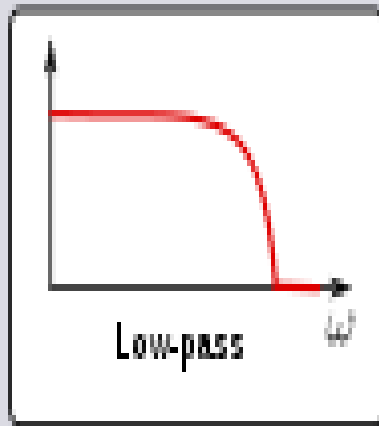
Although the output signal is not an exact replica of the original signal, the data can still be deduced from the received signal. Note that although some of the frequencies are blocked by the medium, they are not critical.

Figure 3.20 *Baseband transmission using a dedicated medium*



An example of a dedicated channel where the entire bandwidth of the medium is used as one single channel is a LAN. Almost every wired LAN today uses a dedicated channel for two stations communicating with each other. In a bus topology LAN with multipoint connections, only two stations can communicate with each other at each moment in time (timesharing); the other stations need to refrain from sending data. In a star topology LAN, the entire channel between each station and the hub is used for communication between these two entities.

Low pass, Band pass & High pass filters



Note

Baseband transmission of a digital signal that preserves the shape of the digital signal is possible only if we have a low-pass channel with an infinite or very wide bandwidth.

Note

In baseband transmission, the required bandwidth is proportional to the bit rate; if we need to send bits faster, we need more bandwidth.

Table 3.2 *Bandwidth requirements*

<i>Bit Rate</i>	<i>Harmonic 1</i>	<i>Harmonics 1, 3</i>	<i>Harmonics 1, 3, 5</i>
$n = 1$ kbps	$B = 500$ Hz	$B = 1.5$ kHz	$B = 2.5$ kHz
$n = 10$ kbps	$B = 5$ kHz	$B = 15$ kHz	$B = 25$ kHz
$n = 100$ kbps	$B = 50$ kHz	$B = 150$ kHz	$B = 250$ kHz

Today's Task

Case 2: Low-Pass Channel with Limited Bandwidth

Page no: 72-75 [Forouzan, B. A.. Data Communication and Networking]

Note

If the available channel is a bandpass channel, we cannot send the digital signal directly to the channel; we need to convert the digital signal to an analog signal before transmission.

Note

If the available channel is a bandpass channel, we cannot send the digital signal directly to the channel; we need to convert the digital signal to an analog signal before transmission.

Broadband Transmission (Using Modulation)

- Broadband transmission or modulation means changing the digital signal to an analog signal for transmission.
- Modulation allows us to use a bandpass channel a channel with a bandwidth that does not start from zero..

Figure 3.24 *Bandwidth of a bandpass channel*

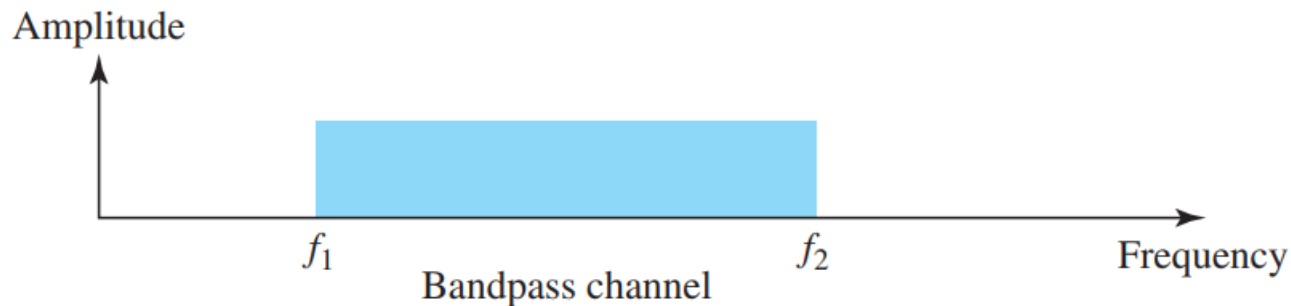
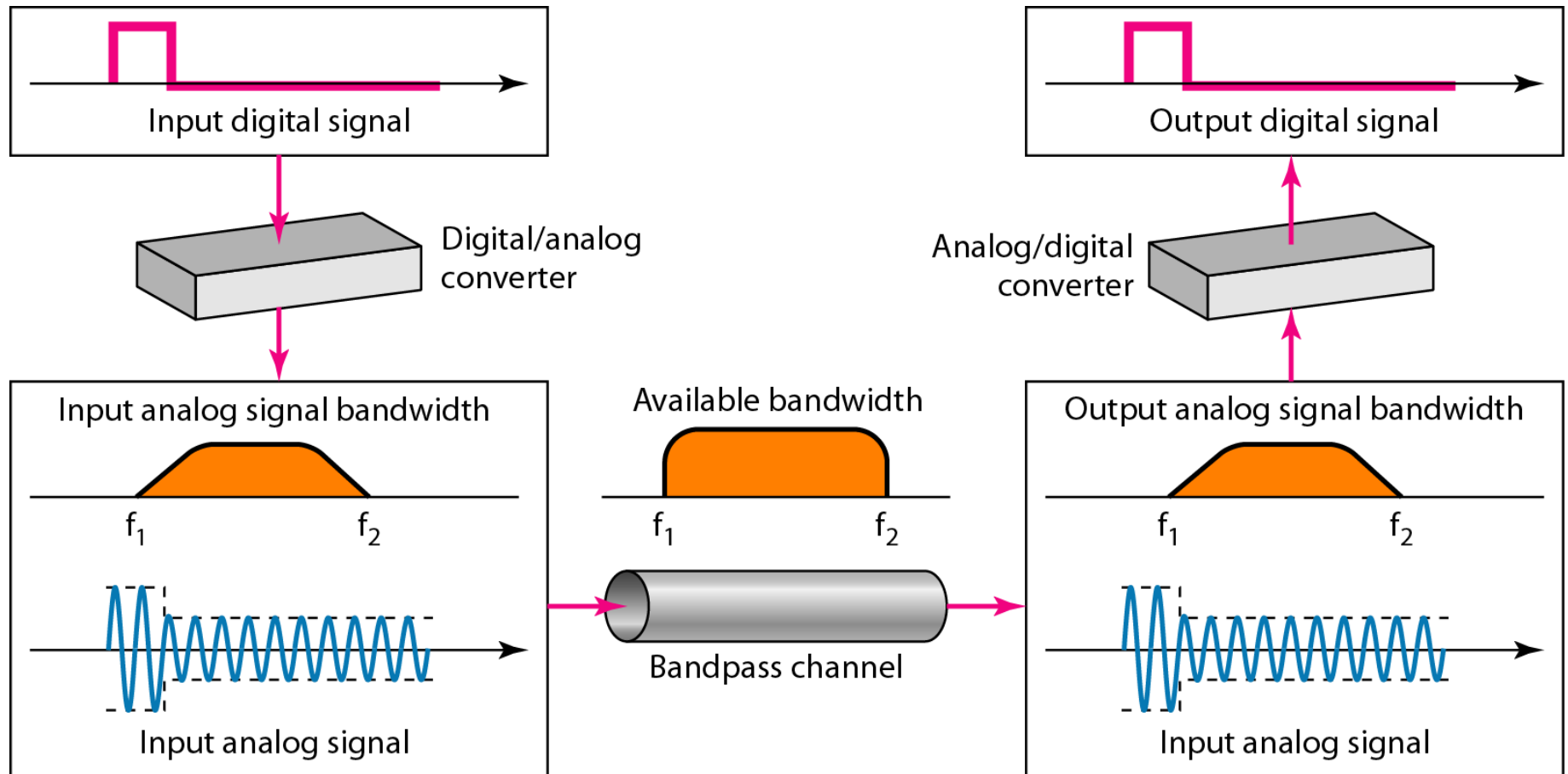


Figure 3.24 *Modulation of a digital signal for transmission on a bandpass channel*



3-4 TRANSMISSION IMPAIRMENT

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are **attenuation**, **distortion**, and **noise**.

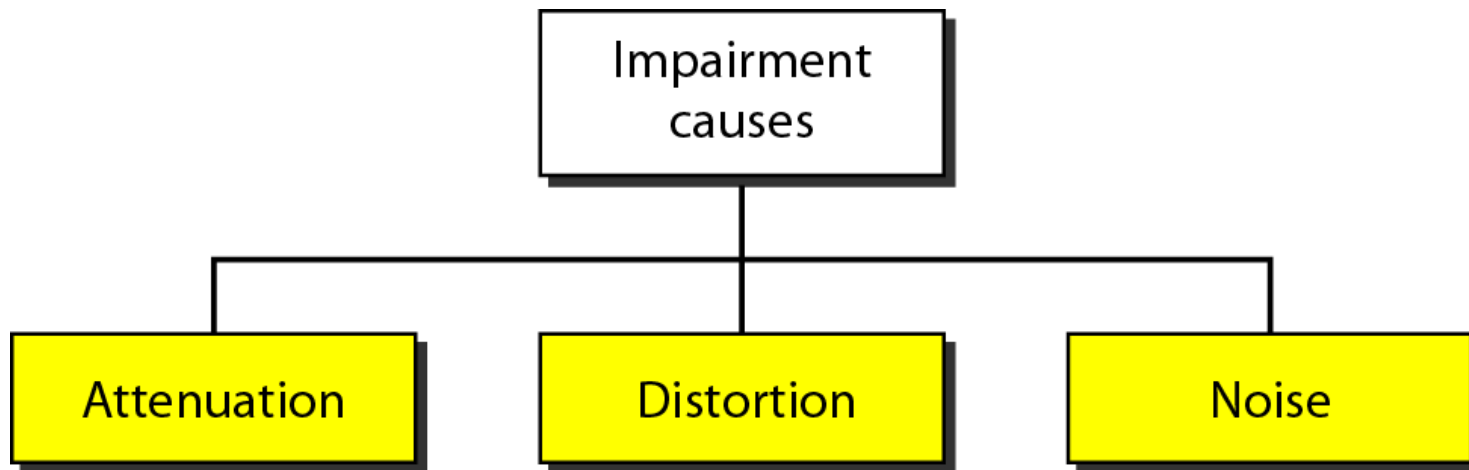
Topics discussed in this section:

Attenuation

Distortion

Noise

Figure 3.25 *Causes of impairment*



Attenuation

- Attenuation means a loss of energy.
- When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium.
- To show that a signal has lost or gained strength, engineers use the unit of the decibel.
- The decibel (dB) measures the relative strengths of two signals or one signal at two different points.

Distortion

- Distortion means that the signal changes its form or shape.
- Distortion can occur in a composite signal made of different frequencies.
- As a result, signal components at the receiver have phases different from what they had at the sender.
- The shape of the composite signal is therefore not the same.

Noise

- Noise is another cause of impairment.
- Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.
- **Thermal noise** is the random motion of electrons in a wire, which creates an extra signal not originally sent by the transmitter.
- **Induced noise** comes from sources such as motors and appliances. These devices act as a sending antenna, and the transmission medium acts as the receiving antenna.
- **Crosstalk** is the effect of one wire on the other. One wire acts as a sending antenna and the other as the receiving antenna.
- **Impulse noise** is a spike (a signal with high energy in a very short time) that comes from power lines, lightning, and so on.

Figure 3.26 *Attenuation*

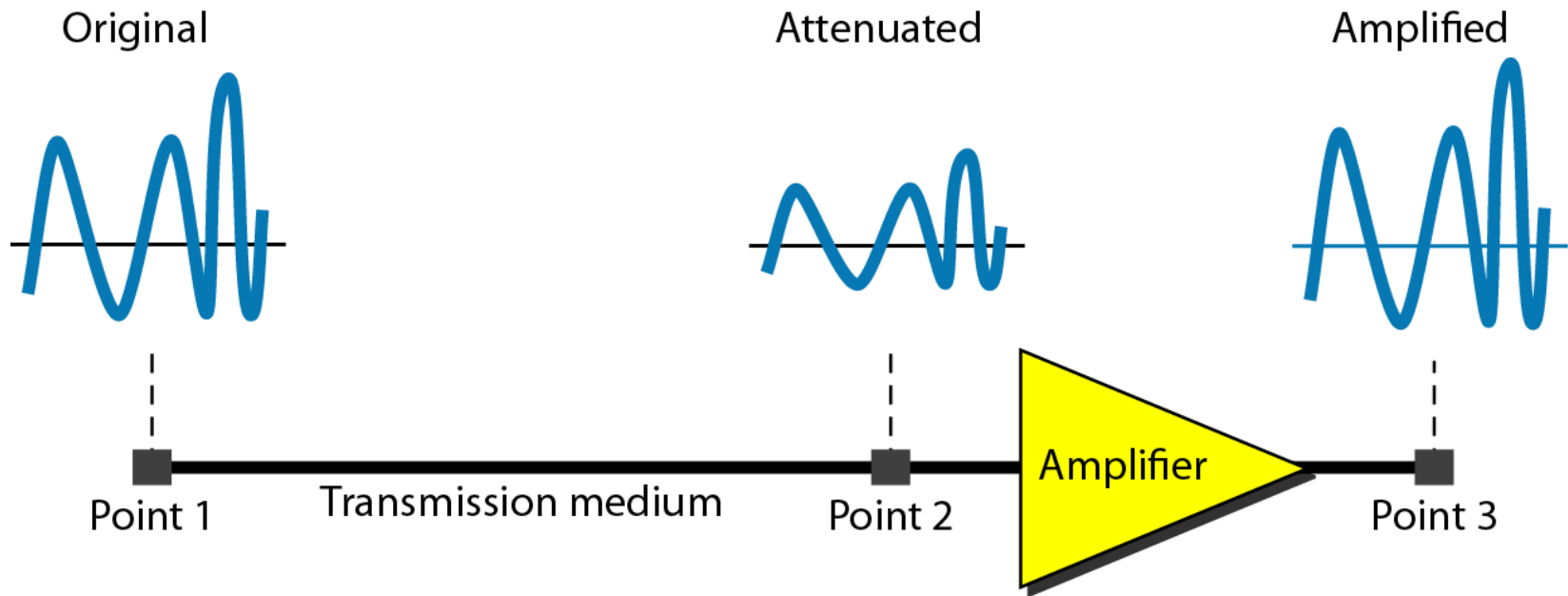


Figure 3.28 *Distortion*

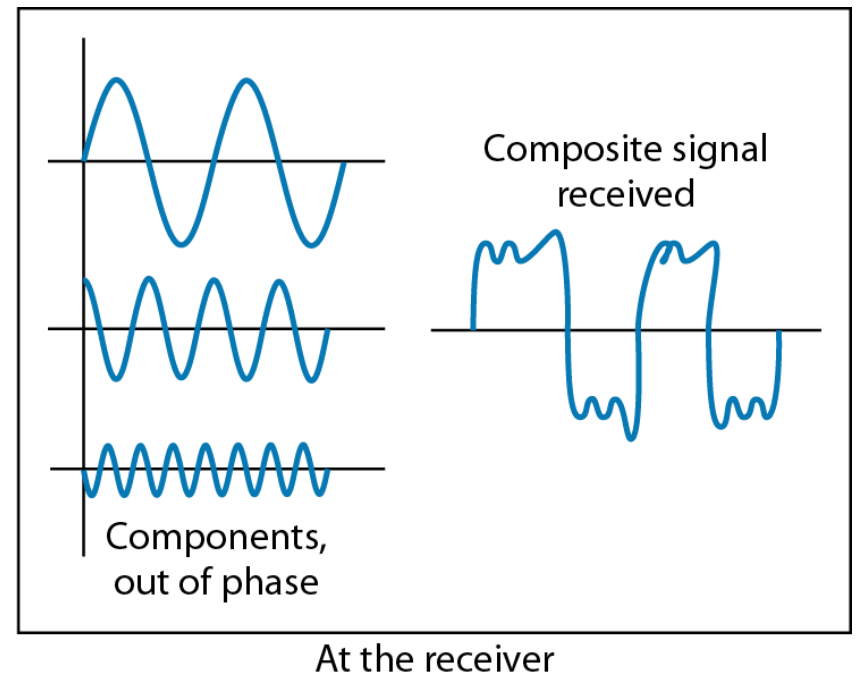
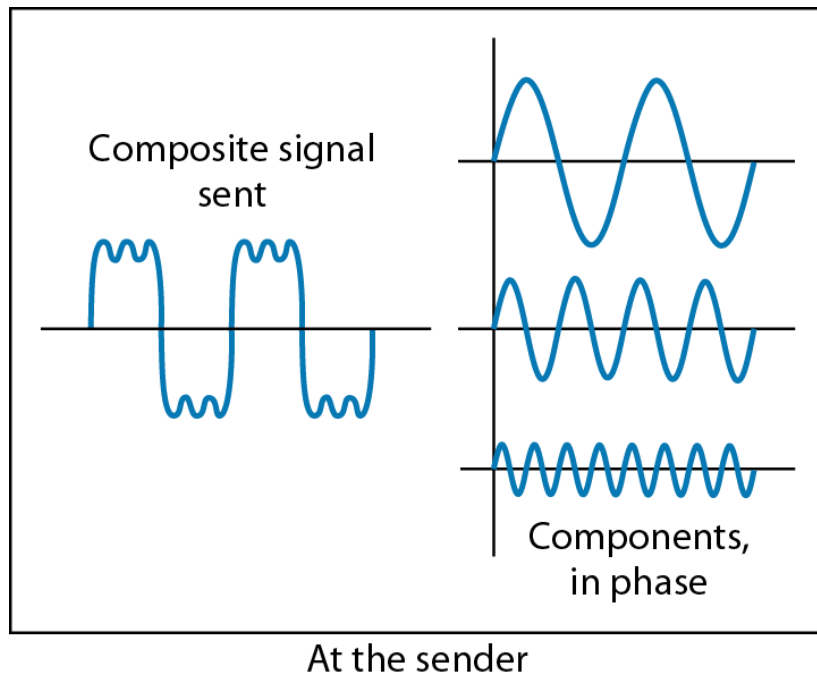


Figure 3.29 *Noise*

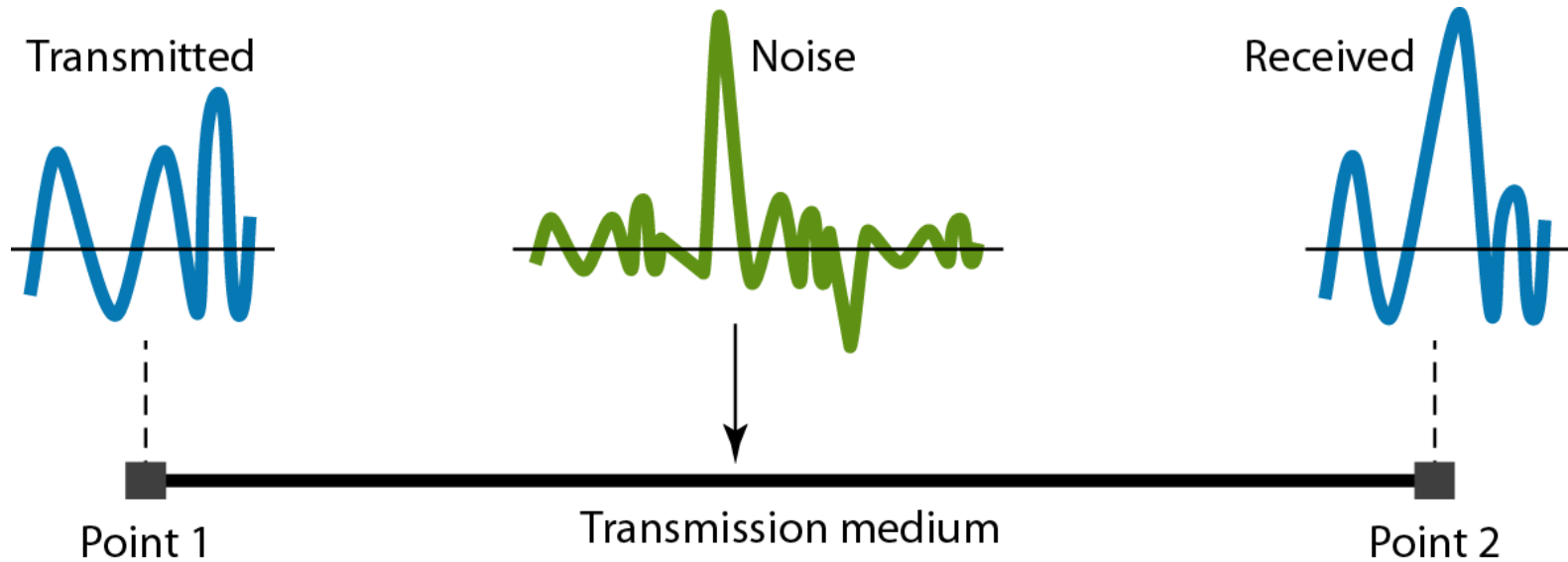
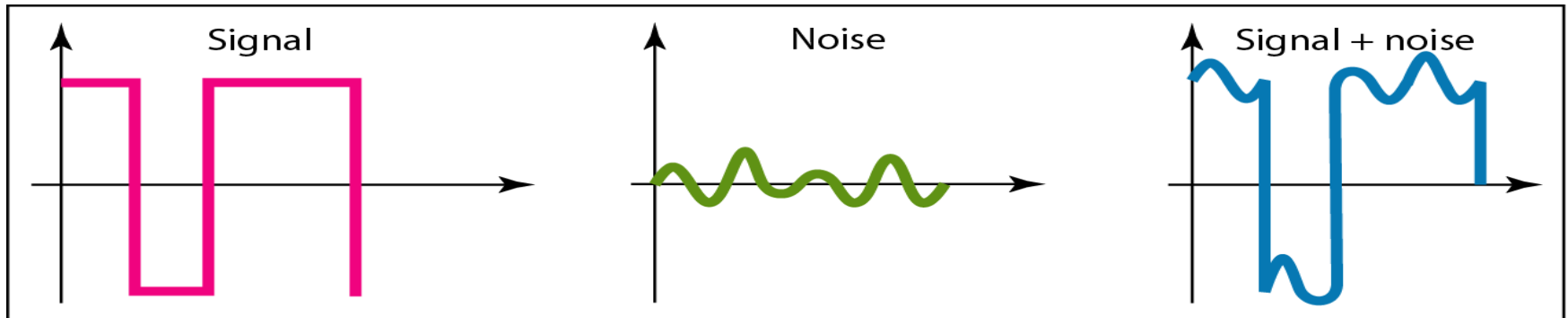
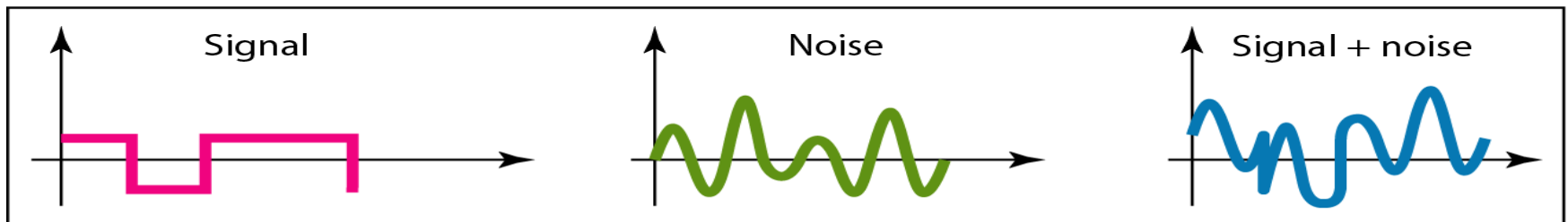


Figure 3.30 *Two cases of Signal to Noise Ratio: a high SNR and a low SNR*

$$\text{SNR} = \frac{\text{average signal power}}{\text{average noise power}}$$



a. Large SNR



b. Small SNR

3-5 DATA RATE LIMITS

A very important consideration in data communications is how fast we can send data, in bits per second, over a channel. Data rate depends on three factors:

1. The bandwidth available
2. The level of the signals we use
3. The quality of the channel (the level of noise)

Topics discussed in this section:

Noiseless Channel: Nyquist Bit Rate

Noisy Channel: Shannon Capacity

Using Both Limits



Note

Increasing the levels of a signal may reduce the reliability of the system.

Note

Increasing the levels of a signal may reduce the reliability of the system.

Two theoretical formulas were developed to calculate the data rate: one by Nyquist for a noiseless channel, another by Shannon for a noisy channel.

Data Rate Calculation

Nyquist
Noiseless Channel

L = signal levels

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

Shannon
Noisy Channel

SNR = signal to noise ratio

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$



Example 3.34

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$



Example 3.35

Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 4 = 12,000 \text{ bps}$$



Example 3.37

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$C = B \log_2 (1 + \text{SNR}) = B \log_2 (1 + 0) = B \log_2 1 = B \times 0 = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.



Example 3.38

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 5000. The signal-to-noise ratio is usually 6000. For this channel the capacity is calculated as



Example 3.38

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as

$$\begin{aligned} C &= B \log_2 (1 + \text{SNR}) = 3000 \log_2 (1 + 3162) = 3000 \log_2 3163 \\ &= 3000 \times 11.62 = 34,860 \text{ bps} \end{aligned}$$

This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.



Example 3.40

For practical purposes, when the SNR is very high, we can assume that $\text{SNR} + 1$ is almost the same as SNR. In these cases, the theoretical channel capacity can be simplified to

$$C = B \times \frac{\text{SNR}_{\text{dB}}}{3}$$

For example, we can calculate the theoretical capacity of the previous example as

$$C = 2 \text{ MHz} \times \frac{36}{3} = 24 \text{ Mbps}$$



Example 3.41

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution

First, we use the Shannon formula to find the upper limit.

$$C = B \log_2 (1 + \text{SNR}) = 10^6 \log_2 (1 + 63) = 10^6 \log_2 64 = 6 \text{ Mbps}$$



Example 3.41 (continued)

The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps, for example. Then we use the Nyquist formula to find the number of signal levels.

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \quad \rightarrow \quad L = 4$$

Note

The Shannon capacity gives us the upper limit; the Nyquist formula tells us how many signal levels we need.

3-6 PERFORMANCE

One important issue in networking is the **performance** of the network—how good is it? We discuss quality of service, an overall measurement . In this section, we introduce terms that we need for future chapters.

Topics discussed in this section:

Bandwidth

Throughput

Latency (Delay)

Bandwidth-Delay Product

Performance

- **Bandwidth:** a range of frequencies within a given band, in particular that used for transmitting a signal.
- **Throughput:** the amount of bits passing through a system or process at given time.
- **Latency (Delay):** the total time it takes a data packet to travel from one node to another.
- **Bandwidth-Delay Product:** the product of a data link's capacity (in bits per second) and its round-trip delay time (in seconds).



Note

In networking, we use the term bandwidth in two contexts.

- ❑ The first, bandwidth in hertz, refers to the range of frequencies in a composite signal or the range of frequencies that a channel can pass.
- ❑ The second, bandwidth in bits per second, refers to the speed of bit transmission in a channel or link.

Figure 3.31 *Filling the link with bits for case 1 Frequency Based. FDM*

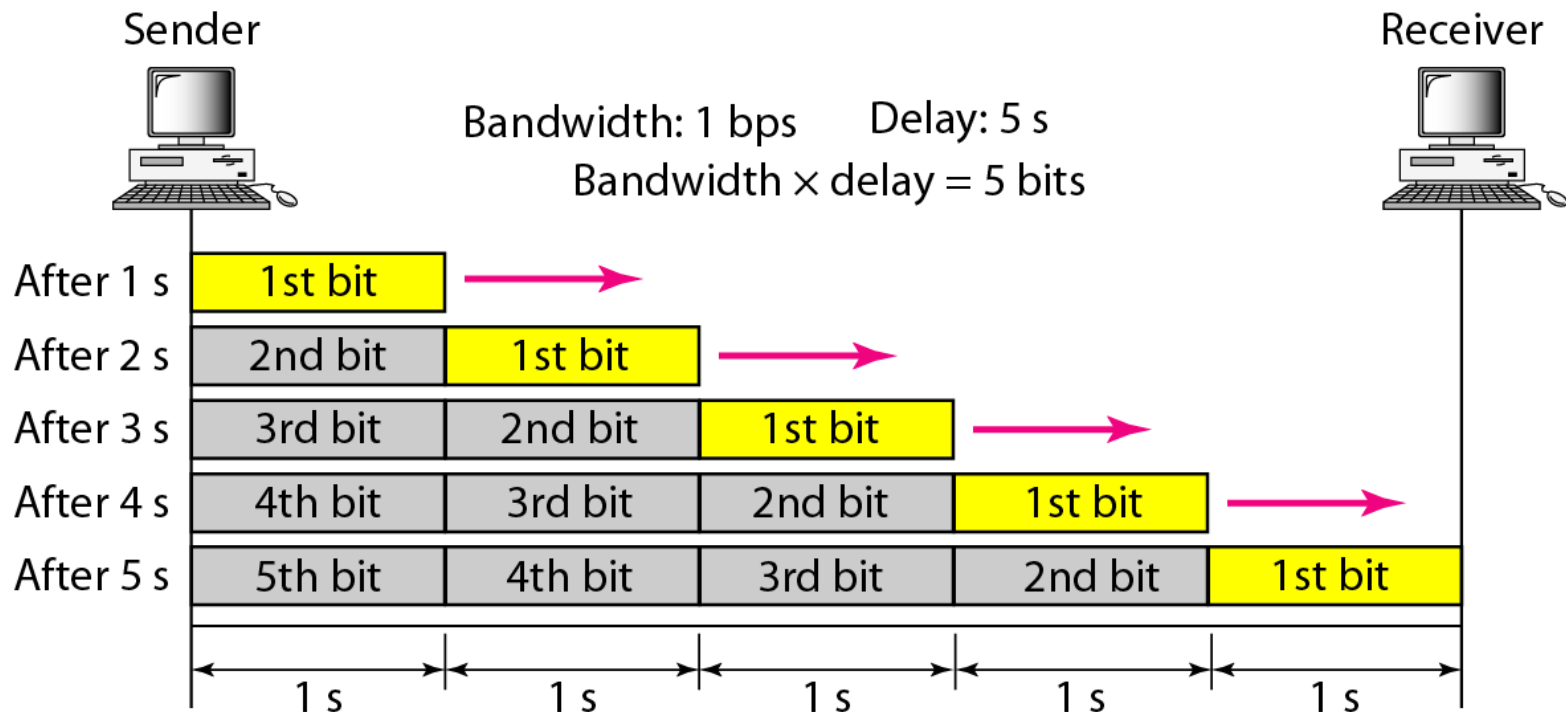
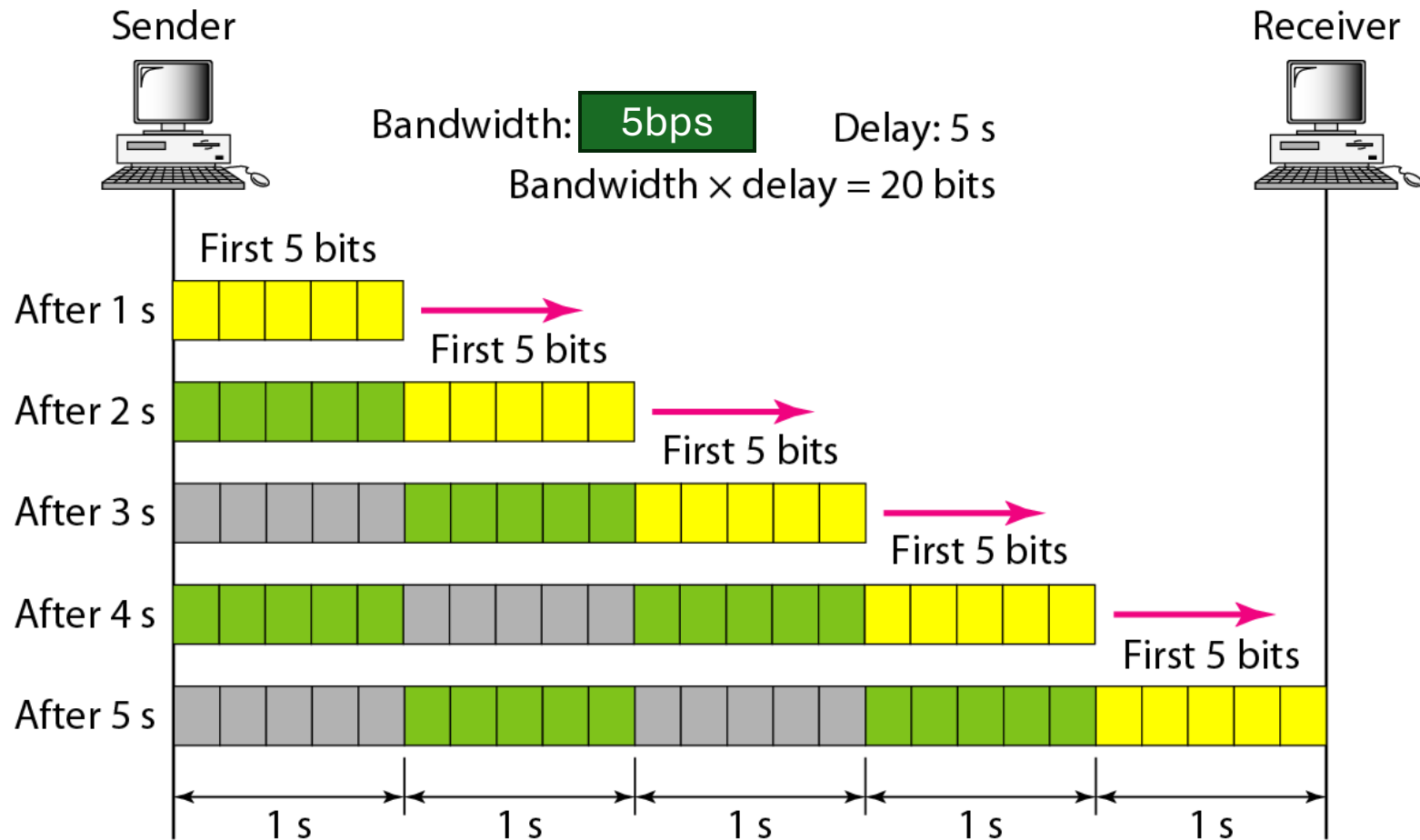
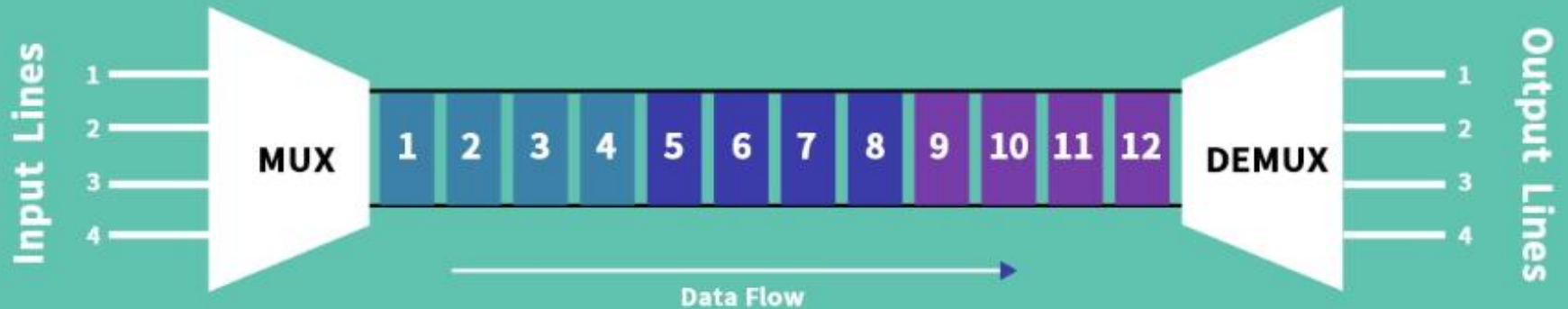


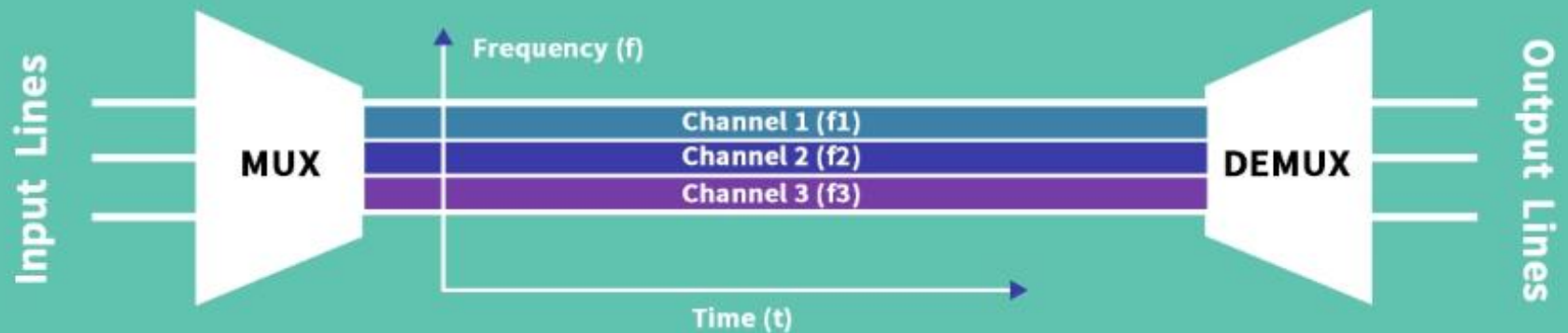
Figure 3.32 *Filling the link with bits in case 2 Time Based. TDM*



Time Division Multiplexing (TDM)



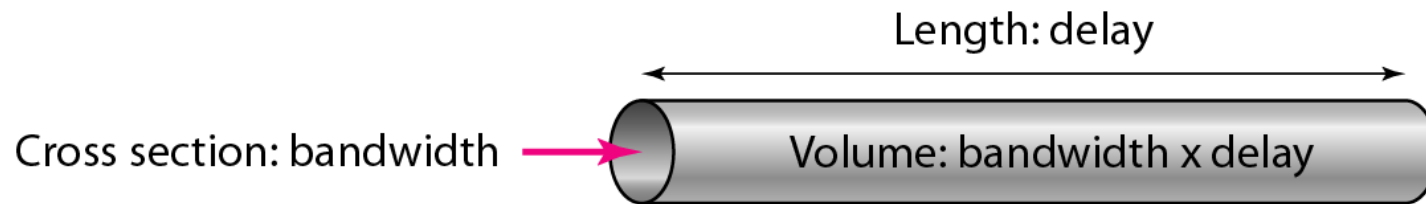
Frequency Division Multiplexing (FDM)



Note

The bandwidth-delay product defines the number of bits that can fill the link.

Figure 3.33 *Concept of bandwidth-delay product*



Bandwidth Utilization & Multiplexing

Note

Bandwidth utilization is the try to use of available bandwidth to achieve specific goals.

Efficiency can be achieved by multiplexing.

Whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared. Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link. As data and telecommunications use increases, so does traffic.

Topics discussed in this section:

Frequency-Division Multiplexing

Wavelength-Division Multiplexing

Synchronous Time-Division Multiplexing

Statistical Time-Division Multiplexing

Figure 6.1 *Dividing a link into channels*

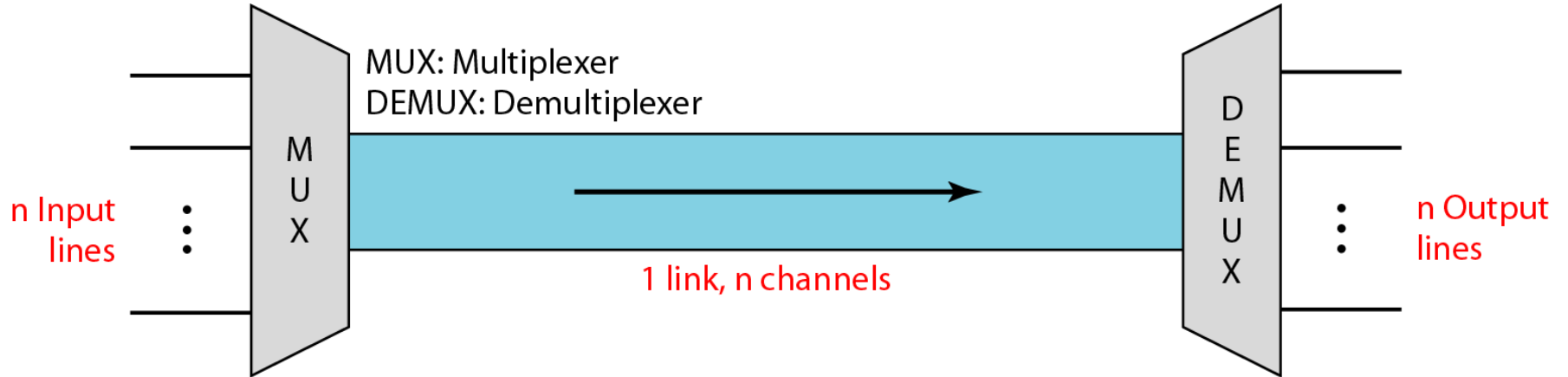


Figure 6.2 *Categories of multiplexing*

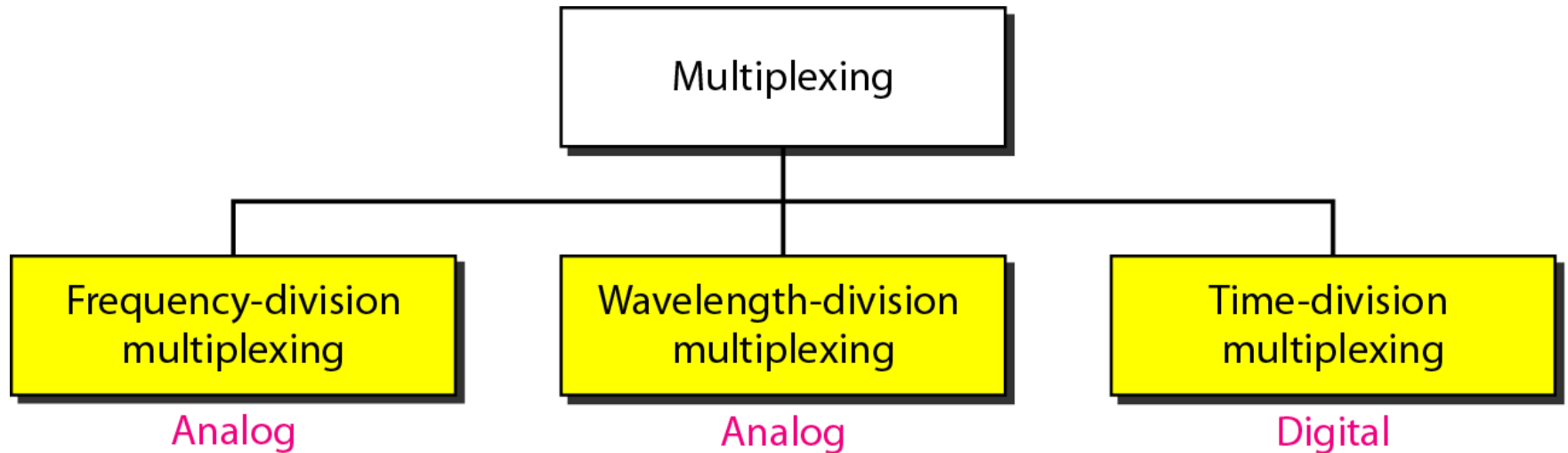


Figure 6.3 *Frequency-division multiplexing*





Note

FDM is an analog multiplexing technique that combines analog signals.

Figure 6.4 *FDM process*

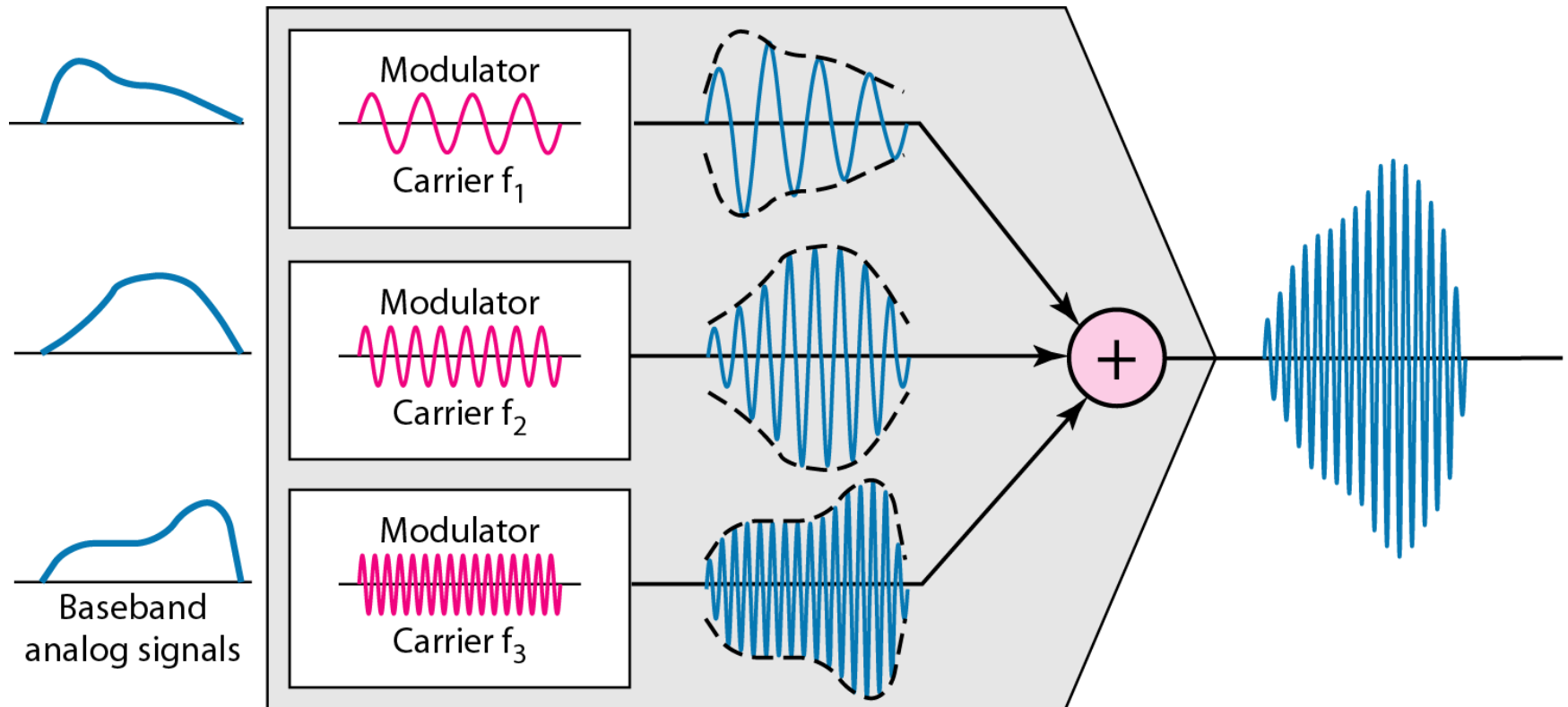
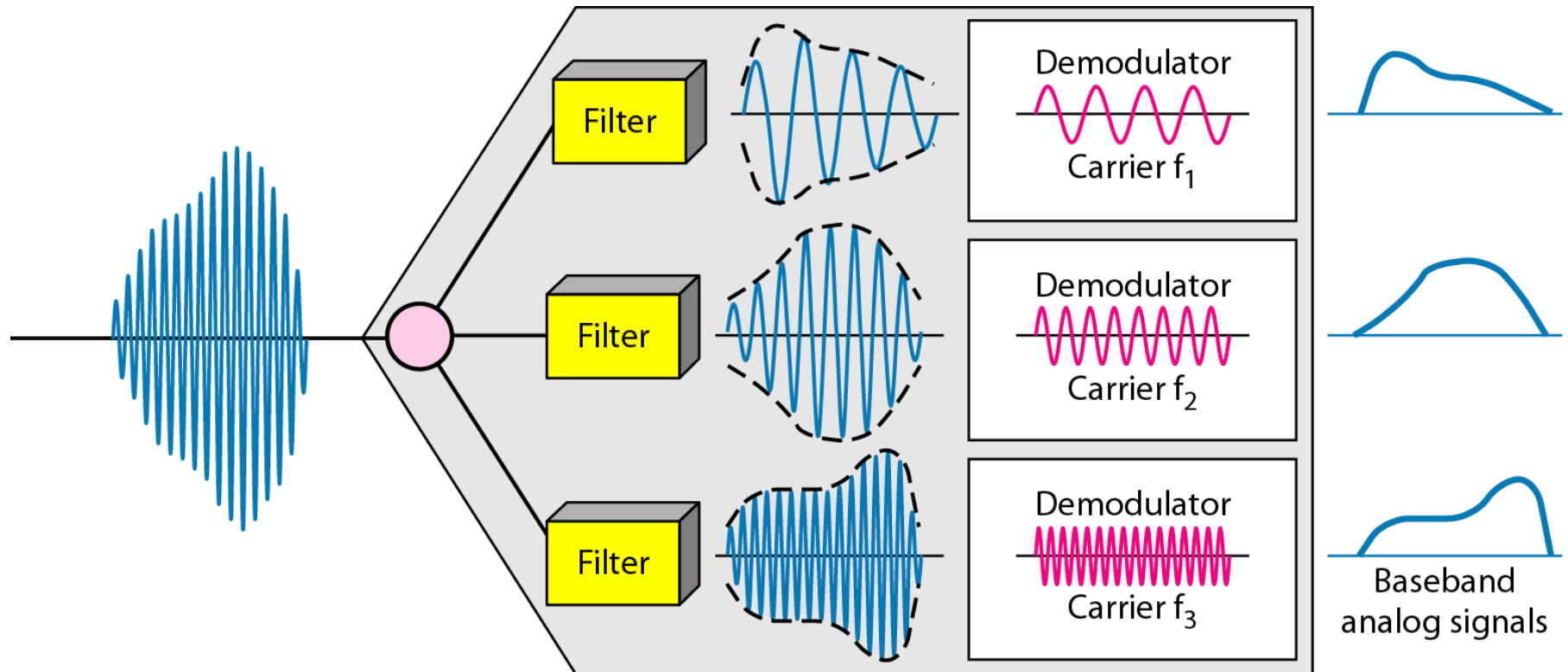


Figure 6.5 *FDM demultiplexing example*





Example 6.1

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.



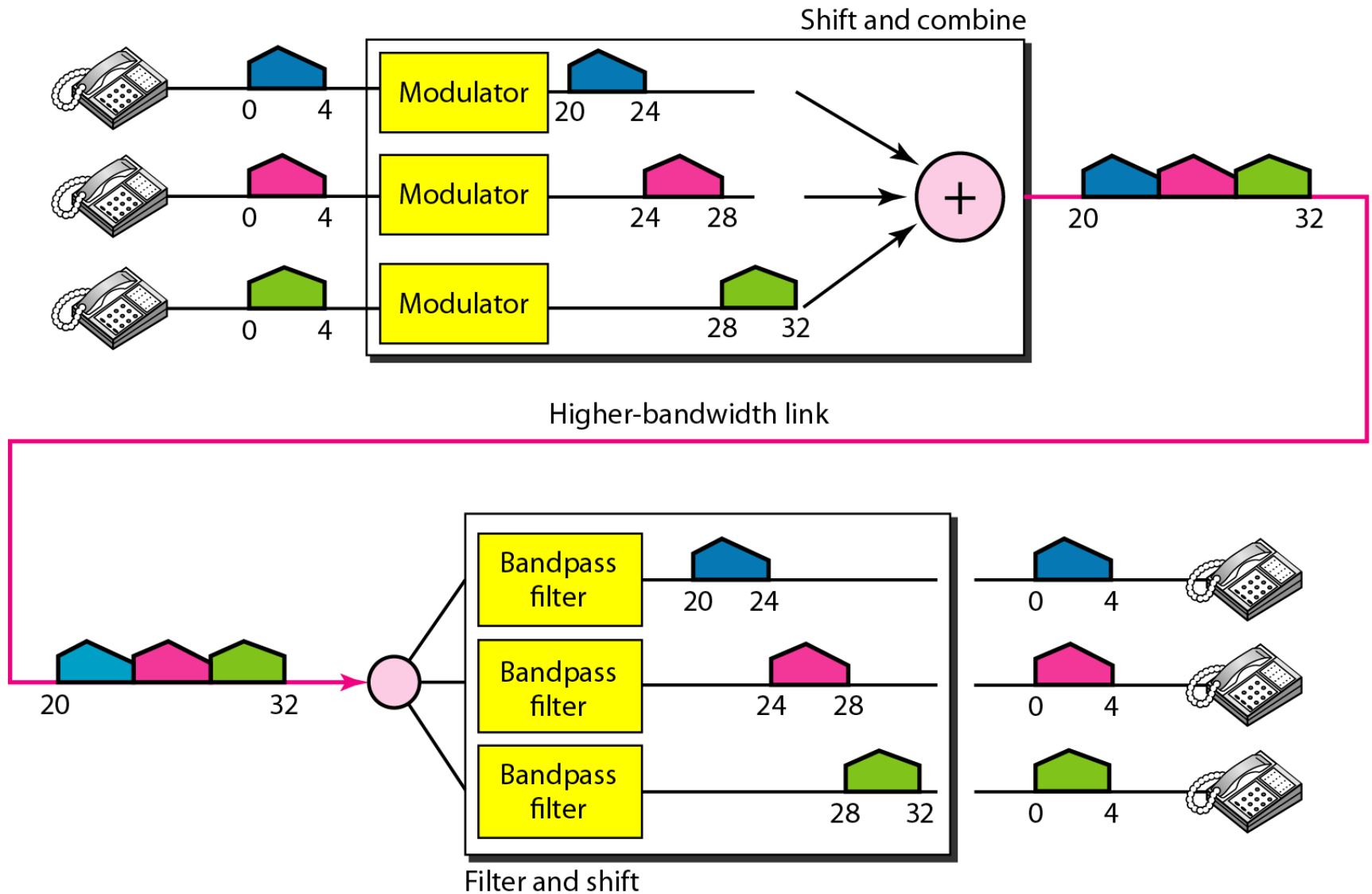
Example 6.1

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.

Solution

We shift (modulate) each of the three voice channels to a different bandwidth, as shown in Figure 6.6. We use the 20- to 24-kHz bandwidth for the first channel, the 24- to 28-kHz bandwidth for the second channel, and the 28- to 32-kHz bandwidth for the third one. Then we combine them as shown in Figure 6.6.

Figure 6.6 *Example 6.1*





Example 6.2

Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?



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Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

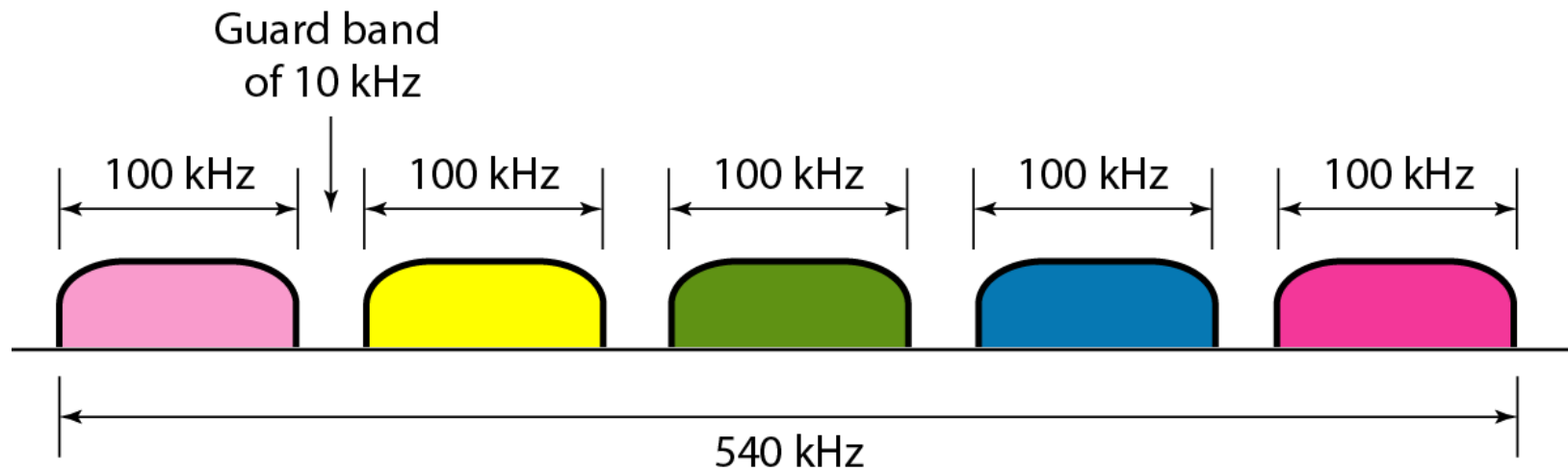
Solution

For five channels, we need at least four guard bands. This means that the required bandwidth is at least

$$5 \times 100 + 4 \times 10 = 540 \text{ kHz},$$

as shown in Figure 6.7.

Figure 6.7 *Example 6.2*





Example 6.3

Four data channels (digital), each transmitting at 1 Mbps, use a satellite channel of 1 MHz. Design an appropriate configuration, using FDM.



Example 6.3

Four data channels (digital), each transmitting at 1 Mbps, use a satellite channel of 1 MHz. Design an appropriate configuration, using FDM.

Solution Given

Number of digital channels = 4

Data rate of each channel = 1 Mbps

Total satellite bandwidth = 1 MHz (analog)

Multiplexing technique = FDM (Frequency Division Multiplexing).



Example 6.3

Solution

The satellite channel is analog. We divide it into four channels, each channel having a 250-kHz bandwidth. Each digital channel of 1 Mbps is modulated such that each 4 bits is modulated to 1 Hz. One solution is 16-QAM modulation. Figure 6.8 shows one possible configuration.

Step 1: Divide the satellite bandwidth (FDM)

Total available bandwidth = **1 MHz**

Using **FDM**, divide it equally among 4 channels:

$$\text{Bandwidth per channel} = \frac{1 \text{ MHz}}{4} = 250 \text{ kHz}$$

So:

- Each digital signal must fit into 250 kHz of analog bandwidth

Step 2: A digital signal cannot be transmitted directly over a satellite channel. It must be modulated onto an analog carrier.

Key relationship:

$$\text{Bit rate} = \text{Symbol rate} \times \text{Bits per symbol}$$

And for an **ideal channel**:

$$\text{Bandwidth} \approx \text{Symbol rate}$$

So:

•With **250 kHz bandwidth**, we can transmit $\approx 250,000$ symbols per second

Step 3: Required data rate per channel

Each channel must transmit:

$$1 \text{ Mbps} = 1,000,000 \text{ bits/sec}$$

But the channel allows only:

$$250,000 \text{ symbols/sec}$$

Step 4: Calculate bits per symbol

$$\begin{aligned} \text{Bits per symbol} &= \frac{\text{Bit rate}}{\text{Symbol rate}} \\ &= \frac{1,000,000}{250,000} = 4 \end{aligned}$$

Each symbol must carry 4 bits

Step 5: Choose a modulation scheme

Number of signal levels needed:

$$2^{\text{bits per symbol}} = 2^4 = 16$$

So we need:

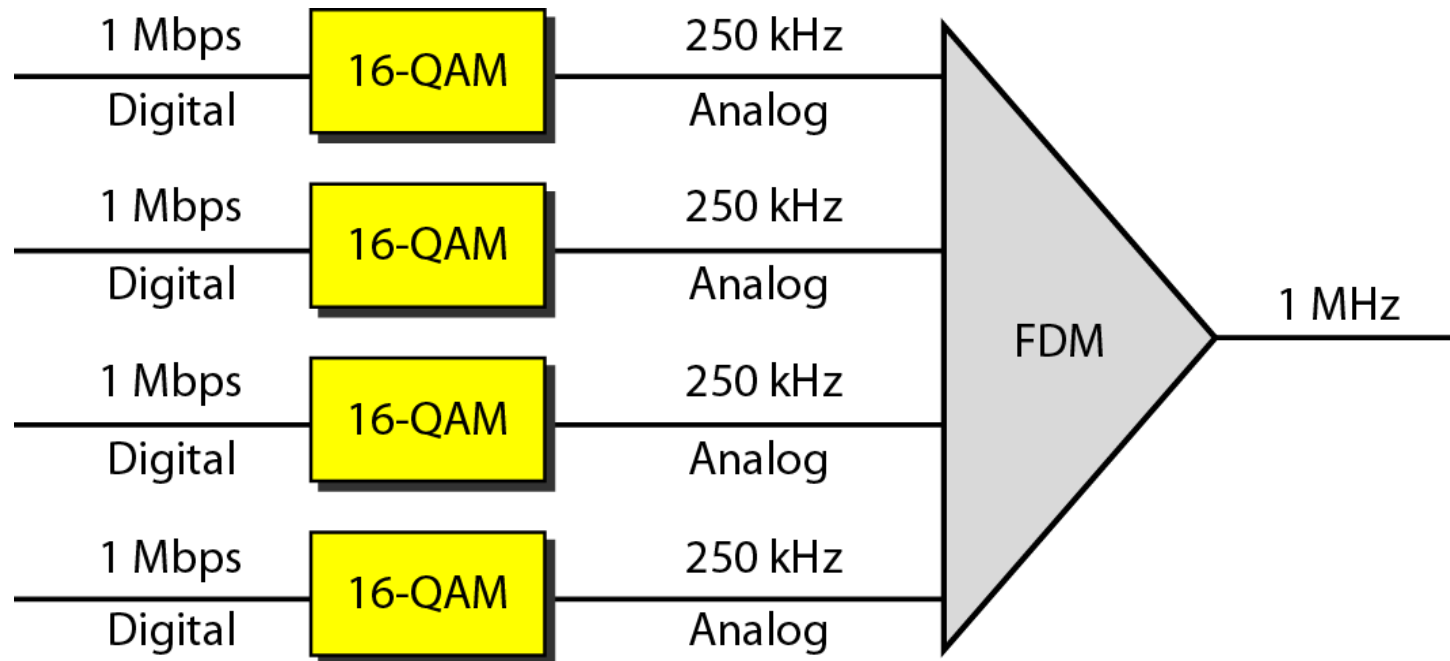
- **16 distinct signal states**

A suitable modulation technique is:

16-QAM (Quadrature Amplitude Modulation)

- Each symbol represents **4 bits**
- Symbol rate = **250 kbaud**
- Bit rate = **$250,000 \times 4 = 1 \text{ Mbps}$**
- ✓ Fits perfectly in **250 kHz bandwidth**

Figure 6.8 *Example 6.3*





The Analog Carrier System

To maximize the efficiency of their infrastructure, telephone companies have traditionally multiplexed signals from lower-bandwidth lines onto higher-bandwidth lines. In this way, many switched or leased lines can be combined into fewer but bigger channels. For analog lines, FDM is used. One of these hierarchical systems used by telephone companies is made up of groups, supergroups, master groups, and jumbo groups (see Figure 6.9)



The Analog Carrier System

In this **analog hierarchy**, 12 voice channels are multiplexed onto a higher-bandwidth line to create a **group**. A group has 48 kHz of bandwidth and supports 12 voice channels.

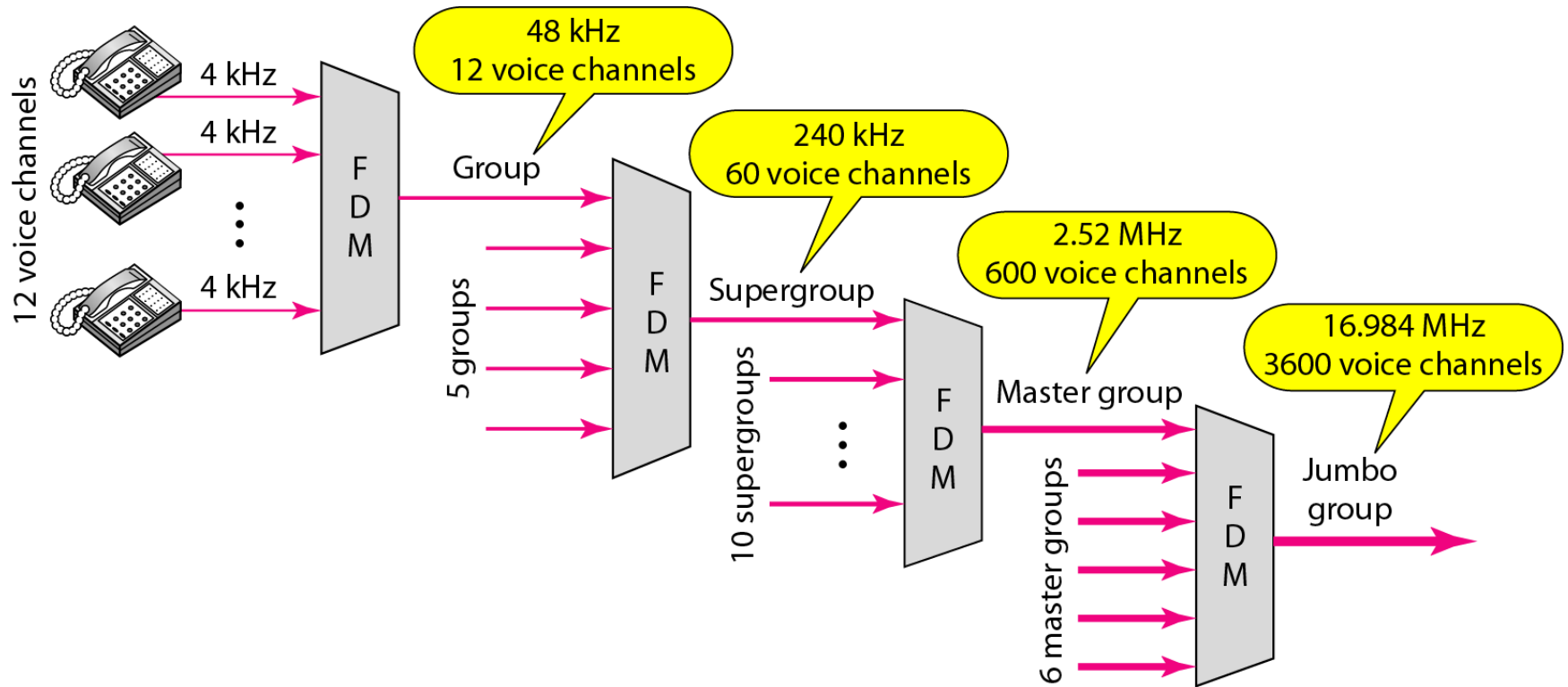
At the next level, up to five **groups** can be multiplexed to create a composite signal called a **supergroup**.

A **supergroup** has a bandwidth of 240 kHz and supports up to 60 voice channels. Supergroups can be made up of either five groups or 60 independent voice channels.

At the next level, 10 supergroups are multiplexed to create a **master group**. A master group must have 2.40 MHz of bandwidth, but the need for guard bands between the supergroups increases the necessary bandwidth to 2.52 MHz. Master groups support up to 600 voice channels.

Finally, six master groups can be combined into a **jumbo group**. A jumbo group must have 15.12 MHz (6×2.52 MHz) but is augmented to 16.984 MHz to allow for guard bands between the master groups.

Figure 6.9 *Analog hierarchy*



Why FDM is Used

- No waiting time for transmission
 - Allows simultaneous transmission of multiple signals
 - Efficient use of available bandwidth
 - Each user gets a dedicated frequency band
-

Applications of FDM in Data Communication

1. Radio Broadcasting

Different radio stations use **different frequency bands**

All stations transmit **simultaneously** over air

 *Example:* FM radio stations (91.1 MHz, 93.5 MHz, etc.)

Applications of FDM in Data Communication

2. Television Broadcasting

- Each TV channel is assigned a **separate frequency band**
- Multiple channels transmitted at the same time

 *Example:* Cable TV networks

Applications of FDM in Data Communication

3. Telephone Systems (Traditional PSTN)

Multiple voice calls are transmitted over a single trunk line

Each call occupies a separate frequency band

 *Example:* Long-distance telephone communication

4. Cable Television (CATV)

Internet data, TV channels, and voice signals are sent together

Each service uses different frequency ranges

 *Example:* Broadband cable connections

Applications of FDM in Data Communication

5. Satellite Communication

- Satellite transponders use FDM to carry:
 - TV channels
 - Radio signals
 - Data communication



Example: DTH television services

6. Mobile Communication (Early Systems)

- Used in 1G and early 2G mobile systems
 - Each user was allocated a fixed frequency band
-

Applications of FDM in Data Communication

Use Case: Cable TV Network

One coaxial cable carries:

- News channels

- Sports channels

- Movies

Each channel occupies a **different frequency band**

All channels reach the TV **at the same time**



Result:

User selects a channel without affecting others.

Advantages and Limitations

Advantages of FDM

- Continuous data transmission
- No time synchronization required
- Simple receiver design

Limitations of FDM

- Requires **guard bands** (wastes bandwidth)
 - Less efficient for bursty data
 - More prone to noise and interference
-

Other Applications of FDM

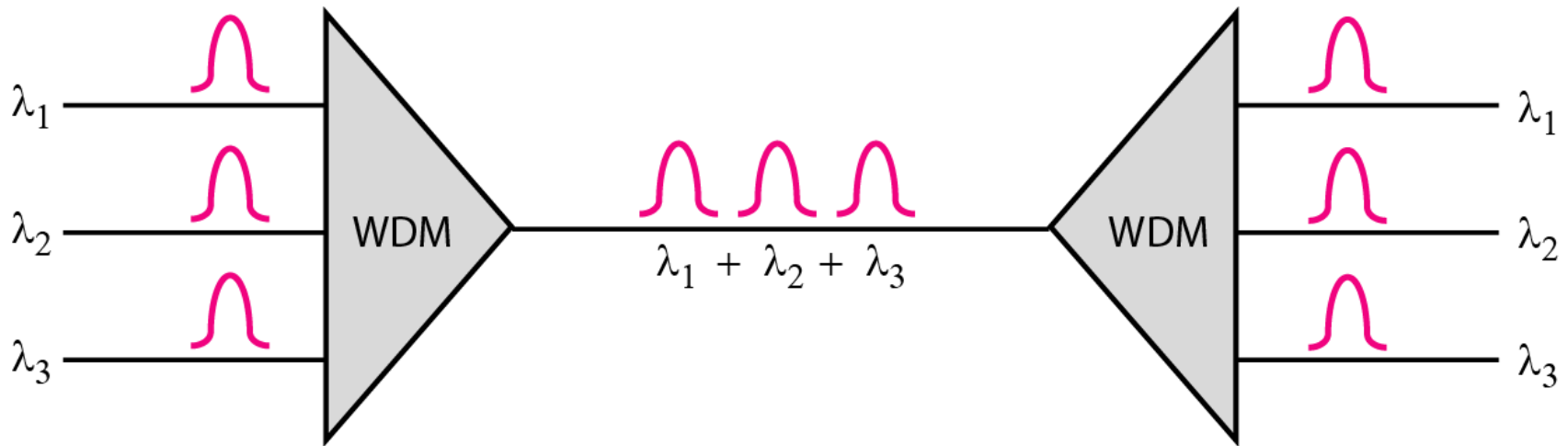
Page Number: *CHAPTER 6 BANDWIDTH UTILIZATION: MULTIPLEXING AND
SPECTRUM SPREADING* **161**



Note

WDM is an analog multiplexing technique to combine optical signals.

Figure 6.10 *Wavelength-division multiplexing*



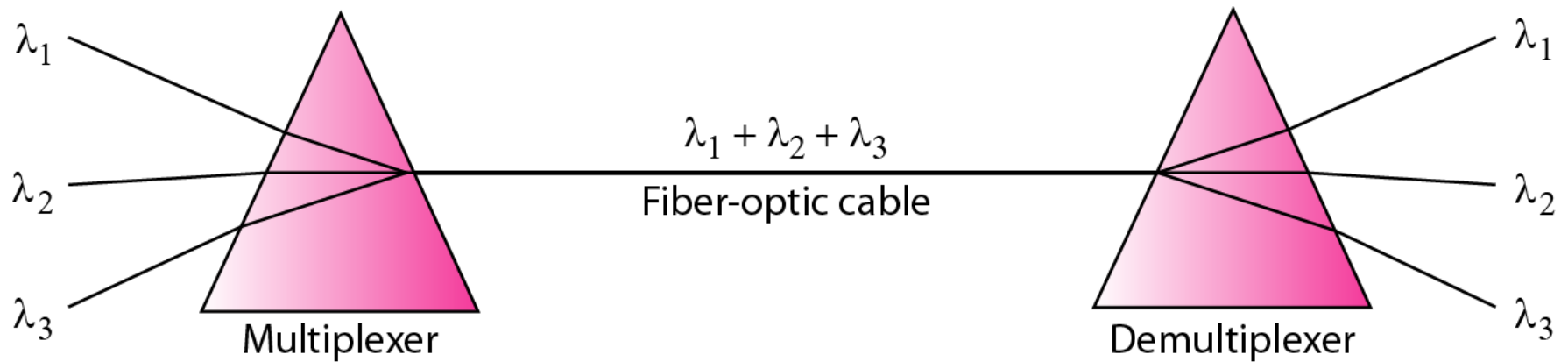
Wavelength Division Multiplexing (WDM)

- **WDM** is a multiplexing technique designed for **fiber-optic communication**.
 - Optical fiber supports **very high data rates**, much higher than metallic cables.
 - Using a fiber for a **single signal wastes available bandwidth**.
 - **Multiplexing** allows multiple data channels to share **one optical fiber**.
 - **WDM is conceptually similar to FDM**, but:
 - FDM uses **electrical frequencies**
 - WDM uses **optical wavelengths (light)**
 - Multiple **narrow bands of light** (different wavelengths) are combined into:
 - **One wider band of light** at the multiplexer
-

Wavelength Division Multiplexing (WDM)

- At the receiver:
 - The **demultiplexer separates** the combined light back into original wavelengths
 - **Prisms** (or equivalent optical devices) are used to:
 - Combine light beams at the **MUX**
 - Split light beams at the **DEMUX**
 - **WDM is an analog multiplexing technique** for optical signals.
 - One major application of WDM is the **SONET optical network**.
 - **Dense WDM (DWDM):**
 - Uses **very close wavelength spacing**
 - Supports a **large number of channels**
 - Provides **very high bandwidth efficiency**
-

Figure 6.11 *Prisms in wavelength-division multiplexing and demultiplexing*



Wavelength Division Multiplexing (WDM)

What is SONET?

SONET stands for **Synchronous Optical Network**.

SONET is like a highway system for data using light, where many data streams travel together in an organized and synchronized way.

- **Is the guard band concept used in WDM like FDM?**

YES, but much smaller

- ◆ **Guard bands in WDM**

- Called **channel spacing**
 - Used between **light wavelengths**
 - Extremely **narrow**, because:
 - Light frequencies are very stable
 - Lasers are highly precise
-

Wavelength Division Multiplexing (WDM)

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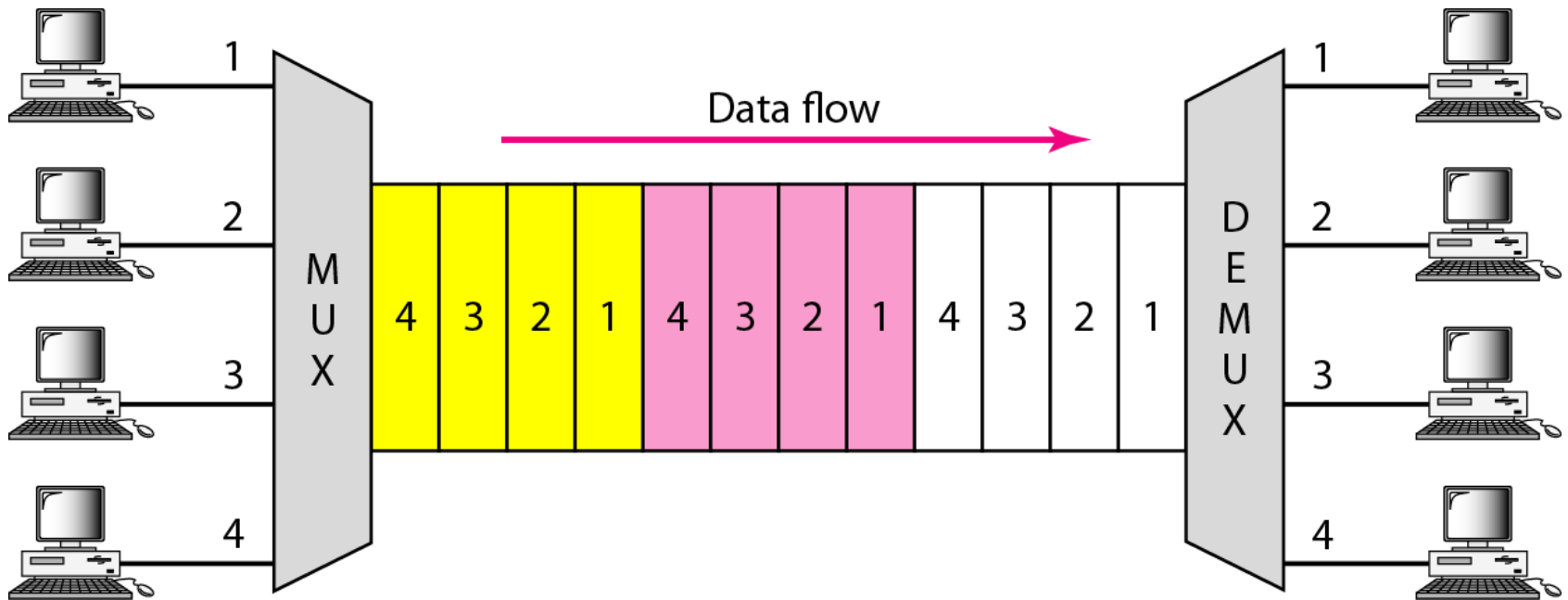
- Called **channel spacing**
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 - Extremely **narrow**, because:
 - Light frequencies are very stable
 - Lasers are highly precise
-



Note

**TDM is a digital multiplexing technique
for combining several low-rate
channels into one high-rate one.**

Figure 6.12 *TDM*



Time Division Multiplexing (TDM)

- TDM is a digital multiplexing technique.
 - It allows multiple connections to share the high bandwidth of a single link.
 - Unlike FDM, bandwidth is not divided — time is divided.
 - Each connection uses the entire channel bandwidth, but only for a short time.
 - Data from different sources are transmitted sequentially in time slots.
 - The same physical link is used, but it is divided by time, not frequency.
 - TDM involves multiplexing only, not switching.
 - Data from each source always goes to a fixed destination.
-

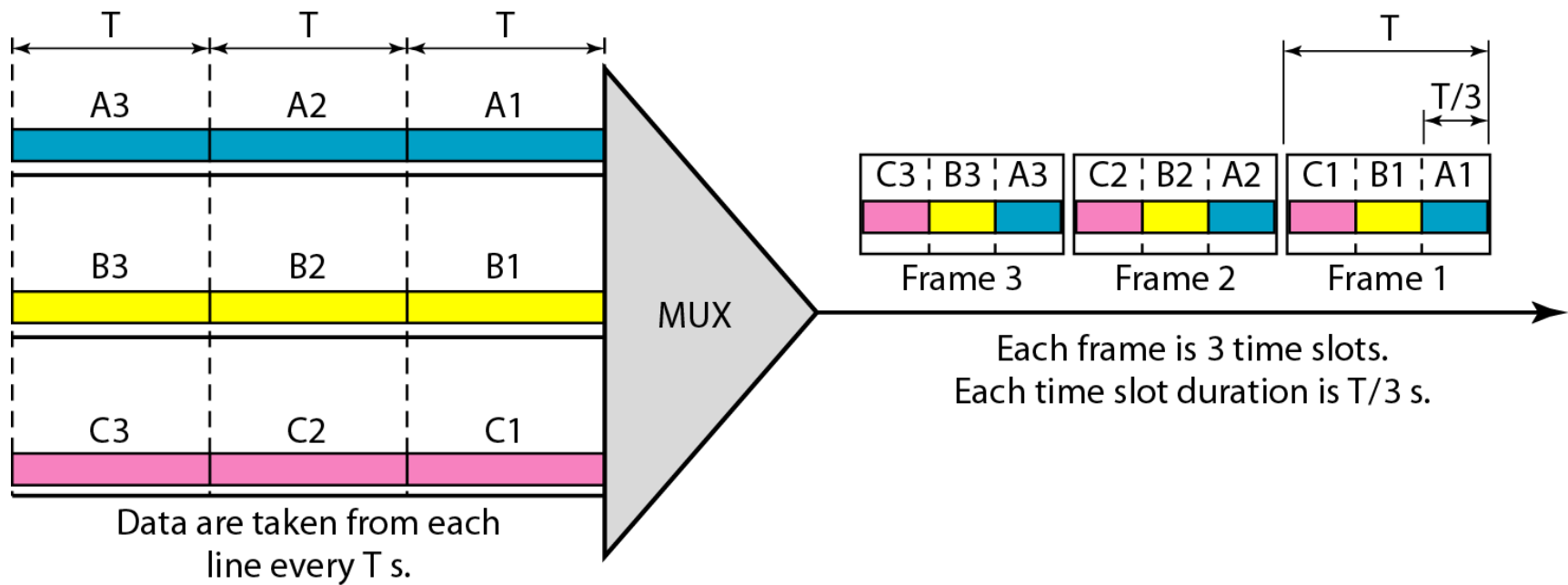
Time Division Multiplexing (TDM)

- Primarily used for digital data transmission.
 - Analog data can also be used after:
 - Sampling
 - Conversion to digital form
 - TDM combines several low-rate channels into one high-rate channel.
 - TDM is classified into:
 - **Synchronous TDM**
 - **Statistical TDM**
-

Note

In synchronous TDM, the data rate of the link is n times faster, and the unit duration is n times shorter.

Figure 6.13 *Synchronous time-division multiplexing*



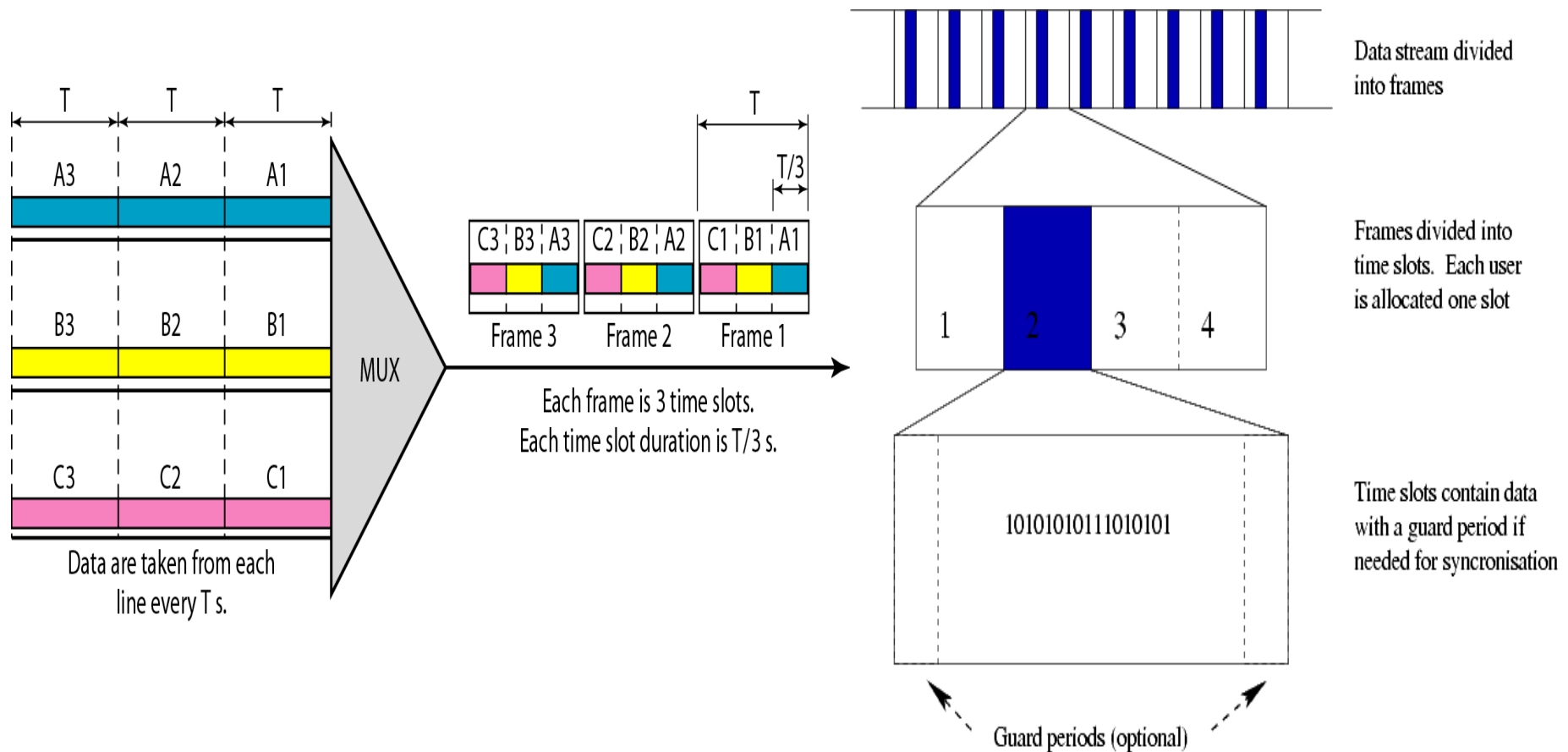
Synchronous Time Division Multiplexing (TDM)

- Synchronous TDM is a digital multiplexing technique.
 - Data from multiple input connections are collected into a frame.
 - If there are n input connections:
 - Each frame is divided into n fixed time slots
 - One time slot is permanently assigned to each input line
 - Time slots are transmitted in a round-robin (cyclic) order.
 - Even if a source has no data, its time slot remains reserved.
 - The output link data rate must be n times the data rate of each input connection.
 - Time slots together form a frame, and frames repeat continuously.
 - Synchronous TDM involves multiplexing only, not switching.
 - Used to combine several low-rate channels into one high-rate channel.
-

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-

Figure 6.13 *Synchronous time-division multiplexing*



Synchronous Time Division Multiplexing (TDM)

Frame and Time Slot Relationship

Let T = duration of one data unit on an input line

For n input lines:

Time slot duration = T / n

Frame duration = T

Each frame contains:

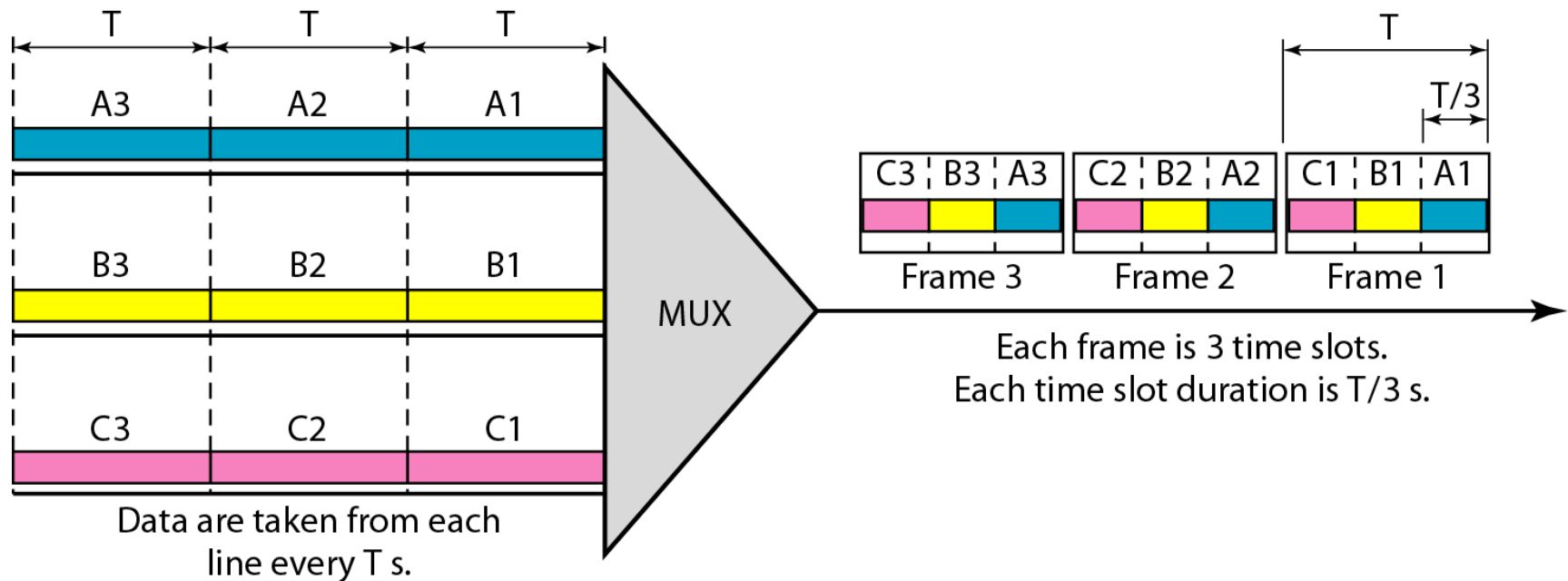
One data unit from each input line

Figure 6.13 shows synchronous TDM with 3 input connections (A, B, C):

- Each frame has 3 time slots (one for A, B, and C).
 - Frame 1 contains: A1, B1, C1
 - Frame 2 contains: A2, B2, C2
 - Frame 3 contains: A3, B3, C3
 - Data from each input line is taken once every T seconds.
 - Each time slot duration = $T / 3$ seconds.
 - The output data rate is 3 times the data rate of each input line.
 - Data units after multiplexing are shorter in duration, allowing higher transmission speed.
-

Example 6.5

In Figure 6.13, the data rate for each input connection is 3 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of (a) each input slot, (b) each output slot, and (c) each frame?





Example 6.5

In Figure 6.13, the data rate for each input connection is 1 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of (a) each input slot, (b) each output slot, and (c) each frame?

Solution

We can answer the questions as follows:

- a. The data rate of each input connection is 1 kbps. This means that the bit duration is $1/1000$ s or 1 ms. The duration of the input time slot is 1 ms (same as bit duration).*



Example 6.5 (continued)

- b. The duration of each output time slot is one-third of the input time slot. This means that the duration of the output time slot is $1/3$ ms.*
- c. Each frame carries three output time slots. So the duration of a frame is $3 \times 1/3$ ms, or 1 ms. The duration of a frame is the same as the duration of an input unit.*

Example 6.6

Figure 6.14 shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find (a) the input bit duration, (b) the output bit duration, (c) the output bit rate, and (d) the output frame rate.

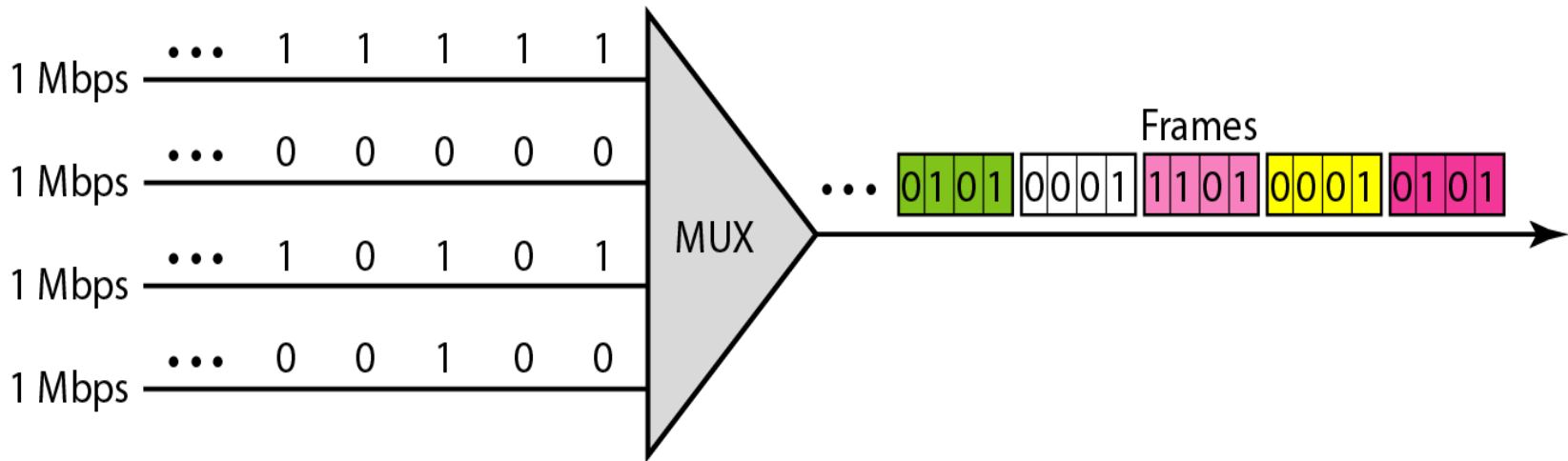
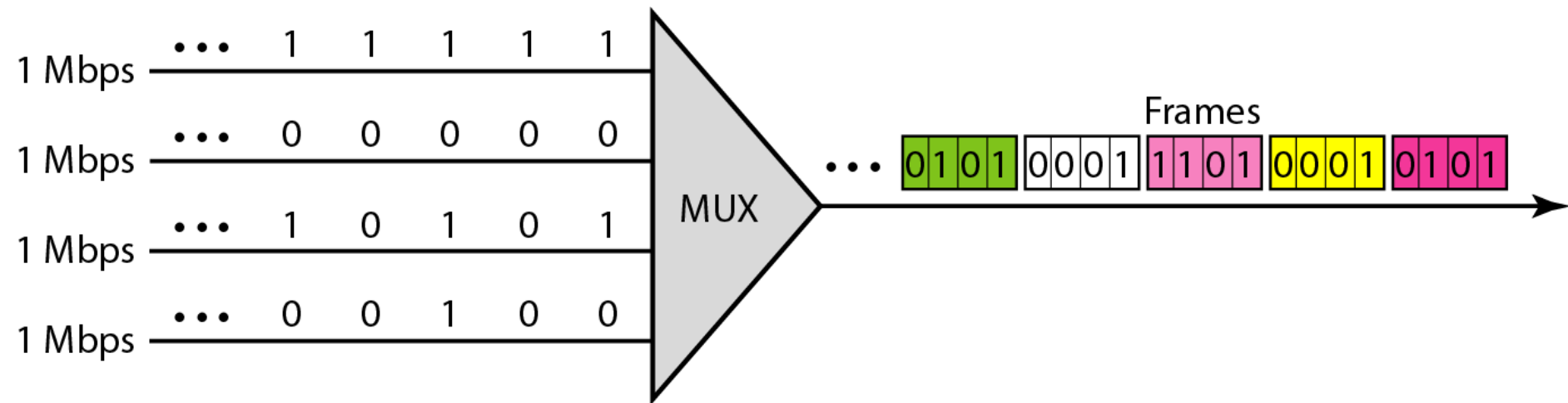


Figure 6.14 *Example 6.6*





Example 6.6

Figure 6.14 shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find (a) the input bit duration, (b) the output bit duration, (c) the output bit rate, and (d) the output frame rate.

Solution

We can answer the questions as follows:

- a. The input bit duration is the inverse of the bit rate:
 $1/1 \text{ Mbps} = 1 \mu\text{s}.$*
- b. The output bit duration is one-fourth of the input bit duration, or $\frac{1}{4} \mu\text{s}.$*

Example 6.6 (continued)

- c. *The output bit rate is the inverse of the output bit duration or $1/(4\mu\text{s})$ or 4 Mbps. This can also be deduced from the fact that the output rate is 4 times as fast as any input rate; so the output rate = $4 \times 1 \text{ Mbps}$ = 4 Mbps.*
- d. *The frame rate is always the same as any input rate. So the frame rate is 1,000,000 frames per second. Because we are sending 4 bits in each frame, we can verify the result of the previous question by multiplying the frame rate by the number of bits per frame.*



Example 6.7

Four 1-kbps connections are multiplexed together. A unit is 1 bit. Find (a) the duration of 1 bit before multiplexing, (b) the transmission rate of the link, (c) the duration of a time slot, and (d) the duration of a frame.

Solution

We can answer the questions as follows:

- a. The duration of 1 bit before multiplexing is $1 / 1 \text{ kbps}$, or 0.001 s (1 ms).*
- b. The rate of the link is 4 times the rate of a connection, or 4 kbps.*



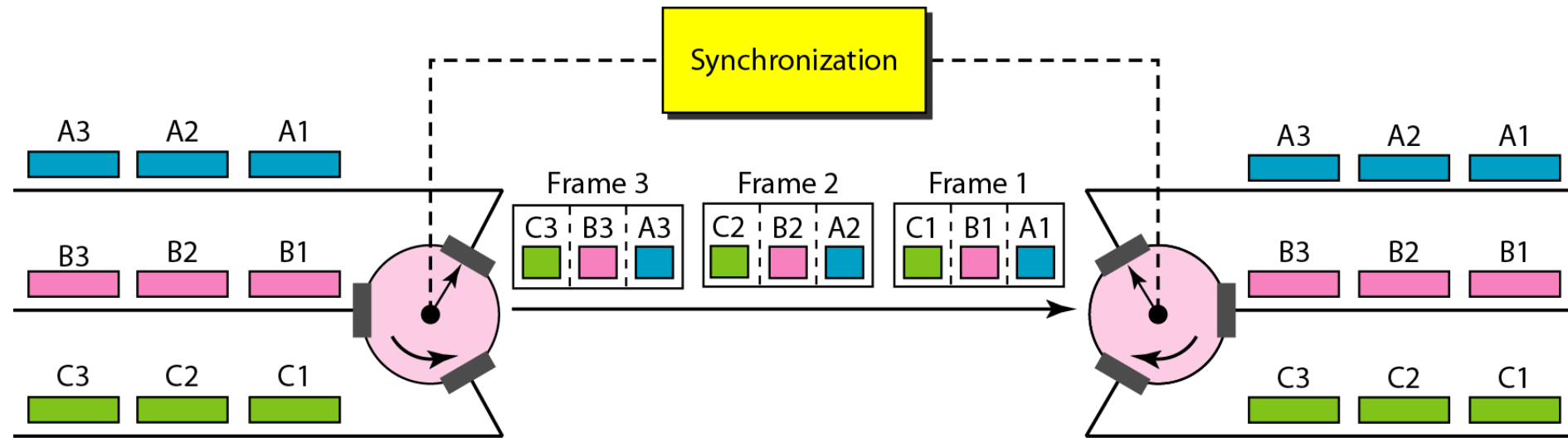
Example 6.7 (continued)

- c. The duration of each time slot is one-fourth of the duration of each bit before multiplexing, or $1/4$ ms or $250\text{ }\mu\text{s}$. Note that we can also calculate this from the data rate of the link, 4 kbps. The bit duration is the inverse of the data rate, or $1/4$ kbps or $250\text{ }\mu\text{s}$.*
- d. The duration of a frame is always the same as the duration of a unit before multiplexing, or 1 ms. We can also calculate this in another way. Each frame in this case has four time slots. So the duration of a frame is 4 times $250\text{ }\mu\text{s}$, or 1 ms.*

Interleaving

- TDM can be visualized as two fast-rotating switches, one on the multiplexing side and the other on the demultiplexing side.
- The switches are synchronized and rotate at the same speed, but in opposite directions.
- On the multiplexing side, as the switch opens in front of a connection, that connection has the opportunity to send a unit onto the path.
- This process is called **interleaving**.
- On the demultiplexing side, as the switch opens in front of a connection, that connection has the opportunity to receive a unit from the path.

Figure 6.15 *Interleaving*





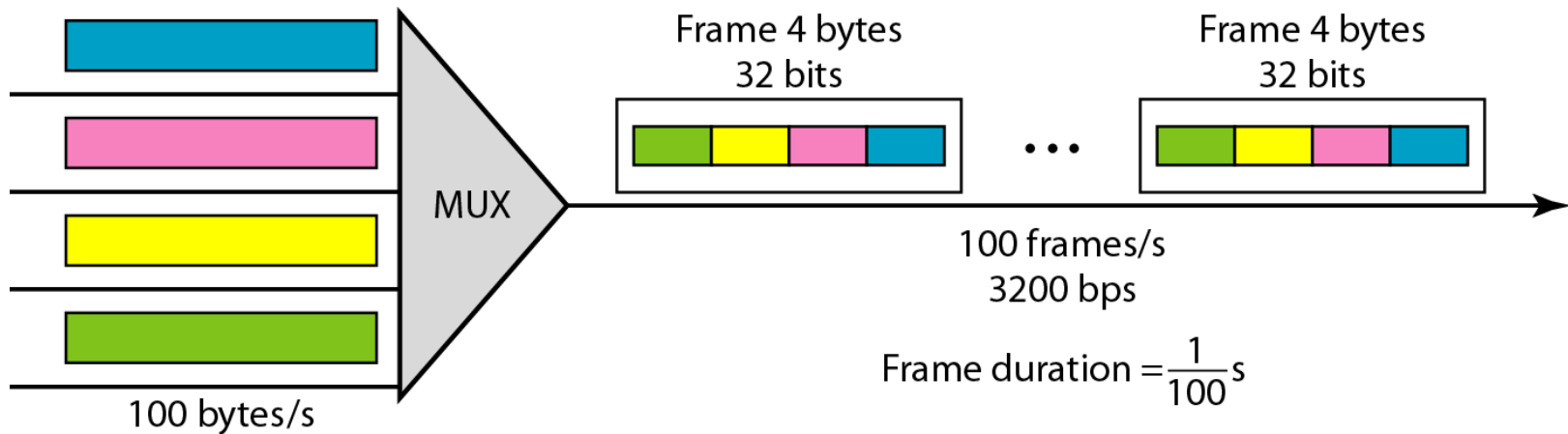
Example 6.8

Four channels are multiplexed using TDM. If each channel sends 100 bytes /s and we multiplex 1 byte per channel, show the frame traveling on the link, the size of the frame, the duration of a frame, the frame rate, and the bit rate for the link.

Solution

The multiplexer is shown in Figure 6.16. Each frame carries 1 byte from each channel; the size of each frame, therefore, is 4 bytes, or 32 bits. Because each channel is sending 100 bytes/s and a frame carries 1 byte from each channel, the frame rate must be 100 frames per second. The bit rate is 100×32 , or 3200 bps.

Figure 6.16 *Example 6.8*



Problems and Solution of TDM

Empty Slots

Synchronous TDM is not as efficient as it could be. If a source does not have data to send, the corresponding slot in the output frame is empty. Figure 6.18 shows a case in which one of the input lines has no data to send and one slot in another input line has discontinuous data.

The first output frame has three slots filled, the second frame has two slots filled, and the third frame has three slots filled. No frame is full.

Data Rate Management

One problem with TDM is how to handle a disparity in the input data rates. In all our discussion so far, we assumed that the data rates of all input lines were the same. However, if data rates are not the same, three strategies, or a combination of them, can be used. We call these three strategies **multilevel multiplexing**, **multiple-slot allocation**, and **pulse stuffing**.

Figure 6.18 *Empty slots*

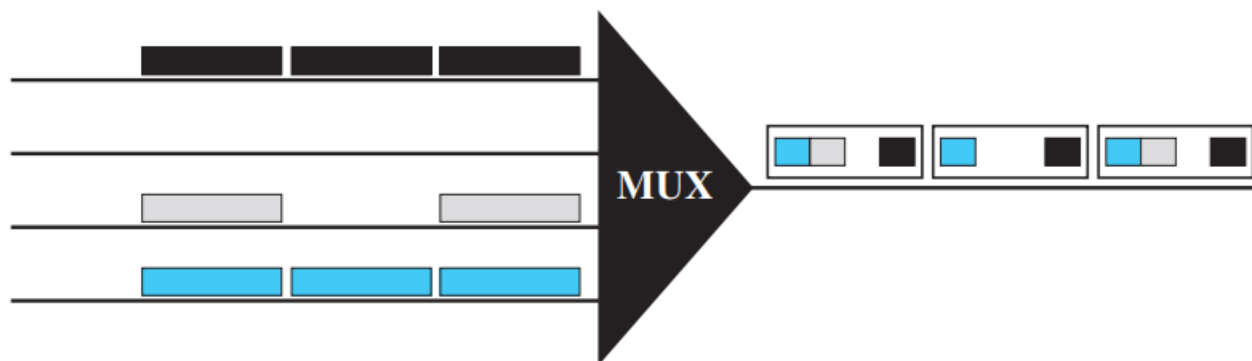


Figure 6.19 *Multilevel multiplexing*

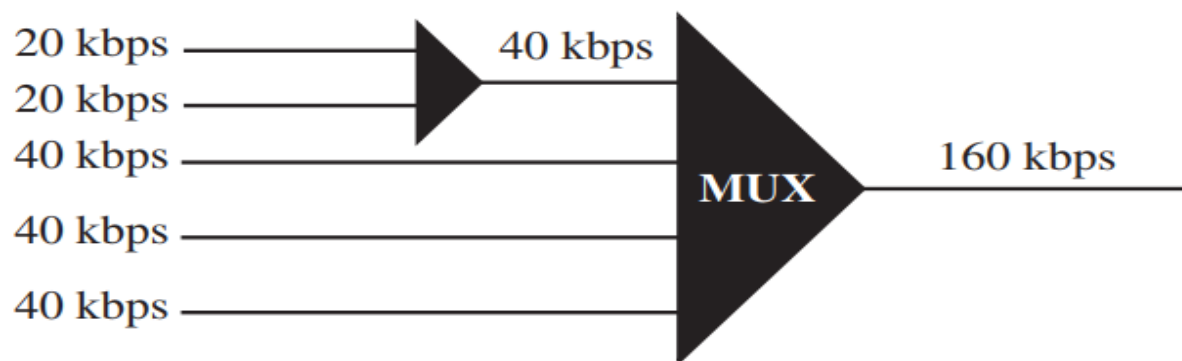


Figure 6.20 *Multiple-slot multiplexing*

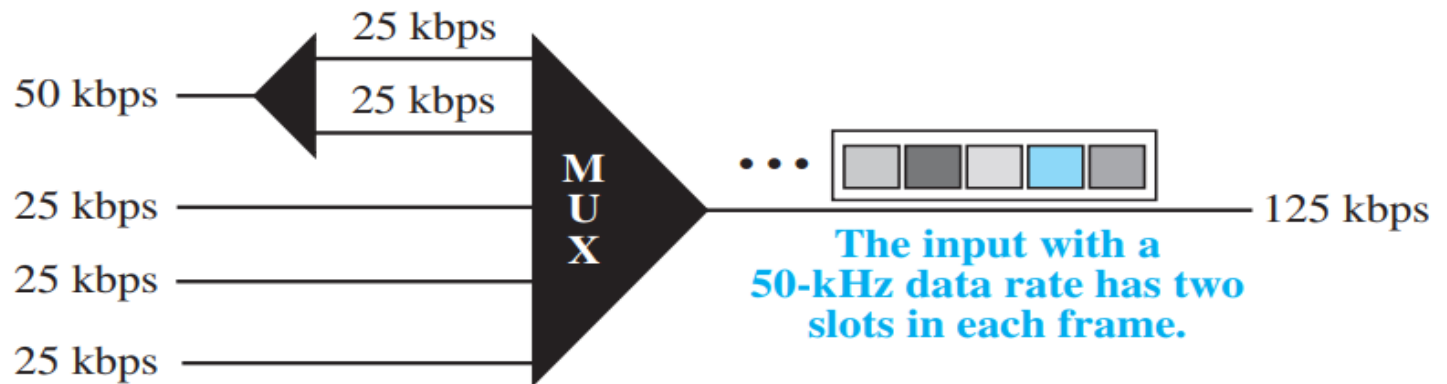
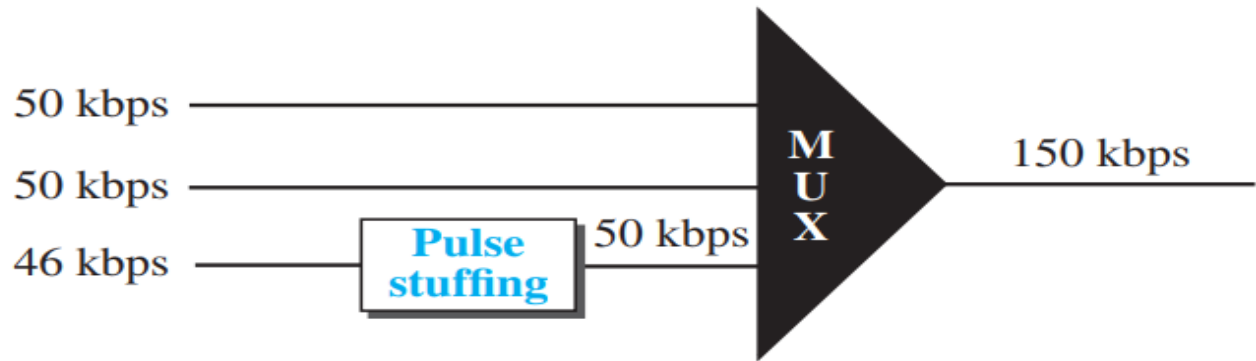
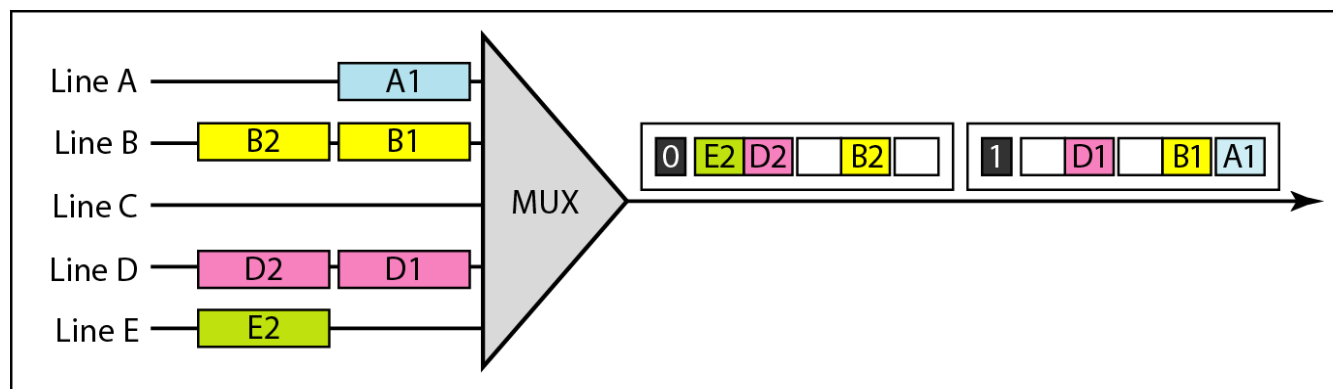


Figure 6.21 *Pulse stuffing*

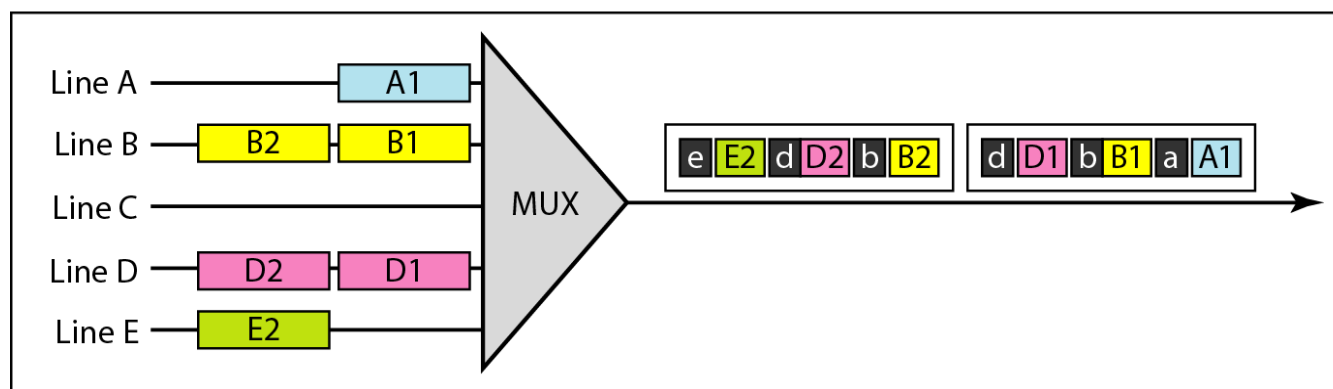


Statistical Time Division Multiplexing

Figure 6.26 *TDM slot comparison*



a. Synchronous TDM



b. Statistical TDM

Statistical Time Division Multiplexing (Statistical TDM)

- Statistical TDM is an improved form of time-division multiplexing.
 - It overcomes the inefficiency of synchronous TDM.
 - In synchronous TDM:
 - Each input line has a fixed, reserved time slot
 - Slots may remain empty if a line has no data
 - In statistical TDM:
 - Time slots are dynamically allocated
 - A slot is assigned only when an input line has data to send
 - The number of slots per frame is less than the number of input lines.
 - This results in better bandwidth utilization.
 - The multiplexer scans input lines in round-robin order.
 - If a line has data:
 - It is assigned a time slot
 - If a line has no data:
 - It is skipped
 - As long as there is data from any input line, no slot remains empty.
-

Thank you